

CONSOLIDATED VERSION



Ultrasonics – Pulse-echo scanners – Part 1: Techniques for calibrating spatial measurement systems and measurement of system point-spread function response



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**Ultrasonics – Pulse-echo scanners –
Part 1: Techniques for calibrating spatial measurement systems and
measurement of system point-spread function response**

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ULTRASONICS – PULSE-ECHO SCANNERS –**Part 1: Techniques for calibrating spatial measurement systems
and measurement of system point-spread function response**

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This Consolidated version of IEC 61391-1 bears the edition number 1.1. It consists of the first edition (2006-07) [documents 87/336/FDIS and 87/343/RVD] and its amendment 1 (2017-07) [documents 87/650/FDIS and 87/653/RVD]. The technical content is identical to the base edition and its amendment.

In this Redline version, a vertical line in the margin shows where the technical content is modified by amendment 1. Additions are in green text, deletions are in strikethrough red text. A separate Final version with all changes accepted is available in this publication.

International Standard IEC 61391-1 has been prepared by IEC technical committee 87: Ultrasonics.

This publication has been drafted in accordance with the ISO/IEC Directives, Part 2.

Terms in **bold** in the text are defined in clause 3.

This standard is intended to be published in two or more parts:

- Part 1 deals with techniques for calibrating spatial measurement systems and measurement of system point-spread function response;
- Part 2 will deal with measurement of system sensitivity, dynamic range, and low-contrast resolution.

The committee has decided that the contents of the base publication and its amendment will remain unchanged until the stability date indicated on the IEC web site under "<http://webstore.iec.ch>" in the data related to the specific publication. At this date, the publication will be

- reconfirmed,
- withdrawn,
- replaced by a revised edition, or
- amended.

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INTRODUCTION

An ultrasonic pulse-echo scanner produces images of tissue in an ultrasonic **scan plane** by sweeping a narrow pulsed beam of **ultrasound** through the section of interest and detecting the echoes generated at tissue boundaries. A variety of **ultrasonic transducer** types are employed to operate in a transmit/receive mode for the ultrasonic signals. Ultrasonic scanners are widely used in medical practice to produce images of many soft-tissue organs throughout the human body.

This standard describes test procedures that should be widely acceptable and valid for a wide range of types of equipment. Manufacturers should use the standard to prepare their specifications; the users should employ the standard to check specifications. The measurements can be carried out without interfering with the normal working conditions of the machine. Typical **test objects** are described in the annexes. The structures of the **test objects** have not been specified in detail, rather suitable types of overall and internal structures are described. The specific structure of a **test object** should be reported with the results obtained using it. Similar commercial versions of these **test objects** are available.

The performance parameters specified and the corresponding methods of measurement have been chosen to provide a basis for comparison with the manufacturer's specification and between similar types of apparatus of different makes, intended for the same kind of diagnostic application. The manufacturer's specification should allow comparison with the results obtained from the tests in this standard. Furthermore, it is intended that the sets of results and values obtained from the use of the recommended methods will provide useful criteria for predicting the performance of equipment in appropriate diagnostic applications. This standard concentrates on measurements of images by digital techniques. Methods suitable for inspection by eye are covered here as well. Discussion of other visual techniques can be found in IEC 61390 [1] ¹⁾.

Where a diagnostic system accommodates more than one option in respect of a particular system component, for example the **ultrasonic transducer**, it is intended that each option be regarded as a separate system. However, it is considered that the performance of a machine is adequately specified, if measurements are undertaken for the most significant combinations of machine control settings and accessories. Further evaluation of equipment is obviously possible but this should be considered as a special case rather than a routine requirement.

1) Figures in square brackets refer to the Bibliography.

ULTRASONICS – PULSE-ECHO SCANNERS –

Part 1: Techniques for calibrating spatial measurement systems and measurement of system point-spread function response

1 Scope

This International Standard describes methods of calibrating the spatial measurement facilities and **point-spread function** of ultrasonic imaging equipment in the ultrasonic frequency range 0,5 MHz to 15 MHz. This standard is relevant for ultrasonic scanners based on the pulse-echo principle of the types listed below:

- mechanical sector scanners;
- electronic phased-array sector scanners;
- electronic linear-array scanners;
- electronic curved-array sector scanners;
- water-bath scanners based on any of the above four scanning mechanisms;
- 3D-volume reconstruction systems.

2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

~~IEC 61102:1991, Measurement and characterisation of ultrasonic fields using hydrophones in the frequency range 0,5 MHz to 15 MHz~~

IEC 60050-801:1994, *International Electrotechnical Vocabulary – Chapter 801: Acoustics and electroacoustics*

IEC 60050-802:2011, *International Electrotechnical Vocabulary – Part 802: Ultrasonics*

IEC 61685:2001, *Ultrasonics – Flow measurement systems – Flow test object*

IEC 62127-1:2007, *Ultrasonics – Hydrophones – Part 1: Measurement and characterization of medical ultrasonic fields up to 40 MHz*

3 Terms and definitions

For the purposes of this document, the ~~following~~ terms and definitions given in IEC 60050-801:1994, IEC 60050-802:2011, IEC 62127-1:2007 and the following apply. See also related International Standards, Technical Specifications and Technical Reports for definitions and explanations [1] [2] [3] [4] [34] [35] [36] [37] [38] [39]. ~~[1-5]~~

3.1

A-scan

class of data acquisition geometry in one dimension, in which echo strength information is acquired from points lying along a single **beam axis** and displayed as amplitude versus time of flight or distance

3.2

acoustic coupling agent (also, coupling agent)

a material, usually a gel or other fluid, that is used to ensure acoustic contact between the transducer and the patient's skin, or between the transducer and the surface of a sealed test object

3.3

acoustic working frequency

arithmetic mean of the frequencies f_1 and f_2 at which the amplitude of the acoustic pressure spectrum is 3 dB below the peak amplitude

(See 3.4.2 of IEC 61102)

3.4

automatic time-gain compensation

ATGC

automatic working time gain control based on the observed decrease in echo amplitudes due to the attenuation in ultrasonic pulse amplitude with depth

3.5

axial resolution

minimum separation along the beam axis of two equally scattering volumes or **targets** at a specified depth for which two distinct echo signals can be displayed

3.6

backscatter coefficient

mean acoustic power scattered in the 180° direction by a specified object with respect to the direction of the incident beam, per unit solid angle per unit volume, divided by the incident beam intensity. For a volume filled with many scatterers, the scatterers are considered to be randomly distributed. The mean power is obtained from different spatial realisations of the scattering volume

NOTE **Backscatter coefficient** is commonly referred to as the differential scattering cross-section per unit volume in the 180° direction

3.7

backscatter contrast (normalized)

difference between the **backscatter coefficients** from two defined regions divided by the square root of the product of the two **backscatter coefficients**

3.8

beam axis

the longitudinal axis of the pulse-echo response pattern of a given **B-mode scan line**, a pulse-echo equivalent to the transmitted beam axis of IEC 61828 [2]

3.9

B-scan

class of data acquisition geometry in which echo information is acquired from points lying in an ultrasonic **scan plane** containing interrogating ultrasonic beams. See **B-mode** below.

NOTE B-scan is a colloquial term for **B-mode** scan or image. (See 3.10)

3.10

Brightness-modulated display

B-mode

method of presentation of **B-scan** information in which a particular section through an imaged object is represented in a conformal way by the scan plane of the display and echo amplitude is represented by local brightness or optical density of the display

[IEC 60854: definition 3.18, modified]

3.11

displayed dynamic range

ratio, expressed in decibels, of the amplitude of the maximum echo that does not saturate the display to the minimum echo that can be distinguished in the display under the scanner test settings

3.12

elevational resolution

minimum separation perpendicular to the ultrasonic **scan plane** of two equally scattering **targets** at a specified depth for which two distinct echo signals can be displayed. Often used here informally for slice thickness for purposes of 3D-scanning

3.13

field-of-view

area in the ultrasonic **scan plane** which is insonated by the **ultrasound beam** during the acquisition of echo data to produce one image frame

3.14

frame rate

number of sweeps comprising the full-frame refresh rate that the ultrasonic beam makes per second through the **field-of-view**

3.15

gain

ratio of the output to the input of a system, generally an amplifying system, usually expressed in decibels

3.16

grey scale

range of values of image brightness, being either continuous between two extreme values or, if discontinuous, including at least three discrete values

[IEC 60854: definition 3.14]

3.17

lateral resolution

minimum separation of two **line targets** at a specified depth in a **test object** made of **tissue-mimicking material** for which two distinct echo signals can be displayed. The **line targets** should be perpendicular to the scanned plane; the separation between the **targets** should be perpendicular to the beam-alignment axis

3.18

line-spread function

LSF

characteristic response in three dimensions of an imaging system to a high-contrast line target

3.19

line target

cylindrical reflector whose diameter is so small that the reflector cannot be distinguished by the imaging system from a cylindrical reflector with diameter an order of magnitude smaller, except by signal amplitude. The backscatter from a standard **line target** should be a simple function of frequency over the range of frequencies studied

3.20

M-mode time-motion mode

method of presentation of **M-scan** information in which the motion of structures along a fixed **beam axis** is depicted by presenting their positions on a line which moves across a display to show the variation with time of the echo

3.21

M-scan time-motion scan

class of acquisition geometry in which echo information from moving structures is acquired from points lying along a single beam axis. The echo strength information is presented using an **M-mode** display

3.22

nominal frequency (of a transducer)

intended **acoustic working frequency** of a transducer as quoted by the designer or manufacturer

[adapted from definition 3.7 of IEC 60854]

3.23

pixel

picture element

smallest spatial unit or cell size of a digitized 2-dimensional array representation of an image. Each **pixel** has an address (x-and y-coordinates corresponding to its position in the array) and a specific brightness level

NOTE **Pixel** is a contraction of 'picture element'.

3.24

point target

reflector whose scattering surface dimensions are so small that it cannot be distinguished (except by signal amplitude) by the imaging system from a similar **target** whose scattering surface is an order of magnitude smaller. The backscatter cross section of a standard point **target** should be a simple function of frequency over the range of frequencies studied.

3.25

point-spread function PSF

characteristic response in three dimensions of an imaging system to a high-contrast **point target**.

NOTE For most ultrasound systems, an individual ultrasound **PSF** cannot be used as the overall system impulse response, due to changes in the **PSF** with depth, with other positions in the region of use and with system focal and frequency settings. The problem is solved by PSF mapping – see Annex D.

3.26

scan line

one of the component lines which form a **B-mode** image on an ultrasound monitor. Each line is the envelope-detected **A-scan** line in which the echo amplitudes are converted to brightness values

3.27

scan plane

a plane containing the **ultrasonic scan lines**

[IEC 61102: definition 3.38, modified]

3.28

side lobe

secondary beam, generated by an **ultrasonic transducer**, that deviates from the direction of the main beam. Usually, the intensity of the **side lobes** is significantly less than that of the central axis beam

NOTE The presence of **side lobes** may be responsible for introducing artifactual echoes into the **ultrasound** image.

3.29

slice thickness

thickness, perpendicular to the **ultrasonic scan plane** and at a stated depth in the **test object**, of that region of the **test object** from which acoustic information is displayed

3.30

speckle pattern

image pattern or texture, produced by the interference of echoes from the scattering centres in tissue or **tissue-mimicking material**

3.31

spot size

the –6 dB width or otherwise specified width of the **PSF** or **LSF**

3.32

target

an object to be interrogated by an **ultrasound** beam

NOTE Examples of **targets** are:

- a) a device specifically designed to be inserted into the ultrasonic field to serve as the object on which the radiation force is to be measured;
- b) a scatterer or ensemble of scatterers giving rise to a signal within the effective ultrasonic beam;
- c) a wire or a filament in a **test object**.

3.33 **test object**

device containing one or more groups of object configurations embedded in a **tissue-mimicking material** or another medium

3.34 **test object scanning surface**

surface on the tissue-mimicking **test object** recommended for transducer location during a test procedure

3.35

time-gain compensation

TGC

change in amplifier gain with time, introduced to compensate for loss in echo amplitude with increasing depth due to attenuation in tissue

3.36

tissue-mimicking material

material in which the propagation velocity (speed of sound), reflecting, scattering and attenuating properties are similar to those of soft tissue for **ultrasound** in the frequency range 0,5 MHz to 15 MHz.

[See 6.4 and Annex D of IEC 61685]

3.37

transmitted ultrasound field

three-dimensional distribution of ultrasound energy emanating from the ultrasonic transducer

3.38

ultrasonic scan line

for automatic scanning systems, the beam-alignment axis either for a particular **ultrasonic transducer** element or for a single or multiple excitation of an **ultrasonic transducer** or of an **ultrasonic transducer element group**

[IEC 61157: definition 3.27, modified]

3.39

ultrasonic transducer

device capable of converting electrical energy to mechanical energy within the ultrasonic frequency range and/or reciprocally capable of converting mechanical energy to electrical energy

[IEC 61102: definition 3.58]

NOTE For the purposes of this standard, **ultrasonic transducer** is taken to refer to a complete assembly that includes the transducer element or elements and mechanical and electrical damping and matching provisions.

3.40

ultrasonic transducer element group

group of elements of an **ultrasonic transducer** which are excited together in order to produce a single acoustic pulse

[IEC 61102: definition 3.60]

3.41

ultrasound

acoustic oscillation whose frequency is above the high-frequency limit of audible sound (conventionally 20 kHz)

[IEV 801 21-04, modified]

3.42

ultrasound beam (pulse-echo response pattern)

region adjacent to the transducer face from which an echo signal from a specified **target** may be detected for the test settings of the scanner and with the scanner operating in a non-scanning mode. This term should be distinguished from the **transmitted ultrasound field**

3.43

voxel

smallest spatial unit or cell size of a digitized 3-dimensional array representation of an image. Each **voxel** has an address (x, y, and z-coordinates) corresponding to its position in the array, and a specific brightness and/or color value

3.44

working liquid

a mixture of water and other solvent that adjusts the speed of sound to 1 540 m/s

[See also 6.4 and Annex D of IEC 61685:2001]

3.45

accuracy

closeness of agreement between a test result and the accepted reference value

[SOURCE: ISO 5725-1:1994, 3.6]

3.46

axial resolution in a PSF-map

twice the Half-Width-at-Half-Maximum (HWHM) of a function's trace created from a set of increasing **pixel** values, commencing near zero and terminating at the first maximum value

(centre of the **PSF**) and representing the leading edge of the echo signal from a point reflector located on the main beam axis

Note 1 to entry: The **axial resolution in a PSF map** differs from the **axial resolution** specified by 3.5. It is used for the **PSF**-mapping only to simplify the data acquisition.

Note 2 to entry: A detailed explanation of the **axial resolution in the PSF-map** measuring method is in D.6.1.4.

Note 3 to entry: The axial resolution mainly depends on the ultrasound frequency used, not on sonograph construction.

Note 4 to entry: **Axial resolution in a PSF-map** is expressed in metres.

3.47

brightness

luminance as perceived by the human visual system

[SOURCE: IEC 62563-1:2009, 3.1.2]

3.48

contrast

C

ratio of the difference of the luminance of two image areas, $L_1 - L_2$, divided by the average of the two luminance values:

$$C = 2 (L_1 - L_2) / (L_1 + L_2)$$

[SOURCE: IEC 62563-1:2009, 3.1.6]

3.49

dynamic imaging

real-time imaging

imaging with a frame rate that is high enough to observe moving structures in apparently continuous motion

3.50

elevational resolution in a PSF-map

difference of point-reflector displacements in passing through the scanning plane in an elevational direction, which result in decreases of MER of –6 dB compared to the MER-value in the beam centre

Note 1 to entry: The **elevational resolution in a PSF-map** differs from the **elevational resolution** specified by 3.12. It is used for the **PSF**-mapping only to simplify the data acquisition.

Note 2 to entry: Detailed explanation of the method is in D.6.1.3.

Note 3 to entry: **Elevational resolution in a PSF-map** is expressed in metres.

3.51

overall gain

G_0

basic level of gain that is uniform for the whole scan area but modified by **TGC** relative to the depth of the scan

3.52

profile line

set of **pixel** values ordered along an abscissa according to the sequence during their acquisition

3.53

lateral resolution in a PSF-map

Full-Width at Half-Maximum (FWHM) of the **PSF**, measured in a lateral direction

Note 1 to entry: The **lateral resolution in a PSF-map** differs from the **lateral resolution** specified by 3.17. It is used for the **PSF**-mapping only to simplify the data acquisition.

Note 2 to entry: Detailed explanation of the method is in D.6.1.2.

Note 3 to entry: **Lateral resolution in a PSF-map** is expressed in metres.

3.54

measuring grid

matrix of points specified by Cartesian coordinates x_i and z_j defined in a plane parallel to the scanning plane

Note 1 to entry: Each point determines the position (x_i, z_j) in which individual measurement of **PSF** is performed.

Note 2 to entry: The step Δx is defined as an increment $x_{i+1} - x_i$ in the lateral direction. The step Δz is defined as an increment $z_{j+1} - z_j$ in the axial direction.

3.55

performance evaluation

tests performed to assess specific absolute performance of the object tested

Note 1 to entry: Typical times for ultrasound-system **performance evaluation** are at pre-purchase evaluation, new- and repaired-system acceptance testing, at time of performance difficulties, and at end-of-useful-life evaluations.

[SOURCE: IEC TS 62736:2016, 3.5]

3.56

precision

closeness of agreement between independent test results obtained under stipulated conditions

[SOURCE: ISO 5725-1:1994, 3.12]

3.57

scanning window

area on the surface of the **test tank** dedicated for transducer application to obtain a suitable sonogram of the target

Note 1 to entry: It is important that the **scanning window** be covered by flexible foil made of material with similar acoustic properties to the working liquid to avoid ultrasound field reflections and absorption.

Note 2 to entry: The foil flexibility should assure proper acoustical contact of any type of curved transducer.

Note 3 to entry: It is important that the foil covering the scanning-window be tough enough to prevent its damage during coupling the measured transducer to the **scanning window**, to prevent resultant leakage of working liquid from the measuring tank.

Note 4 to entry: The **scanning window** has the identical function as the **test object scanning surface** in the case of tissue-mimicking test objects (see 3.34).

3.58

side-lobe signal

echo signal generated by ultrasound signal transmitted/received in a direction different from the central axis of the transducer

3.59

test tank

tank designed to be suitable for providing specified kind of tests, which is filled with a **working liquid** and equipped with **scanning window(s)**

4 Symbols

A	surface area
A_c	cross-sectional area
a_i	length of the semi-major axes for a given half ($i = 1$ or 2) of the ellipsoid of an ovoid object
$A_{r,max}$	greatest $a_{r,max}$ evaluated for whole measured volume
$a_{r,max}$	MER pixel value evaluated from ROI
$a_{r,max}(x,y,z)$	MER pixel value evaluated from ROI scanned for reflector in position (x,y,z)
b	mean of the lengths of the minor axes of the ellipsoid of an ovoid object
C	contrast
D	diameter of the reflector sphere
f	acoustic working frequency
G_o	overall gain
$I(x,y,z)$	ROI specified in a digital picture of scan stored with reflector in position (x,y,z)
k	circular wave number; ($= 2\pi / \lambda$ in which λ is the wavelength)
M	number of quantization levels defined by $M = 2^m$ where m is number of pixel bits
P	perimeter of cross-section of ovoid object
p_x	pixel size in lateral (azimuthal) direction
p_z	pixel size in axial direction
R	ratio of mean of measured spacings to known spacings (see 7.3.1)
$R_{A,PSF}$	axial resolution in a PSF-map
$R_{E,PSF}$	elevational resolution in a PSF-map
$R_{L,PSF}$	lateral resolution in a PSF-map
R_x	lateral dimension calibration factor (see 7.4.2); ratio of mean filament spacings to known spacings for the horizontal direction
R_y	ratio of mean filament spacings to known spacings for the vertical direction
r	radius of a wire or filament target
V	volume of an ovoid object
$W_{F,HM}$	value of FWHM (full width at half of maximum)
$W_{H,HM}$	value of HWHM (half width at half of maximum)
$W_{F,HM,n}$	normalized $W_{F,HM}$ according to Formula (D.3) in D.6.1.2
$W_{H,HM,n}$	normalized $W_{H,HM}$ according to Formula (D.3) in D.6.1.2
Z_m	characteristic acoustic impedance of a wire or filament material
Z_w	characteristic acoustic impedance of the surrounding medium (working liquid or tissue-mimicking material)
λ	ultrasound wavelength in the working liquid, calculated from the nominal frequency of the transducer used
ε	$1-(b/(2a))^2$ eccentricity of an ellipsoid or an ovoid object
σ	backscattering cross-section for a point-like target
ATGC	automatic time-gain compensation
FWHM	full width at half of maximum
HFHM	half width at half of maximum

LUT	look-up table
MER	maximum echo received
PSF	point-spread function
RF	radio frequency
ROI	region of interest
TGC	time-gain compensation
US	ultrasound

5 General conditions

The tests should be performed within the following ambient conditions:

- temperature $23\text{ °C} \pm 3\text{ °C}$;
- relative humidity 45 % to 75 %;
- atmospheric pressure 86 kPa to 106 kPa.

This standard permits the use of test objects of various constructions. Therefore it is essential that the following data of the test object be reported. The following standard choices are recommended:

- a) medium: either **working liquid** or **tissue-mimicking material** [6]
- b) use of coupling gel: thin layer or gel with adapted sound velocity
- c) geometry (one of the models given in Annex A, B or C, where needed with a different spacing between **targets**).

For the medium **working liquid**, the following properties are required:

- speed of sound = $(1\,540 \pm 15)\text{ m/s}$;
- low attenuation $(< 0,1\text{ f dB cm}^{-1}\text{ MHz}^{-1})$;
- negligible scattering (see IEC 61685).

For adjusting the speed of sound in working liquid, see [7, 8].

For the medium **tissue-mimicking material** [9], the following properties are required:

- speed of sound = $(1\,540 \pm 15)\text{ m/s}$;
- attenuation $(0,5 \pm 0,05)\text{ f dB cm}^{-1}\text{ MHz}^{-1})$ in the frequency range used in the tests;
- scattering (moderate, no value imposed).

Note: Where an ultrasound system is designed for particular applications where the mean speed of sound is different from 1 540 m/s, a medium with that design speed of sound should be employed and that change reported with the results.

For tissue-mimicking properties, see also 6.4 and Annex D of IEC 61685:2001 .

Tissue-mimicking material is usually protected by a thin cover. Its thickness and acoustic properties (attenuation and sound velocity) should be reported if these influence the measurement.

The transducer is usually coupled to the cover of **tissue-mimicking material** by an **acoustic coupling agent** (ultrasound gel). If the layer is thin (compared to the wavelength) its influence can be ignored. For a thick layer, for example as needed for a curved-array transducer, the sound velocity of the **acoustic coupling agent** shall be equal to $(1\,540 \pm 15)\text{ m/s}$.

Sound velocity of a medium has two different effects: if it is larger then 1 540 m/s, the axial distances in the medium are rendered proportionally shorter and the focus of the transducer

moves away from the transducer. If the sound velocity is lower, the opposite occurs. The effect on the focus becomes more important for transducers with a high numeric aperture. Therefore the use of the correct sound velocity ($1\,540 \pm 15$ m/s, to which ultrasonic systems are standardized) is essential in Clauses 6 and 7, dealing with geometrical distortions. In Clause 8, dealing with the **PSF**, a deviation can be tolerated for not too high numeric apertures.

In describing scanning procedures with “horizontal” and “vertical”, it is assumed that a **test object** is insonated from above, and that the image on the scanner is oriented correspondingly.

6 Techniques for calibrating 2D-measurement systems

6.1 Test methods

To carry out the test procedures, the following items are required:

- a) tissue-mimicking **test objects** containing **targets** at accurately specified positions;
- b) tissue-mimicking **test object** containing a 3D-object of accurately specified dimensions;
- c) a tank equipped with target holder to position the target at accurately specified positions and containing degassed **working liquid**.

The specifications of these devices are given in ~~the~~ Annexes A, B, C and D.

6.2 Instruments

6.2.1 General

The equipment specified in this subclause has been selected to permit testing of ultrasonic scanners in clinical usage. The devices described will ensure that the data collection and analysis will be objective and reproducible.

6.2.2 Digitizers

While some spatial measurements can be made with long-existing digital callipers, for more generally applicable, objective, reproducible data, the **ultrasound** images obtained for testing should be digitally encoded. Many modern **ultrasound** imaging devices produce digital images from the scan converter that can be used for these measurements and are most closely representative of the displayed images. Such measurements can be employed well by hospital-based users with some digital measurement expertise. For spatial measurements this procedure is directly applicable. For **PSF** and **LSF** measures it is necessary, however, to have a characteristic curve of linear echo amplitude at the transducer as a function of the digital image values or to create a sparse representation of that curve by use of calibrated reflectors as described in 7.2.1.2 of [19]. In some systems, rf scan-line data are available. Such data are more accurate for precision measurements in which the linear signal amplitude is important. Measurements made with rf data should be clearly indicated as such and the level at which they came from the system documented. For those machines that do not produce digital images, a frame grabber may be used to acquire and digitize **ultrasound** images. This digitizer requires adequate spatial resolution (at least 512×512 **pixels**) and sufficient **grey scale** (at least 256 grey shades). Also, adequate image analysis software should be used to perform the simple measurements described below on the digital **ultrasound** images of the **test objects**. The digitizer shall exhibit a linearity producing spatial uncertainty of $<1\%$ over 75 % of the image dimension measured, signal level (grey scale) linearity $<3\%$ of full range and signal level stability over a year of $<5\%$ of full range.

The digital imaging software should allow the user to be able to place the cursor at any location on the screen and obtain the pixel address (i.e. row, column coordinates). This will allow the user to calibrate the digital image to actual distances recorded in the **ultrasound** images. Once the digitizer is calibrated, digitized ultrasound images can be subjected to more sophisticated

software analysis than that which is possible directly on the ultrasound display. The digital imaging software should allow the reading of the grey value at any pixel address.

To calibrate the digital image pixel distance (i.e. calibrate the digitizer relative to the **ultrasound** imaging system):

- a) Scan an image of a **test object** containing appropriate **working liquid**. Make a note of the magnification level for this image and make all subsequent measurements and comparisons with the same level of magnification.
- b) Measure the known distances between the positions of two wires or filaments at a distance of about 75 % of the screen size with the electronic callipers to confirm that the calliper measured distance corresponds to the actual distance. The measurement should be performed for a pair of wires or filaments connected by a vertical line and for a pair of wires or filaments connected by a horizontal line. In case deviations are found the scanner should be adjusted before proceeding further. If adjustment is not possible the actual distances shall be used in d).
- c) Digitize the scanned image and use the imaging software to measure the distances in pixels between pairs of wires or filaments by obtaining the pixel address of each wire or filament location, and subtracting to obtain the distances in pixels. Repeat for several different locations, checking both vertical and horizontal distances.
- d) Measure the distance in pixels for various positions of wires, using various directions of the connecting line with respect to the vertical. Divide each distance by the actual distance in mm. Average these ratios; this average ratio is the pixels per millimetre calibration for your digitizer. Once this calibration has been performed, this ratio can be used to compute relative distances in all subsequent digitized images for the particular scanner and magnification used.

See [10].

6.2.3 Tissue-mimicking test objects

Tissue-mimicking **test objects** shall contain structures that allow the following types of measurement to be made:

- a) linear;
- b) curvilinear;
- c) circumferential;
- d) area;
- e) volume;
- f) image distortion;
- g) **M-mode** calibration.

Examples of tissue-mimicking **test objects** are given in Annex B.

6.3 Test settings

6.3.1 General

The many combinations of scanner settings and transducers make it impracticable to carry out tests for all of them. Tests are therefore carried out for each **ultrasonic transducer**, with two settings, one which provides a complete image and one which provides the highest resolution of the **test objects**. The focusing of the ultrasonic beam should be extended over as large a range as possible, to achieve the best resolution over all visible **targets**.

The **test object**, containing an array of filaments as in Figure A.1, is used for the procedures described below.

6.3.2 Display settings (focus, brilliance, contrast)

The focus is made sharp and the brilliance and contrast controls are turned to their lowest positions. The brilliance is now increased until the echo-free zone at the side of the image becomes the minimum perceptible shade of grey. The contrast control is then increased to make the image contain the greatest range of grey shades possible. The focus is then checked for sharpness. If it requires further adjustment, the whole procedure is repeated.

6.3.3 Sensitivity settings (frequency, suppression, output power, gain, TGC, ATGC)

- a) The **nominal frequency** of the **ultrasonic transducer** is noted.
- b) If there is a suppression- or reject-control, it is adjusted to allow the smallest possible signals to be displayed.
- c) The output power and **gain** are adjusted to present the images of the **target** filaments as the smallest visible points on the display.
- d) The **time-gain compensation (TGC)** controls are arranged to present the images of the **target** with equal brilliance over the image. For scanning in **working liquid** the TGC slope should be close to zero

6.3.4 Final optimisation

A final optimisation of the image may be carried out by a small change in the suppression level, **gain** or output power.

When **automatic time-gain compensation (ATGC)** is an option in a scanner, tests should be carried out in this mode of operation. The **test object** is imaged with **ATGC** enabled, the image being optimised using any control which still functions manually, e.g. the overall **gain** or output power.

6.3.5 Recording system

The digital acquisition of **ultrasound** images allows for objective measurements and also allows the images to be saved for comparison at a later date. A major advantage of digital recording is that images are not subjected to the degradation that occurs with either photographic or video recording systems.

6.4 Test parameters

6.4.1 General

Techniques are described in this standard for the following types of measurement:

- linear;
- curvilinear;
- circumferential;
- area;
- volume;
- display and recording distortion;
- **M-mode** calibration.

Transmitted intensity should be low enough to avoid pulse distortion due to non-linear propagation (see IEC 61102). A list of all factors influencing the operation of the scanner for example transducer, frequency, sensitivity control settings, focusing, image processing option shall be made. These data are to be recorded in sufficient detail so as to allow the test to be repeated exactly at a later date by another operator and shall accompany the measuring results.

6.4.2 Measurement accuracy (linear, curvilinear, circumferential, area)

To assess the accuracy of the measurement system of a scanner, the wires or filaments in the **test object** shown in Figure A.1 or Figure A.2 are imaged with the sensitivity adjusted to make the displayed echoes as sharp as possible. If the test object is sealed, a coupling agent shall be used. An **ultrasound** image is obtained and digitized of the set of filament **targets** situated at the middle of the typical working range for the **ultrasonic transducer** assembly being used. Other factors that may affect the value of the resolution are also noted, for example the image processing options of the scan converter or focusing. The procedure is repeated for the other **ultrasonic transducers** of the scanner.

Measurements are made in straight lines on the screen of lengths approximately equal to 75 % of the displayed range. Using image analysis software, a linear brightness profile is obtained along each dimension. Distances are measured “from peak to peak” of the wire or filament brightness profiles. (In case the measuring results are noisy, the position of a peak value is replaced by the midpoint between the –3 dB points and that action noted.) These measurements are carried out along at least a vertical and a horizontal line in Figures A.1, and A.2 and, when possible, along near-vertical directions in the **field-of-view**. The average percentage error is tabulated for each length in each direction. The process is repeated for the available display scales.

To evaluate the accuracy of measurements of curved lines and cross-sectional areas, closed figures having an area approximately 0,75 of the **field-of-view** are traced centrally on the display. The circumferences and areas are measured and the percentage errors calculated. The tracing is done point-to-point, so that a polygon-shaped region is traced. The circumference and area of the polygon are measured. Additional measurements are made with two smaller figures (areas 0,1 and 0,25 of the **field-of-view**) located at the top and bottom of the display. This process is repeated for the available scales of the display.

For rigorous testing, sources of variability should be evaluated. This holds true for both short-term and long-term variation in measurement and analysis procedure. Short-term testing (same-day) should be repeated multiple times from setup to final analysis. In manual testing, an operator should repeat the tests over a short period of time and then the tests should be repeated by several operators.

6.4.3 Display and recording of image distortion

Scan the two-dimensional, regular array of filaments of the **test object** shown in Figure A.2, so that their echoes are seen with equal brilliance throughout the **field-of-view**. Select wires or filaments located horizontally and vertically from the centre throughout the **field-of-view**. Measure on the digitized image the distance from the centre of each wire or filament to the centre of a reference wire or filament located at approximately the centre of the **field-of-view**. Calculate and tabulate the percentage errors.

Observe the directly viewed image of the array of filaments to check that any distortion (i.e. failing orthogonality) of dimensions in the display is less than 3 %.

6.4.4 M-mode calibration

6.4.4.1 General

An **M-mode** facility exists on most real-time scanners. A partial assessment of its performance can be carried out using the **test objects** described in Annex A.

6.4.4.2 Spatial measurements (scrolling A-scan line)

Performing an **M-mode** scan with the **ultrasound beam** directed at wires or filaments in a resolution **test object**, as described earlier for the **B-mode**, enables the measurement errors of a system to be ascertained.

Distortions of the display or record are checked and recorded using the array of **target** filaments in the **test object** as is done for a **B-mode** image.

The accuracy of the time-axis calibration of the **M-mode** trace can be checked by injecting bursts of **ultrasound** into the **ultrasonic transducer** using an external pulse generator and transducer at accurately known intervals, for example 1 ms bursts at 200 ms intervals.

Measurement checks should be carried out for the digitized image of the **M-mode** trace on the display screen. The errors in measurement should be less than 3 %.

6.4.4.3 Tissue-thickness M-mode

For tissue-thickness **M-mode**, the system measures changes in relative thickness of moving tissue. Thus, evaluation of the accuracy of these measurements requires a tissue-like phantom that can be compressed and relaxed at pre-determined positions, and the ability to compare the compressed and relaxed thicknesses with the **M-mode** readings. Also, it should be possible to compress and relax the phantom at different rates. A deformable sponge phantom might be useful for this measurement. It is important to have the ability to test the synthetic **M-mode** at various depths, so the phantom should allow for re-positioning the transducer for different **target-to-transducer** distances. Also, the test should be repeated for each of the **M-mode** sweep speeds.

7 Methods for calibrating 3D-measurement systems

7.1 General

Three-dimensional (3D) imaging systems exist which are only used for visualization, while others include measurement capabilities. Since 3D-reconstruction of volumes is achieved in different ways, it is important to examine the volume reconstruction method and its associated problems, and evaluate the accuracy of the reconstructed images. This discussion is limited to measurement of dimensional accuracy of reconstruction. Measurement of 3D-system resolution will be discussed in document IEC 61391-2 dealing with system resolution and sensitivity.

7.2 Types of 3D-reconstruction methods

7.2.1 General

True 3D-imaging requires that the imaging system assemble data in a 3-dimensional **voxel** matrix. This matrix is usually composed of data from a stack of ultrasonic **scan planes** that contain the **target** volume. The 3D-imaging system stores the information as a 3-dimensional matrix. The spatial density of data-matrix points depends on the number of **ultrasonic scan lines** within each ultrasonic **scan plane**, the pulse length, and the number and spacing of ultrasonic **scan planes** that make up the elevational (depth) dimension. The manner in which the volume of interest is scanned is important, since reconstruction accuracy will depend on how well physical distances are preserved. The distance between successive ultrasonic **scan planes** should be constant, but it should be less than the elevational resolution (**slice thickness**) of the **ultrasonic transducer**. Otherwise, interpolation of adjacent ultrasonic **scan planes** will be included in the reconstructed volume. This interpolation can lead to errors in the dimensions of the reconstructed volumes.

Once the 3D-matrix is constructed, data may be obtained along any dimension within that volume. For example, if multiple images of xy-**ultrasonic scan-plane** data are collected to form a 3D-volume, the resulting 3D-matrix can be “sliced” normal to the y-axis in the xz plane, producing C-scan slices from the resulting data. Also, the resulting 3D-reconstruction can be rotated in space, so as to be viewed and measured from angles that were not available in the original ultrasonic **scan planes**.

[See [11, 12] for reviews of 3D-scanning techniques.]

3D-volume-matrix acquisition and reconstruction are performed in two basic ways:

- a) reconstruction by external positioning;
- b) sequential reconstruction.

Each method has its own characteristics, strengths, and problems.

7.2.2 3D-volume reconstruction methods

7.2.2.1 External positioning methods

Reconstruction methods for 3D-volume by external positioning methods use a reference point and a coordinate reference frame, and all dimensions and positions within the 3D-volume matrix are recorded with respect to the reference coordinates. In these types of systems there is usually a scanning framework, into which the scanning volume is inserted, and the transducer is usually motorized and rides on a rail at a constant speed. Other forms of rigid support may be used that restrain the **ultrasonic transducer** to maintain its 3D-coordinates accurately. This type of reconstruction method is the most accurate and reliable, but is subject to some problems due to initial positioning of the **ultrasonic transducer's** support frame, motor speed deviations, or changes in the positioning system during the data-collection phase. A variant of external positioning systems exists, in which the transducer is guided manually and its position and direction are sensed with respect to a reference coordinate system.

[See [13].]

7.2.2.2 Sequential positioning methods

Sequential positioning and reconstruction methods use different techniques but are usually based on attachment of subsequent **scan planes** to the 3D-matrix based on the position of the previous plane. One such method uses the rate of change of image speckle in at least one dimension [14, 15]. This encoding involves several assumptions that are not always valid. One assumption is that the motion is either a purely linear sweep or a purely angular sweep. In some commercial implementations, a simple, uniform, linear sweep speed is assumed. Tests are therefore important to demonstrate the capabilities and limitations of the measurements under laboratory conditions. Problems occur if the transducer is not moved at a uniform speed, or if the transducer angle shifts from its previous orientation. The reconstruction scheme, in most cases, cannot compensate for these shifts in reference plane orientation, and the reconstructed volume will contain inaccuracies.

7.3 Test parameters associated with reconstruction problems

7.3.1 Reconstruction by external positioning

For reconstruction by external positioning-system testing, either water-based or tissue-mimicking-based **test objects** may be used.

Test parameters: From the reconstructed volume, measure the reconstructed lengths in all three Cartesian coordinate directions and compare with the physical object dimensions along the same coordinate directions. Check the **ultrasonic transducer** orientation with respect to frame and point of reference; speed of motor; distance between ultrasonic **scan planes**; and Cartesian dimensions of reconstructed volume. Procedures for measuring the following parameters are described below.

- a) linear dimensions;
- b) areas;
- c) perimeters of areas;
- d) volumes.

7.3.2 Sequential plane reconstruction systems

Systems in which 3D-spatial encoding is based on the image speckle require a **test object** with relatively uniform speckle backscatter and structures to allow tests of the accuracy and precision of the registration. With certain **ultrasound** systems not only should the accuracy of measurements over long distances be tested, but also the uniformity of the distance scale. Many sequential encoding systems require additional tests due to imprecise position encoding. When such imprecision exists, the direction in an image can be distorted by local jumps or retardation of recorded position in the direction of scanning.

Test parameters: from the reconstructed volume, measure the reconstructed lengths in all three Cartesian coordinates and compare with the physical object dimensions along these same coordinates. Procedures for measuring the following parameters are described below.

- a) linear dimensions (axes);
- b) areas;
- c) perimeter of areas;
- d) volumes.

7.3.3 Test instruments (phantoms) for evaluation of 3D-reconstruction accuracy

7.3.3.1 Filament phantom (water-filled)

For the reconstruction method by external positioning, a filament **test object**, filled with **working liquid** as described in Annex A, Figure A.1 or Figure A.2, may be used. Since the system does not depend on speckle correlation to place the ultrasonic **scan planes** into the 3D-matrix, the filament matrix is a useful structure for testing reconstruction accuracy. For each of the rows of filaments, the distance r from the image location of a filament to a reference filament is measured and plotted against the known distance r' in the phantom. The maximum and r.m.s. deviations of the measured filament positions from the linear regression lines of r on r' are measured, as well as the slopes of the fitted lines.

If the volume measurement algorithm of the system under test can work with sharp corners and flat surfaces, the accuracy of measurement of well-defined volumes are best tested with a filament **test object** as in Figure A.1 or A.2.

7.3.3.2 Tissue-mimicking phantom

A second **test object** that can be used to test both types of systems is shown in Annex B. Figures B.1 to B.4 show different views of ovoid-shaped **tissue-mimicking material** structures with minimal specular boundaries, set in a **tissue-mimicking material** matrix. Such a **test object** is volumetric and the **targets** are defined in the image by differences in **backscatter contrast**. Only the **grey scale** texture and average signal level define the borders. This reliance on volumetric backscatter rather than specular reflectors or **point targets** allows evaluation of image formation, display and measurement aspects of the ultrasound system's reconstruction of volumetric **targets** imaged in the body.

7.4 Test methods for measurement of 3D-reconstruction accuracy

7.4.1 General

These methods are for 3D-measurements with conventional 2D-scanners as well as 3D-scanners. For high quality 3D-imaging and testing thereof, the separation of the recorded image planes should be less than the width of the ultrasonic **scan plane** thickness at its focus, ideally, less than one half the elevational (ultrasonic **scan plane**) focal width. If controls for scan-plane separation are provided, such settings should be employed.

7.4.2 Measurement methods and accuracy using the filament test object

Volumetric measurements from two orthogonal 2D-images: 3D-measurements are often made with simple 2D-imaging systems by acquisition of two orthogonal views of a roughly spherical object, measurement of the three major axes of the object, and calculation of the spherical or ellipsoidal volume by appropriate equations therefore or by the ultrasound system. Either volumetric calculation method can be tested on one of the images from the stack of the previous paragraph by measuring the diameters of the assumed sphere passing through four or more filaments in the phantom and then repeating the measurements on the same or adjacent image, assuming that the two were acquired at 90-degree angles to each other. The calculation of the sphere with cross-section equal to the circle measured is then calculated, compared with the volume of the assumed sphere, and the error computed.

Tests for correction of angulation of scan plane in the elevational direction and for verification of the measurement algorithm from a 3D-sweep parallel to filaments: Perform a 3D-scan with the central acquired image planes normal to the filaments and the scan direction parallel to the filaments, i.e., with the transducer in View B of Figure A.1. If the system allows rotation of the transducer in an arc in the elevational direction, perform the scan in that way and, if possible, display the reconstructed volume with all the reconstructed images normal to the filaments. If that reformatting is not an option in a sector scanner, correct the separation of the filaments for the known viewing angle. If not already so done in the same way in 6.4.2, perform the measurements of 6.4.2 on the first, middle and last image of the 3D-set and document their errors and variances. The ratios of mean filament spacings to known spacings for the horizontal and vertical directions are referred to as the lateral- and longitudinal-dimension calibration factors, R_x and R_y respectively. Check that the mean spacings of groups of filaments that should have the same spacings are the same in each of the image planes within 1 %.

See [16].

For calibration of the **ultrasonic scan-plane** separation in 3D-imaging and for assessing distortion of images reconstructed (from a volume data set) with one axis in the **ultrasonic scan-plane's** thickness direction, the transducer is moved or rotated slowly in the elevational (ultrasonic **scan-plane's** thickness) direction normal to the filaments. That is, the transducer is moved from left to right (Transducer View A) on Figure A.1. This motion is performed in accordance with directions for this type of 3D-scanning as provided by the **ultrasound** system manufacturer. Often only a linear translation or a sector sweep is allowed but it is instructive to deviate from the instructions to see the amount of error generated.

For each of the rows of filaments from this second scan (View A), display reconstructed images that are perpendicular to the filaments. Calculate and report the maximum and r.m.s. deviations of:

- a) the measured filament spacings from their known values;
- b) the measured filament positions for the fitted lines;
- c) the slopes of the fitted lines from the expected values.

To evaluate the accuracy of curved lines and cross-sectional areas, closed figures are traced centrally on the display of curves and areas covering approximately 0,75 of the field-of-view. The lengths, circumferences and areas are measured and the percentages of measured-to-known areas calculated. Volume measurements are tested in these data with the sweep direction orthogonal to the filaments. Mark a known area, A , in the reconstructed image that is perpendicular to the filaments. Mark an **ultrasound** system-indicated length L' along the filaments for the third dimension of a 3D-volume enclosed by the filaments. For Figure A.1 this volume would be a cylindrical rod. Compare the measured volume $A' \times L'$ with the known

volume $A \times L/R_x$, where R_x is the lateral-dimension calibration factor from the second paragraph of this subclause. Further examples of measurements with filament **test objects** are given in [16].

7.4.3 Measurement accuracy using volumetric targets in a backscattering object phantom (Figure B.1) with a 2D-scanner

7.4.3.1 General

In these measurements, the transducer is rotated and tilted to find a circular cross-section of each of the **targets** in the 3D-**test object** that can be fully imaged in one view. For each **target**, move and adjust the image plane to find the minor axis, the largest diameter where the object still appears circular. The calliper markers are placed on the ends of the largest vertical (axial) diameter, b_y , obtainable through that circular cross-section and the value is noted on the calliper read-out. A horizontal (lateral) diameter, b_h , is measured in a similar way. The average of those two diameter measures is labelled b .

Rotate the transducer through 90° and find and measure the longest dimension of the ellipsoid, referred to as a , where a is the sum of the lengths of each half of the egg, $a_1 + a_2$. These procedures are repeated for the smaller 3D-object, if possible. The results obtained are tabulated in Table 2. Compare the measured values with the known values of the diameters given in Table 1.

7.4.3.2 Perimeters

Using an image of a cross-section of an ovoid **target**, begin to measure at a desired point on the perimeter of the **target** to be measured. Calliper markers are placed all along the selected image of the 3D-object until the starting point is reached. On average, the separation of the calliper markers should be no more than 1/20 of the estimated length of the perimeter except when an ellipse, or at least a curved line, fit is performed by the **ultrasound** system. The equation for the perimeter [17] of each of the two half-ellipses is approximately:

$$P = \frac{\pi}{\sqrt{2}} \left[a_i^2 + \left(\frac{b}{2} \right)^2 \right]^{1/2} \quad (1)$$

where

a_i is either a_1 or a_2 , the semi-major axes for a given half of the ellipsoid;

b is the mean minor axis of the ellipsoid.

The perimeter of the entire egg-shaped object is the sum of the perimeters of the two halves. The perimeter of the circular cross-section is $2\pi b$. See Table 1 for expected values for the two objects in Figures B.1 and B.2.

7.4.3.3 Areas

On virtually all machines, a value for the enclosed (cross-sectional) area is calculated from the same measurement points defined in the perimeter measurements (see 7.4.3.2).

The measured area values should be compared against the known areas for the 3D-object cross-sections. In the largest elliptical and circular cross-sections, respectively, the areas are [17]

$$A_c = 0,79 b (a_1 + a_2) = \pi b (a_1 + a_2) / 4 \text{ and } A_c = 0,79 b^2 = \pi b^2 / 4$$

The surface area of the ellipsoids is given by:

$$A = 2 \pi \left(\frac{b}{2} \right)^2 + \pi a_1 \left(\frac{b}{2 \varepsilon_1} \right) \arcsin \varepsilon_1 + \pi a_2 \left(\frac{b}{2 \varepsilon_2} \right) \arcsin \varepsilon_2 \quad (2)$$

where

ε_1 is the eccentricity $(1 - (b/(2a_1))^2)$;

ε_2 is the eccentricity $(1 - (b/(2a_2))^2)$.

See Table 1 for expected values for the two objects in Figure B.1.

7.4.3.4 Volumes

Volume measurements or relative measurements of objects of consistent shape can be made by measuring maximum linear dimensions on three orthogonal axes. In the case of 3D-ellipsoidal objects [17], the actual volume is given as:

$$V = 0,52 (a_1 + a_2) b^2 = \frac{4}{3} \pi \frac{(a_1 + a_2)}{2} \left(\frac{b}{2}\right)^2 \quad (3)$$

The volume can be determined by measuring individually the two lengths, a and b , where $a = a_1 + a_2$. Averaging two perpendicular measures of b is appropriate. These measurements can be made from two image planes, with the transducer rotated 90° between them. Note this calculation method is essentially correct only for volumes approximating two half-ellipsoids with circular cross-sections. (For estimating the volume of any mass that resembles an ellipsoid with orthogonal axes a , b , and c , equation 3 would have $V = 0,52abc$). See Table 1 for expected values for the two objects in Figure B.3.

For applicability to other shapes and probably increased accuracy, a series of images is obtained from equally spaced ultrasonic **scan planes**. For high accuracy, the separation of the ultrasonic **scan planes** should be less than the width of the ultrasonic **scan-plane** thickness at its focus, ideally, less than half that focal width. In the conceptually simplest calculation, the volume is estimated by considering it to be composed of a number of cylinders of base area equal to that measured in a ultrasonic **scan plane** and height equal to the separation of the planes, i.e., multiply the cross-sectional area of the 3D-object in each plane by the scan separation and add these volumes for each slice. This calculation can be tedious by hand. More sophisticated volume-measurement algorithms are implemented on most 3D cross-section imaging systems and should be tested.

Table 1 – Expected values for the two ellipsoidal objects in Figure B.3

	Half the perimeter ^a cm	Perimeter cm	Area cm ²	Surface area cm ²	Volume cm ³
<i>Small object</i>					
Long half	3,01	6,02	3,82	14,51	4,58
Short half	1,65	3,31	1,70	9,94	2,04
Total	4,66	9,32	5,51	24,45	6,61
Cross-section	2,83	5,65	2,54		
<i>Large object</i>					
Long half	5,35	10,7	16,5	63,6	55,0
Short half	4,16	8,3	11,0	69,9	36,6
Total	9,5	19,0	27,5	133,6	91,6
Cross-section	7,9	15,7	19,6		
The perimeters and half-perimeters here are calculated from the exact elliptical integrals, rather than from Equation 1.					
^a Half the perimeter is a test of curved path length					

Table 2 – Suggested table of reported values

	Half the perimeter		Perimeter		Area		Surface Area		Volume	
	cm	%	cm	%	cm ²	%	cm ²	%	cm ³	%
<i>Small object</i>										
Long half										
Short half										
Total										
Cross-section										
<i>Large object</i>										
Long half										
Short half										
Total										
Cross-section										
Enter the measurement then that measurement as a percent of the expected value from Table 1. This form should be filled out for each of the measurement algorithms under study. It is instructive to fill out all or parts of the table from values calculated from the linear dimensions, a_1 , a_2 , and b , using equations 1, 2 and 3, as well as from any other measurement method provided on the imaging system. Similar measurements are performed in 3D-fetal biometry [18]. This table may also be used for curved line, area and volume measurements from the filament test objects.										

8 Measurement of point-spread and line-spread functions (high-contrast spot size)

8.1 General

High-contrast resolution characteristics of most imaging systems can be characterized by measurement of the **point-spread function (PSF)**, which is the characteristic response of the imaging system to a high-contrast **point target** or the similar **line-spread function (LSF)**. For most optical systems, the **PSF** is singular, symmetric and isotropic. Thus, a measurement of the **PSF** usually is sufficient to characterize the system's impulse response.

Unlike optical imaging systems, **ultrasound** produces a **PSF** and **LSF** that is neither singular nor isotropic. Further, the **ultrasound PSF** and **LSF** are asymmetrical, having different axial and lateral dimensions, and they also vary with distance from the transducer (i.e. depth in the

image). Thus, many different measurements of the **PSF** and **LSF** at different positions and depths must be performed to obtain representative values of the system's imaging performance at specific positions along the beam axis.

An individual **PSF** or **LSF** cannot be used as the system's impulse response. Due to changes in the **PSF** and **LSF** with depth, there are different levels of resolution, depending on the position within the **ultrasound** scan.

Another complication arises from the way **ultrasound** behaves with high-contrast **targets** versus low-contrast **targets**. At high contrast, **ultrasound** can image very small structures, such as thin wires, filaments or the tip of a needle, or an air microbubble. Each of these will produce a characteristic smear that is associated with the **PSF** or **LSF**. For low-contrast **targets**, such as tissue, however, **ultrasound** images do not produce images of individual point **targets**, but rather, form a grey **speckle pattern** that appears as a grainy background, whose characteristics arise from the coherent nature of the **ultrasound beam**. A measurement of the dimensions of the **PSF** for an ultrasound system at a given depth describes the high-contrast resolution of the system at that depth. For low-contrast resolution measurements, a tissue-mimicking phantom is used, and the contrast-detail performance is sometimes measured. In this document, only the high-contrast behaviour of the **ultrasound PSF** or **LSF** will be measured. Low-contrast measurements will be described in another standard (See future IEC 61391-2).

In this document **PSF** and **LSF** are measured at two levels: FWHM (full width at half maximum), i.e. at –6 dB from the maximum, and at –20 dB. This requires that the relation between grey level and signal intensity be known (see 8.4.5).

NOTE Resolution on a two-target phantom (axial, lateral, or elevational) can be approximated from the **PSF** or **LSF** by adding two **PSF**- or **LSF**-curves as functions of distance between them and identifying the separation where there is a dip in the combined curves of –6 dB. Rigorously, the rf signal at each point in the **PSF** or **LSF**, and for each assumed target, must be summed to more accurately simulate the two-target response.

8.2 Test methods

To carry out the test procedures, the following items are required:

- a) **test objects**, as described in Annex A and Annex C, containing **targets** at accurately specified positions;
- b) a tank containing degassed liquid and, optionally, movable targets as described in Clause C.4 and D.5.4.2;
- c) digitized images, as described in 6.2.2 and 6.3.5.

The specifications of these devices are given in Annexes A and C.

8.3 Instruments

8.3.1 General

The instruments specified in this subclause have been selected to permit testing of real-time ultrasonic scanners without requiring electronic signals to be output from or input to the scanners.

8.3.2 Test object and tank

A **test object** should contain structures that allow the following machine features to be measured:

- a) **axial resolution**;
- b) **lateral resolution**;
- c) scan **slice thickness**;

Examples of **test objects** are given in Annexes A and C.

A tank of degassed **working liquid** is required. Under circumstances that allow a deviation of the speed of sound (see 5) the **working liquid** can be replaced by water to facilitate handling.

8.3.3 Image digitizer

An image digitizer as described in 6.2.2 is required.

8.4 Test settings

8.4.1 General

The many combinations of scanner settings and transducers make it impossible to carry out tests for all of them. Tests are therefore carried out for specified settings and for each transducer. The scanner is set up using the procedures outlined below. The focusing of the ultrasonic beam should be extended over as large a range as possible i.e. to achieve the best average resolution over all visible **targets**. A setting should be specified by a test instruction for each test, if it differs from the general recommendations. See D.5.2.

8.4.2 Display settings (focus, brilliance, contrast)

The focus is made sharp and the brilliance and contrast controls are turned to their lowest positions. The brilliance is now increased until the echo-free zone at the side of the image becomes the minimum perceptible shade of grey by observing the grey-shade bar on the display. The contrast control is then increased to make the image contain the greatest range of grey shades possible. The focus is then checked for sharpness; if it requires further adjustment, the whole procedure is repeated. If the system allows multiple focal zones, activate all of them for the desired range of measurements.

8.4.3 Sensitivity settings (frequency, suppression, output power, gain, TGC, ATGC)

- a) The **nominal frequency** of the scanner's transducer is noted.
- b) If there is a suppression or reject control, it is adjusted to allow the smallest possible signals to be displayed.
- c) The output power and **gain** are set to the minimum values consistent with displaying an adequate image.
- d) The **ATGC** controls are adjusted to remove the **ATGC**.
- e) The overall **gain** control is increased until the signal from the **target** is clearly visualized on the display.

8.4.4 Recording system

Digitized images will be used, as described in 6.2.2 and 6.3.5.

8.4.5 Calibration of system characteristic curve

A well-established method for calibrating the system characteristic curve (output signal as a function of acoustic-pressure signal at the transducer) is with planar boundaries of graded reflectivity [19, 20]. A simple, calibration phantom provides specular reflectors located parallel to each other and normal to the intended path of the **ultrasound beam** at different ranges from the transducer position in a **tissue-mimicking material**. At each range at least two reflectors are provided, with different reflectivities.

In its simplest form one plate has a 10 dB lower reflection coefficient than the other. This is achieved by constructing one plate of 342 stainless steel and the other of plexiglas. The steel is located on each step of one staircase (A), and the plexiglas on each step of another, adjacent staircase (B). At usual diagnostic frequencies, the first step is 1 cm below the surface

followed by two additional steps at 2 cm and 5 cm depth. These steps are followed by two steps, one at 10 cm depth and another at 18 cm depth. The space above the steps is filled with **tissue-mimicking material**. The speed of sound, c , the density, ρ , and the attenuation coefficient slope, α/f (where α is the attenuation coefficient at the working frequency, f) of this **tissue-mimicking material** shall be as follows: $c = (1\,540 \pm 6)$ m/s; $\rho = (1,05 \pm 0,05)$ g/cm³; $\alpha/f = (0,5 \pm 0,05)$ dB cm⁻¹MHz⁻¹.

Echo signals due to backscatter in the **tissue-mimicking material** should be at least 20 dB weaker than the echo signal from the weakest reflecting interface at all frequencies in the tested waveforms.

In use to calibrate a system's sensitivity control, gain or output, this phantom is imaged at a range at or beyond the transducer's focal zone and at a gain where the weaker reflector, on staircase B, is approximately 6 dB above the electronic noise level. If signal levels are too high, choose a deeper step and reduce the TGC as much as possible. Signal levels (10 dB apart) are recorded from the steps at that range on staircases A and B, respectively. Then the system's sensitivity control is increased until the signal from the step on staircase B (plate B) is equal to the signal level from the step on staircase A (plate A) at the previous control setting. Thus, the signal from plate A is now 10 dB above that for plate B at the first setting. By repeating this process, the entire signal dynamic range can be calibrated in 10-dB steps. By recording settings of the system's sensitivity control, this control is thus calibrated relatively as well. Interpolation of those control settings will allow calibration of signal changes at finer increments.

Another method for constructing the phantom is used to produce weaker signals lying more within the linear range of most diagnostic **ultrasound** systems. This method should be compared against the primary method for accuracy. The secondary method is to fabricate the steps on one staircase from high-density polyethylene (LB-861) and the steps on the other staircase from low-density polyethylene (NA-117)²⁾. The high-density material has an impedance of $2,33 \times 10^6$ kgm⁻²s⁻¹, while the impedance of the low-density material is $1,79 \times 10^6$ kgm⁻²s⁻¹ [21]. The reflection coefficients of these interfaces should differ by 10 dB in a tissue-mimicking gel. More staircases with steps having other reflection coefficients can be employed to produce even weaker echoes and finer gradations, e.g., [22]. However, these also shall be calibrated against the primary method. With low echogenicities, the temperature of the medium becomes critical and shall be controlled sufficiently precisely to maintain the relative echogenicity within $\pm 0,5$ dB.

Even more convenient, when available, is electrical signal injection, either directly, or by acoustically coupling another transducer, with a calibrated transmitter and receiver to the transducer under test [23]. Connectors for direct injection are expensive and system-specific, while acoustic coupling must be performed carefully. However, if referenced to the primary method above, signal injection can be used in practice.

Point targets for **PSF** measurements: A second acoustic approach is based on **targets** of different sizes and should also be referenced to the primary method above. As scattering cross-section increases with frequency, this type of **target** emphasizes the high-frequency part of the signal from the scanner. However, the same emphasis occurs for signals from distributions of small scatterers, so it is not unrealistic to test a scanner in this way.

Theory indicates that, for **targets** much smaller than the wavelength, **targets** of all diameters have the same frequency dependence. Hence, it is possible to make a series of reflectors with well-defined scattering ratios. A series of sphere diameters increasing by 26 %, such as, in microns, 10 – 12,6 – 15,87 – 20 – 25,2 – 31,75 – 40 – 50,4 – 63,5 and 80 would give reflectivities in steps of 6 dB, spanning a range of 54 dB. This approach has been implemented

²⁾ Both materials are available from USI Corporation, Marlboro, MA, among others. They are examples of suitable products available commercially. This information is given for the convenience of users of this document and does not constitute an endorsement by IEC of these products.

with dense distributions of small scattering particles, but practicalities of implementation are not well documented for individual particles.

Similarly, a series of flat-ended wires can provide calibrated reflectivities. Here the ratio of diameters should be $\sqrt{2} = 1,41$, to obtain a step of 6 dB. For 5 MHz, diameters in the range 200 μ to 1 600 μ could span a range of 36 dB with a precision of circa 1 dB.

8.5 Test parameters

8.5.1 General

To measure **point-** or **line-spread functions**, an appropriate **target** is scanned in a test tank. A standard point **target** is the tip of a flat-ended wire viewed on end; a standard line **target** is a filament or wire, typically lying normal to the direction of ultrasound propagation. The dimensions of the **point-spread function (PSF)** in each of the three directions are measured for complete, detailed system-performance evaluation, usually in a developmental-, production, or research setting. **Line-spread functions (LSFs)** are employed in the field and should be used in specifications to be tested in the field. **LSFs** are recommended over **resolution**, as measured with multiple **targets**, because fewer targets are required and spatial spread-function measurements are required at many locations in the **field-of-view** and are dependent on a system's focal- and frequency-settings. Actual **resolution** with target pairs is sometimes preferred for visual measurements and when non-linear signal processing, such as signal differentiation may be employed. **PSF** and **LSF** measurement on a line normal to and through the **ultrasound beam axis** are closely related to the imaging system's high-contrast **resolution** and to each other when scanned in a direction along the same line at the same depth in a phantom. **PSF**, **LSF** and **resolution** each can be derived from a full set of measurements of the other, if the characteristic curve of the ultrasound system is calibrated. See the definition of **spot size**, used in this standard to relate to the width of a **PSF** or **LSF** in a way more self-explanatory to those not familiar with the spread-function terminology.

PSFs and **LSFs** currently yield relative measurements. Quantitative measurements indicating the sensitivity of the imaging system to a specified or standard **point target** or point pressure source can be obtained [24-28] but experience with these techniques is too limited for standardization at this time. These measurements are closely related to the high-contrast **axial resolution** and axial **LSF** of the imaging system for a specific position along the beam (i.e. depth in a phantom).

For all such measures, the sensitivity and angulation of the scanner is adjusted so that the maximum signal from the **target** is clearly visualized during each test.

If a **line target** is used, oriented perpendicularly to the ultrasonic **scan plane**, the cross-section of the line with the ultrasonic **scan plane** constitutes the **lateral line-spread function**, but this configuration has two limitations:

- a) the **line-spread function** can only be measured in the ultrasonic **scan plane**;
- b) the intensity is sensitive to the orientation of the ultrasonic beam perpendicularly to the **line target**.

To overcome this limitation a spherical **target** has been used. However, now interference between ultrasonic paths inside the sphere leads to complicated patterns of angular scattering [24] unless the sphere is extremely small and a relatively weak reflector. The complications generated by interference and multiple reflections inside the spherical target may be solved by time-domain analysis of the received echo when a larger and/or highly reflective sphere is used. See D.5.4.2.

An alternative **test object** for the measurement of resolution is a single filament (or wire) mounted vertically (in the direction of the ultrasound beam) on a micromanipulator which allows the filament to be moved through the **field-of-view** (Figure C.3). This permits the **spot size** to be measured at any point in the **field-of-view**. It also removes the problem, encountered with

an array of **targets**, of aligning all filaments perpendicularly to the beam at the same time to record a maximum signal from each.

When the point is much smaller than the wavelength (or than the diameter of the ultrasonic beam at the position of the **target** for measured transducers of large f-numbers), the **point-spread function** is measured independently of the properties of the **target**. Note that the response width from the **PSF** (from a **point target**) and the **LSF** (from a line **target**) are approximately equal. It has been shown that scattering strength depends smoothly on angle and frequency, and means of constructing **targets** have been proposed [25]. Figure C.4 gives two simpler alternatives.

From the theory of the scattering by the front plane (equation 3 of [25].), the value for the back-scattering cross section, σ , is found:

$$\sigma = \left[\frac{k r^2}{2} \left(\frac{Z_m - Z_w}{Z_m + Z_w} \right) \right]^2 \quad (4)$$

where

k is the circular wave number; ($= 2 \pi / \lambda$ in which λ is the wavelength in the medium)

r is the radius of the wire;

Z_m is the characteristic acoustic impedance of the wire material;

Z_w is the characteristic acoustic impedance of the surrounding medium (water).

Scattering is proportional to the square of the frequency and the fourth power of diameter (as long as the diameter of the **target** is so small that no phase differences between various parts of the **target** occur). At small angles the influence of the angle between front plane and the direction of the beam is small [25]. At larger angles observed scattering is diminished. Making the **target** small decreases the influence of the angle.

At oblique incidence of the **ultrasound** on the front end of the wire the scattering strength decreases. The consequence is that for a sector scanner, reorientation of transducer or **target** is necessary, in case quantitative measurements of reflected intensities in various parts of the field are to be compared. Quantitative measurements of beam width remain possible without reorientation.

Figure C.4 shows a sketch of two **point targets** made from dental wire (diameter 0,24 mm). The left **target** is intended for pulsed **ultrasound**; echoes from the wire tip and the supporting tube are received separated in time. The supporting tube (outer diameter 1,0 mm) provides mechanical stability. The right **target** is intended for CW **ultrasound**; the tip of the wire is the only feature inside the **ultrasound beam** (grey) which gives backscattering.

8.5.2 Axial PSF and LSF dimensions and axial resolution

8.5.2.1 General

The dimensions of the **PSF** or **LSF** in the axial direction should be measured at different depths. An appropriate phantom for **PSF** is shown in Figure C.3 or C.4, the latter being required when reverberations from the support wire are obtrusive. **LSF** is measured with a phantom as depicted in Figure A.1, A.2 or C.1 and axial resolution with the phantom of Figure C.2. Measurements can be made in increments of less than 1/3 of the axial **field-of-view** for a total depth depending on the transducer's imaging range. Sources of variability should be identified and documented, as indicated in 6.4.2. Axial resolution at a specific depth can be obtained as the minimum separation 1) of the two filaments in Figure C.2 that are distinguishable before the two filaments first blur together; and 2) quantitatively, at which there is at least a 6 dB drop in the signal between the two filaments.

8.5.2.2 Procedure – axial dimensions

Align the transducer for maximum signal from the **target(s)**. Alignment with the beam normal to any filament **target** is particularly critical. Set the system sensitivity controls so that all the **targets** are clearly visualized, but are not saturated. Make several **B-mode** images or image volumes of the **target** or phantom and digitize the best image. On some phantoms there is a scan window parallel to contained filament **targets**. Make sure that the **ultrasonic transducer** is well seated on the surface. When the transducer's face is curved, the central **scan line** should be normal to the scan window and the **ultrasound beams** should be normal to the filaments.

For **PSF** measurements, a 2D-scan is required for the **point target** in each measured location, unless alignment with the central axis or central plane of the beam or image plane has been accomplished. For **LSF** or **PSF** an image of the **test object** is obtained which contains the imaged targets for the entire depth range of the **ultrasonic transducer**. Use image-processing software to plot a brightness profile in the vertical direction for many successive **scan lines** covering an imaged **target**, moving laterally from left to right, until the maximum amplitude profile of the **target** is obtained. For this maximum amplitude of the **target**, at least, the full-width-at-half-maximum (FWHM) and the –20 dB width are measured and recorded for the given filament depth. Now, measure the axial response width for all the imaged-**target** positions (depths) in the phantom image. Once all the axial response widths are obtained, plot the FWHM or –20 dB width versus depth in the phantom image. This plot shows the variation in axial **PSF** or **LSF** for the imaging system with depth. For most systems, there will be only a small variation in axial **PSF** or **LSF** with depth.

Profile length (**PSF** or **LSF**) at a relative signal level near the expected axial clutter level or clutter level at a certain number of wavelengths is also most illuminating, if artefacts in the phantom allow that measurement.

8.5.3 Lateral PSF- and LSF-width and lateral resolution

8.5.3.1 General

The dimensions of the **PSF** or **LSF** in the lateral direction correspond to the high-contrast **lateral resolution** of the imaging system for a specific position along the beam (i.e. several depths in the phantom). An appropriate phantom for **PSF** is shown in Figure C.3 or C.4, the latter being required when reverberations from the support wire are obtrusive. **LSF** is measured with a phantom as depicted in Figure A.1, A.2 or C.1. Multi-**target** lateral resolution measurements are rarely required and are not addressed here. Measurements can be made over the total axial imaging range of the **ultrasonic transducer** in increments of less than 1/7 of the axial **field-of-view** along a generally central axis and at a distance from a lateral edge of the image of 1/8 of the image width. For volumetric imaging systems in which **PSF** might be dependent on elevational position in the volume (as for a linear array scanned in an enclosed housing), make a measurement at a distance from an elevational edge of the image of 1/8 of the volume thickness and in a corner of the volume at a distance from a lateral edge and an elevational edge of the image of 1/8 of the volume width and thickness, respectively. Sources of variability should be identified and documented, as indicated in 6.4.2.

8.5.3.2 Procedure – lateral dimensions

Align the transducer for maximum signal from the **target(s)**. Alignment with the beam normal to any filament **target** is particularly critical. Set the system sensitivity controls as described in 8.5.2.2. Make several **B-mode** images or image volumes of the **target** or phantom and digitize the best image. Make sure that the **ultrasonic transducer** is well seated on any scanning surface of the **test object**, so that the transducer face is perpendicular to the **test object's** surface.

For **PSF** measurements, a 2D-scan is required for the point **target** in each measured location, unless alignment with the central axis or central plane of the beam or image plane has been accomplished. For **LSF** or **PSF**, an image of the **test object** is obtained which contains the

imaged **targets** for the entire depth range of the **ultrasonic transducer**. Use the image processing software to plot a brightness profile in the horizontal direction for many successive **scan lines** covering an imaged **point target**, moving vertically from top to bottom (above and below the filament cross-section image), until the maximum amplitude profile of the **target** is obtained. For this maximum amplitude of the **target**, as a minimum, the full-width-at-half-maximum (FWHM) and the –20 dB width are measured and recorded for the given filament depth. Profile width at a relative signal level near the expected transducer clutter level or clutter level at a certain linear or angular beam width, is also most illuminating, if artefacts in the phantom allow that measurement.

Now measure the maximum lateral-response width versus depth for all the imaged **PSF** positions (depths) in the phantom image. Once all the lateral-response widths are obtained, plot the FWHM or –20 dB width versus depth in the phantom image. This plot shows the variation in lateral **PSF or LSF** (lateral **spot size**) for the imaging system with depth. For most systems, the **lateral resolution** varies much more with depth than the **axial resolution**.

NOTE The use of the **test object** described in Figure A.2 may be inadequate for some **ultrasonic transducers** which produce such large **side lobes** that they interfere with those of neighbouring filament images. For this case, a **test object** such as that described in Figure A.1 may be used. For that **test object**, the lines of **targets** are angled, so that individual **targets** are rarely positioned along the same horizontal line. A better **test object** is described in Figure C.1, where all filament **targets** are positioned individually, so that there are no laterally neighbouring **targets**. Finally, a single filament (or wire) on a micromanipulator, similar to that described in Figure C.3 but normal to the ultrasound beam, may be used.

8.5.4 Scan slice thickness (elevational PSF and LSF) or elevational resolution

For simple measurements in the field, the **slice thickness** is determined by scanning the thin sheet of scattering **targets** shown in Figure C.5. The **slice thickness** (the same as the elevational spot-size resolution) at each depth should be measured with the sensitivity and display controls in calibrated positions. Consider the maximum displayed range as being divided into five consecutive equal-sized sections. Place the plane of scan to intersect the sheet of **targets** along a line parallel to the top surface of the **test object**. The ultrasonic **scan plane** should also be perpendicular to the **test-object scanning surface**. To measure the **slice thickness** in the middle of the first section, arrange for the line of intersection of the ultrasonic **scan plane** and the **target** sheet to be at that depth. Using visual measurements or image-analysis software, measure the vertical thickness of the image of the sheet of **targets** and calculate the **slice thickness** at the centre of this first section, as illustrated in Figure C.5. Repeat the procedure for the four other sections. With image-analysis software, the width of the image can be measured at any desired level, e.g., FWHM, FWTM, (–6 dB, –20 dB) or clutter below that of the peak of the sheet signal.

For elevational **PSF** with a **point target** as described in Figures C.3 or C.4 or **LSF** using a horizontal wire similar to Figure C.3 or the filament phantom in Figures A.1, A.2 or C.1, a **B-mode** image is made. The **target** is moved perpendicularly to the ultrasonic **scan plane** (or *vice versa*) until the maximum amplitude of the **target** is obtained. The movement is then executed sequentially in two opposite directions along a line perpendicular to the ultrasonic **scan plane**, until two times a certain decrease in intensity is obtained repeatably twice. For example, at FWHM and FWTM (–6 dB and –20 dB), the clutter level and other desired levels may be obtained. The distance between the points on both sides of the ultrasonic **scan plane** is the **elevational resolution** at the measured level of, e.g. –6 dB, or –20 dB.

Measurements can be made over the total axial imaging range of the **ultrasonic transducer** in increments of less than 1/7 of the axial **field-of-view** along a generally central axis and at a distance from a lateral edge of the image of 1/8 of the image width. For volumetric imaging systems in which **PSF** might be dependent on elevational position in the volume (as for a linear array scanned in an enclosed housing), make a measurement at a distance, from an elevational edge of the image, of 1/8 of the volume thickness and in a corner of the volume, at a distance from a lateral edge and an elevational edge of the image, of 1/8 of the volume width and thickness, respectively. Sources of variability should be identified and documented, as indicated in 6.4.2.

Optionally the measurement of **elevational resolution** can be facilitated by using an array of equal **point targets**, placed on a line that makes an oblique angle with the **scan plane**. The **targets** should be placed at such a distance that their images are spaced with a greater distance than the **lateral resolution** of the scanner at that depth. When the **target** placed in the centre has a backscattering cross-section at the required level below that of the other **targets**, the measurement can be executed by obtaining images of equal intensity.

The most accurate and flexible method to derive the complex set of parameters based on the PSF mapping analysis is described in Annex D.

Annex A (normative)

Test objects – Calibration of 2D-spatial measurement systems

NOTE The **test objects** and materials described in this annex are derived from those described in several national reports and draft standards. The **test objects** have been restricted to those that have been demonstrated in practice.

A.1 Test object structures

Figures A.1 and A.2 contain sets of parallel filaments serving as or approximating **line targets**. The filaments are also parallel to the top surface, which serves usually as the scan window. The figures show two possible arrangements of the cross section perpendicular to the filaments. Figure A.2 has the advantage over A.1 of a more regular structure, but more risk of hiding a **target** by **targets** on top of it.

The scales of the test objects shown are for general abdominal imaging at 2 MHz – 5 MHz centre frequencies. The distances between filaments in a row are 2 cm. Accuracy of filament placement should be $\pm 0,2$ mm. For higher-frequency imaging, a finer, 1 cm, filament spacing, overall dimensions equal to 1/3 of the those shown in Figures A.1 and A.2, and $\pm 0,1$ mm placement accuracy are required.

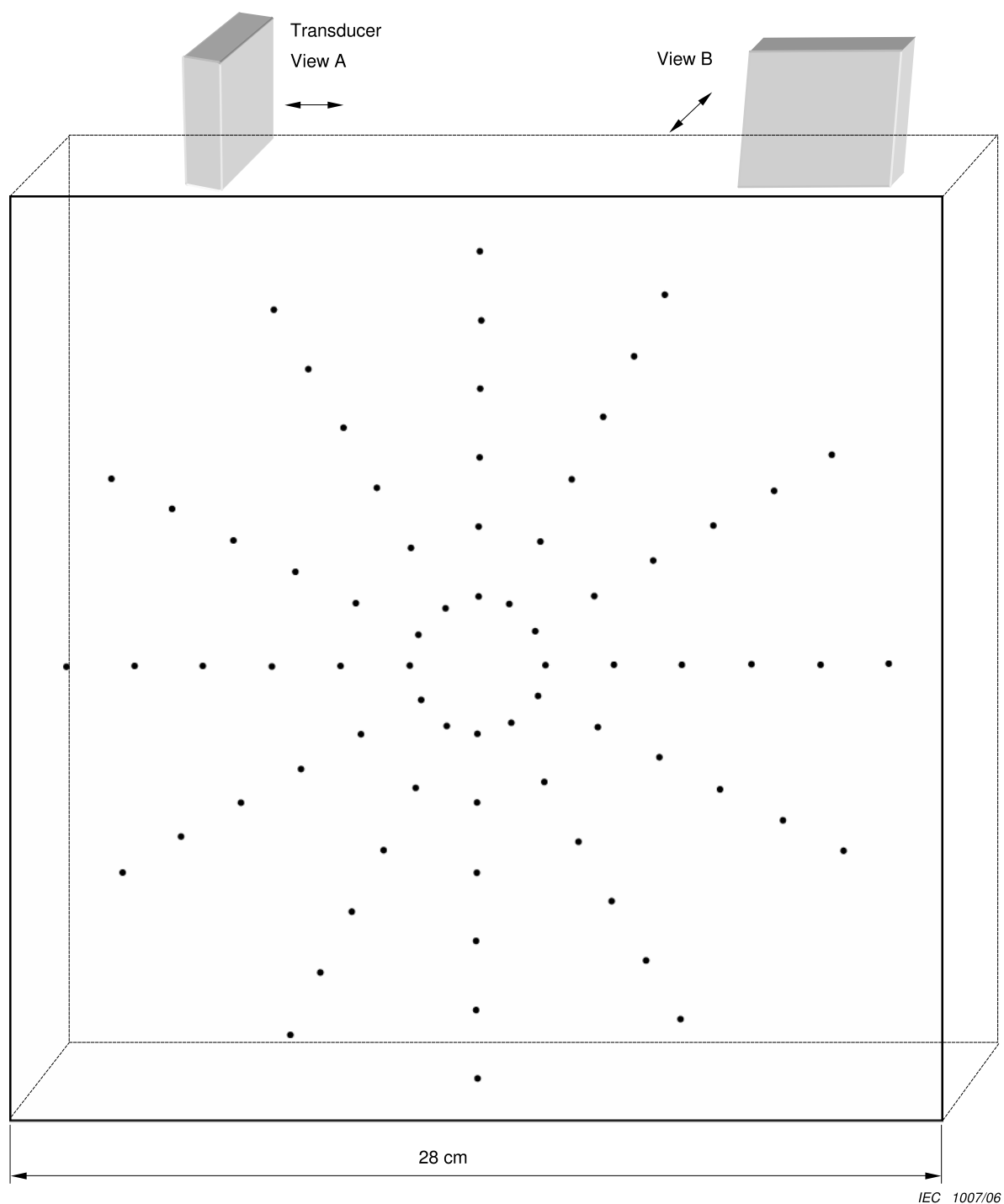
For this standard each **test object** (Figures A.1 and A.2) is designed to be scanned directly in **working liquid**. Nylon filaments in the objects are used to reduce the effect of shadowing. They are typically 0,078 mm in diameter (8X fly-fishing tippet). The tank should be sufficiently large to allow all transducers to be used in all required orientations to the internal structures.

Alternatively, the structures in Figures A.1 and A.2 may be immersed in **tissue-mimicking material** and scanned through an acoustic window of suitable material, such as polythene. A suitable window material should prevent the loss of water from the **tissue-mimicking material**. With **tissue-mimicking material** of typical backscatter, wires rather than nylon filaments should be used for the structures to give stronger echoes. Suitable wires for the **test object** are of stainless steel (type 316) with a diameter of 0,15 mm. These wires are sufficiently thin so that echoes from them are not elongated by internal reverberations at all but the highest diagnostic frequencies. They should be located in each **test object** to within $\pm 0,4$ mm.

These structures are designed with parallel filaments, as in most commercially available phantoms originally designed for calibrating 2D-imagers. They may be employed for 3D-imaging system calibration, two dimensions at a time, by rotating the scanhead 90 degrees about an axis normal to the filaments and lying centrally in the phantom and the imaged volume. A more general use of phantom motion has been published recently [29]. Numerous papers have been published since 1990 on filament phantoms for single view calibration of 3D-imaging and image-guided therapy systems [30].

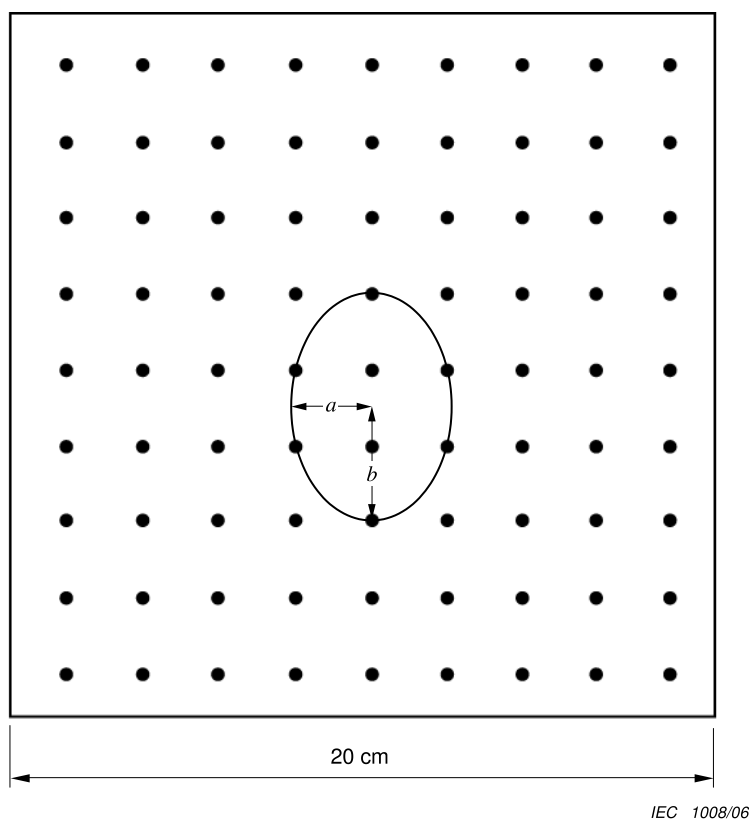
The calibration of **test objects** depends on the sound velocity of the medium (**working liquid** or **tissue mimicking material**). Sound velocity depends on temperature. With non- enclosed liquids, the change of composition by evaporation (e.g. of alcohol) should be considered. The stated margin in sound velocity ($1\,540 \pm 15$) m/s contributes ± 1 % uncertainty to measuring results. In view of variations in sound velocity in the body, this uncertainty is acceptable.

A.2 Test object diagrams



IEC 1007/06

Figure A.1 – Concentric circular arrays of nylon filaments



An imaginary ellipse is shown for curved-line length measurements, such as a perimeter.

Figure A.2 – Regular 2D-array of nylon filaments

Annex B (normative)

Test objects – Measurement and calibration of 3D-image reconstruction accuracy

B.1 Working-liquid-filled filament phantom

The phantoms described in Annex A (Figures A.1 and A.2) are appropriate for 3D as well as 2D measurements.

B.2 Tissue-mimicking filament phantom

A suitable phantom for a filament **test object** containing a backscattering background is shown (Figure A.1) for 2D- and 3D-distance measurement and image-distortion assessment. For 3D-imaging, the scanhead is moved in the elevational (ultrasonic **scan- plane** thickness) direction from left to right on this figure to assess position registration in the **slice thickness** direction. The relative positions of all filaments are maintained within $\pm 0,2$ mm at all locations. For arc length, perimeter, area and volume measurements the imaginary ellipse (or imaginary elliptical rod), $a = \pm 2,7$ cm, $b = \pm 3$ cm, is shown in Figure A.2. The area and circumference (perimeter) of the ellipse are $\pi a b = 25,4$ cm² and approximately $\sqrt{2} \pi(a^2 + b^2)^{1/2} = 17,9$ cm, respectively. Circular area and perimeter measurements and cylindrical rod volume and surface area measurements may be made easily on Figure A.1 and rectangular ones on Figure A.2. Volumes and surface areas of rods are, respectively, the cross-sectional area times the length and two times the cross-sectional area plus the perimeter times the length.

B.3 Tissue-mimicking ovoid target test object

Specification for 3D-, speckle-defined (egg-shaped) objects:

- Attenuation coefficient equal to that of the background material $\pm 0,2$ dB cm⁻¹ MHz⁻¹.
- Speed of sound equal to that of the background material ± 6 m s⁻¹ [9, 31].
- The egg-shaped objects have a **backscatter coefficient** -10 dB ± 2 dB relative to that of the standard background material, whose backscatter coefficient at 3 MHz is 3×10^{-4} cm⁻¹ sr⁻¹ ± 6 dB (with an approximately f^4 dependence from 1 MHz to 15 MHz).

B.4 Spatial measurements from backscatter-defined objects

A suitable **test object** for spatial measurements of backscatter-defined objects is shown in Figures B.1 to B.4. This **test object** is referred to as the 3D-egg **test object** because the measured geometries consist of two half-ellipsoids of revolution (oblate spheroids) joined end to end.

The side view (Figure B.1) shows the long-axis views of the two ellipsoids. A composite of two cross-sectional views is shown in Figure B.2 and projections are illustrated in Figures B.3 and B.4.

NOTE Alternatively, for practical purposes in the hospital, in which a standard phantom is not available, the 3D-object shapes may be constructed of a low -attenuation material, such as open pore reticulated foam [6, 32] and immersed in saline or other fluid, such that the 3D-objects and the fluid meet the above speed-of-sound criteria, the differential attenuation of $0,5$ dB cm⁻¹ MHz⁻¹, and the backscatter coefficient of the 3D-objects (egg) at 3 MHz of 3×10^{-5} cm⁻¹ sr⁻¹ ± 16 dB (with an f^2 to f^4 dependence from 1 MHz to 7 MHz).

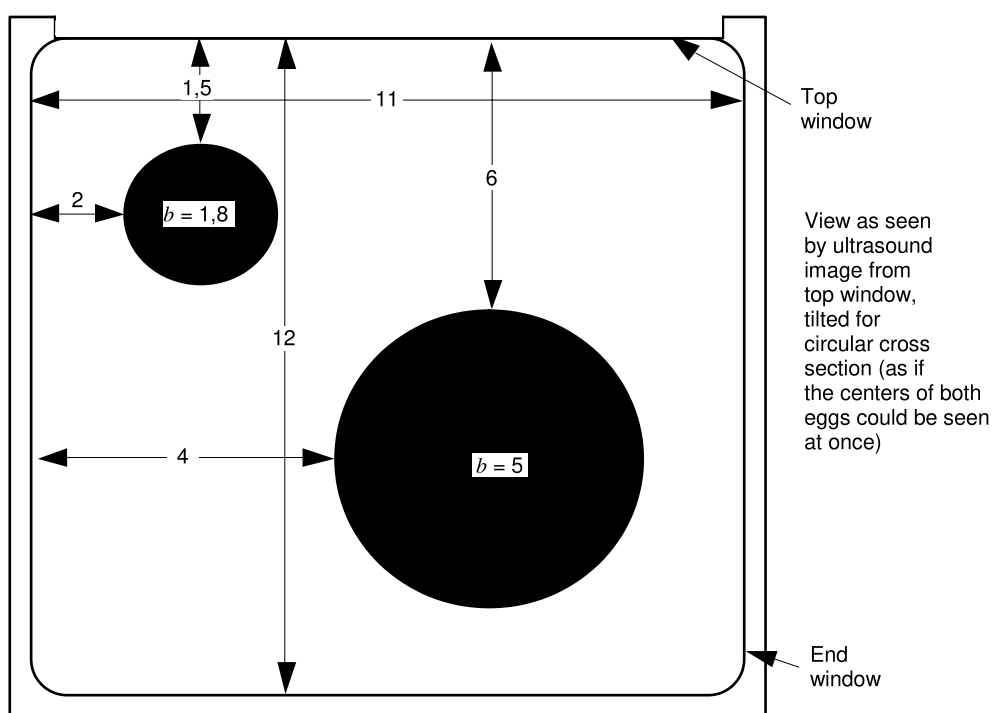
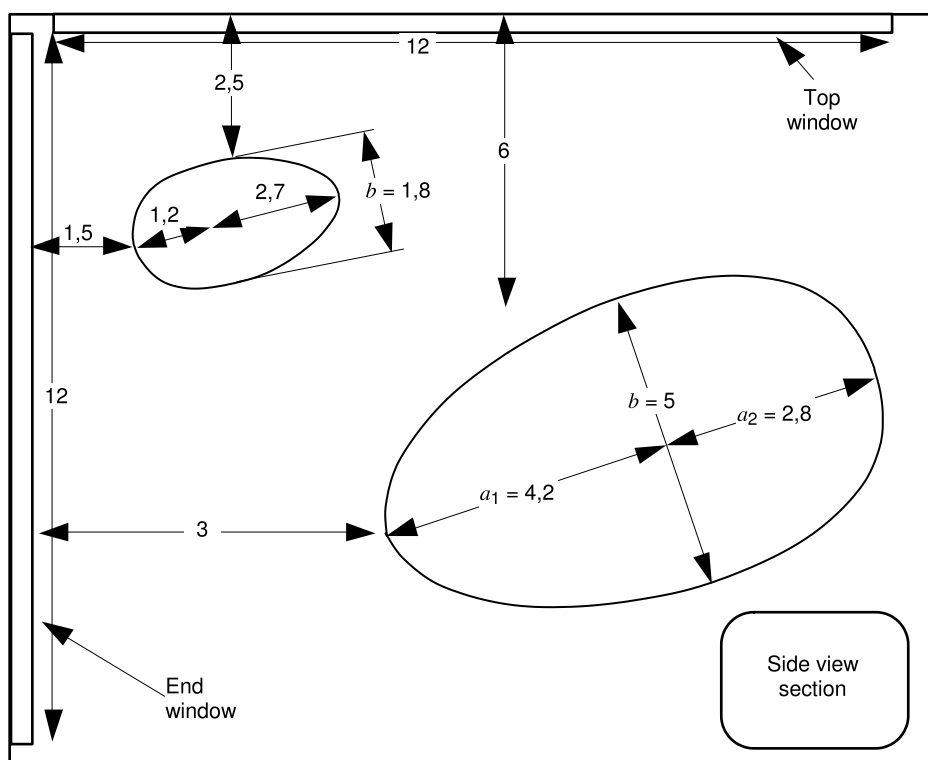
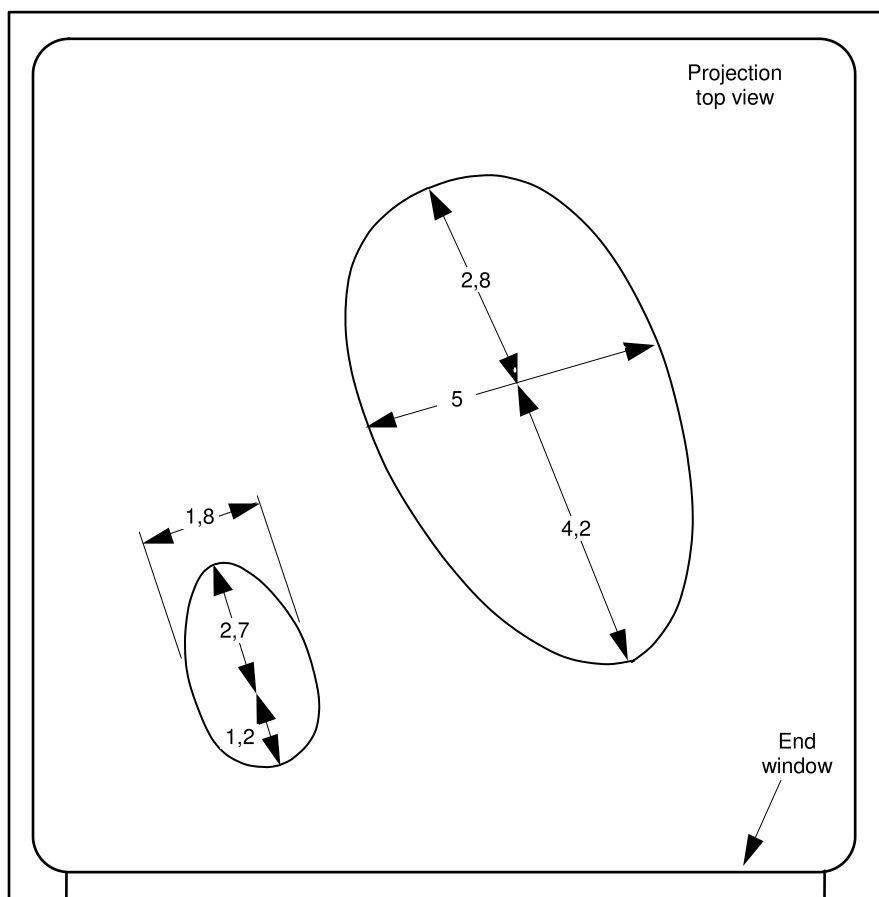


Figure B.2 – Composite of two cross-sectional views of test object shown in Figure B.1



IEC 1011/06

Figure B.3 – Projection view from top of test object shown in Figure B.1

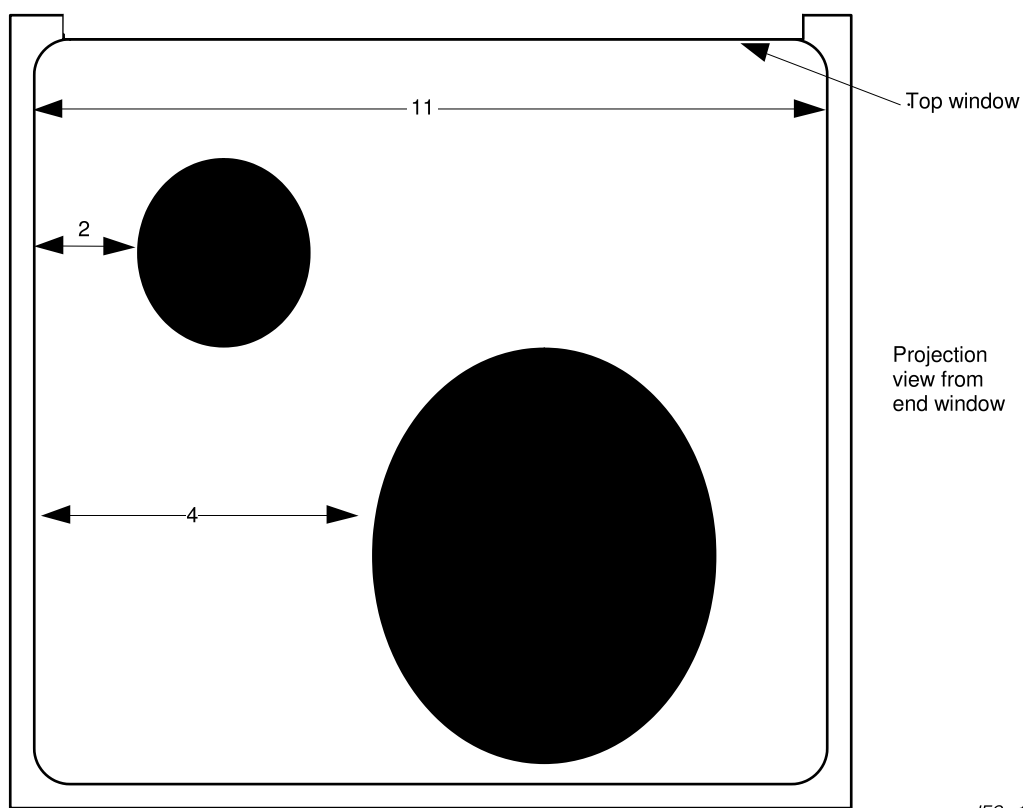


Figure B.4 – Projection view from end window of test object shown in Figure B.1

Annex C (normative)

Test objects – Measurement of point-spread function response

NOTE The **test objects** and materials described in this annex are derived from those described in several national reports and draft standards. The **test objects** have been restricted to those which have been demonstrated in practice.

C.1 Test object structures

For this standard each **test object** (Figures A.1, A.2 and C.1 to C.5) is designed to be scanned directly in **working liquid**. Nylon filaments in the objects are used to reduce the effect of shadowing. They are typically 0,1 mm in diameter. The tank should be sufficiently large to allow all transducers to be used in all required orientations to the internal structures. Under some conditions (see 5) the **working liquid** may be replaced by water.

Alternatively, any of the mentioned structures may be immersed in **tissue-mimicking material** and scanned through an acoustic window of suitable material, such as polythene. A suitable window material should prevent the loss of water from **tissue-mimicking material**. Wires rather than nylon filaments should be used for the structures in the **tissue-mimicking material** to give stronger echoes. Suitable wires for the **test object** are of stainless steel (type 316) with a diameter of 0,15 mm. These wires are sufficiently thin so that echoes from them are not elongated by internal reverberations. They should be located in each **test object** to within $\pm 0,2$ mm.

C.2 Array of filaments in water (Figure C.1)

The two-dimensional array of filaments is supported in a frame. The wires or filaments are as specified in A.1.

C.3 Axial resolution test object (Figure C.2)

Two nylon filaments of typical diameter 0,1 mm are arranged to lie in the same vertical plane (Figure C.2) and at an angle to each other. The filaments can be supported over a range of depths. The volume of the liquid-filled tank should typically be a cube of side length equal to 20 cm.

C.4 Movable single filament or wire in water (Figures C.3, C.4)

The filament is mounted on a micromanipulator (see Figure C.3) which allows it to be moved throughout the **field-of-view** of the scanner. The filament is as specified in A.1. An alternative is to use a fine metal wire mounted as shown in Figure C.4. The diameter of the metal wire shall be much smaller than the beamwidth of the investigated transducer and smaller than $\lambda/2$ for low- f -number transducers.

The use of a movable spherical target for assessing quality parameters derived by **PSF-mapping analysis** is described in Annex D.

C.5 Scanner slice thickness test object (Figure C.5)

- a) A sheet of scatterers makes an angle of 75° to the top surface [33]. Suitable material for the sheet of scatterers has a **backscatter coefficient** more than 50 dB higher than that of the background material and the sheet is less than 0,4 mm thick.
- b) The method of calculating the scan **slice thickness** from the image is discussed in 8.5.4.

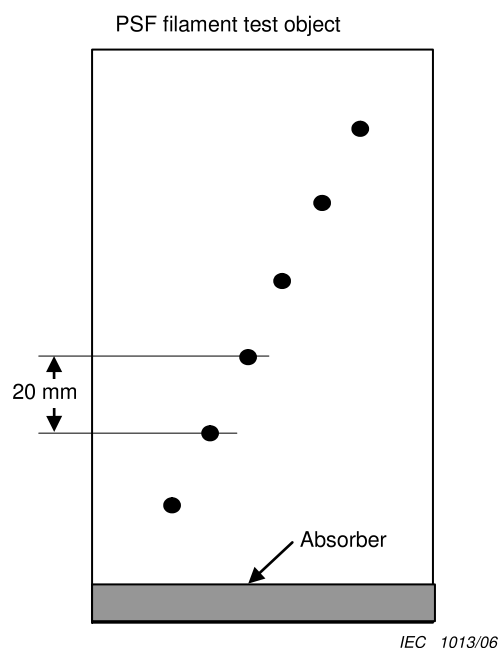


Figure C.1 – Filament test object for measuring the LSF

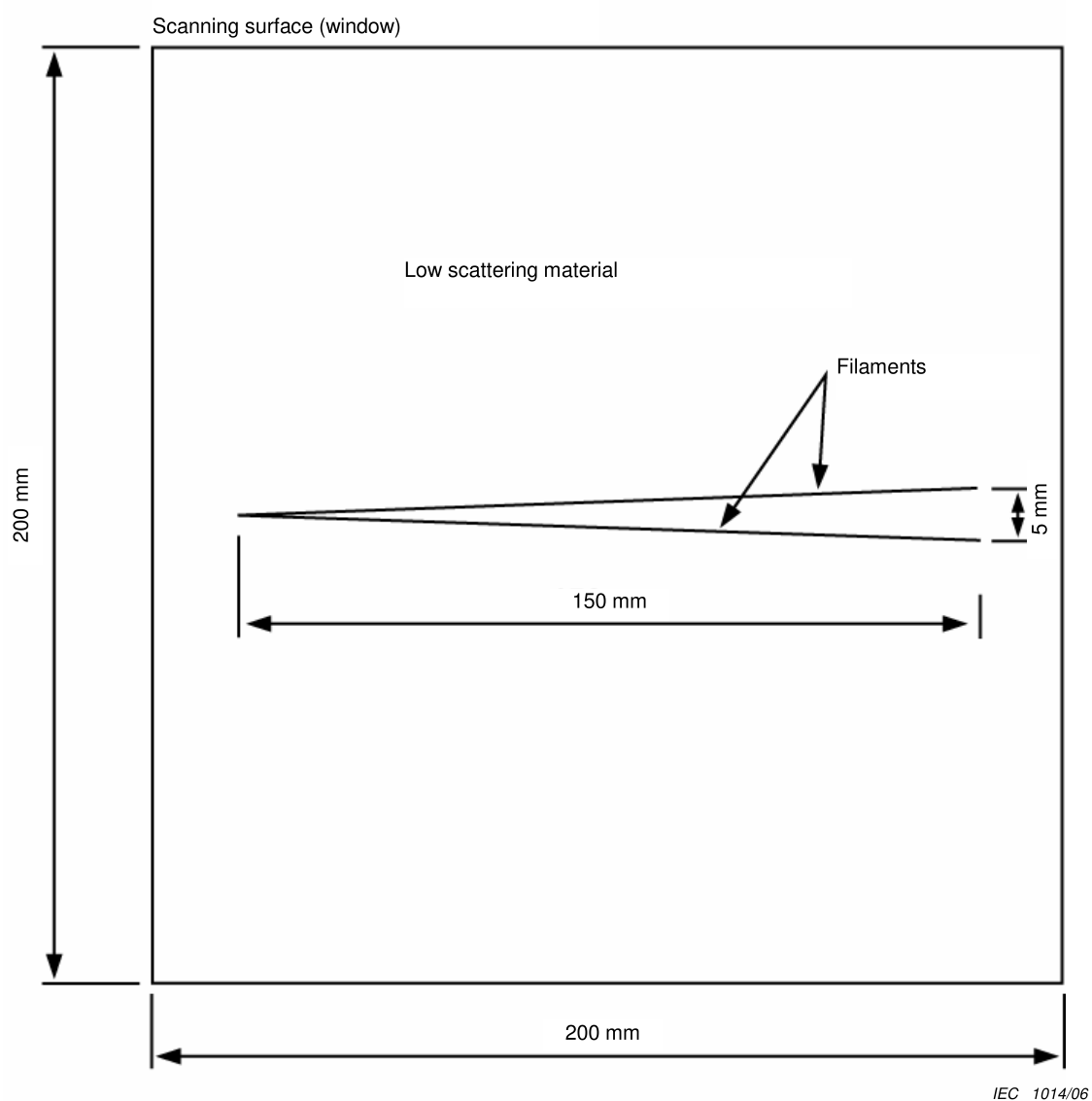
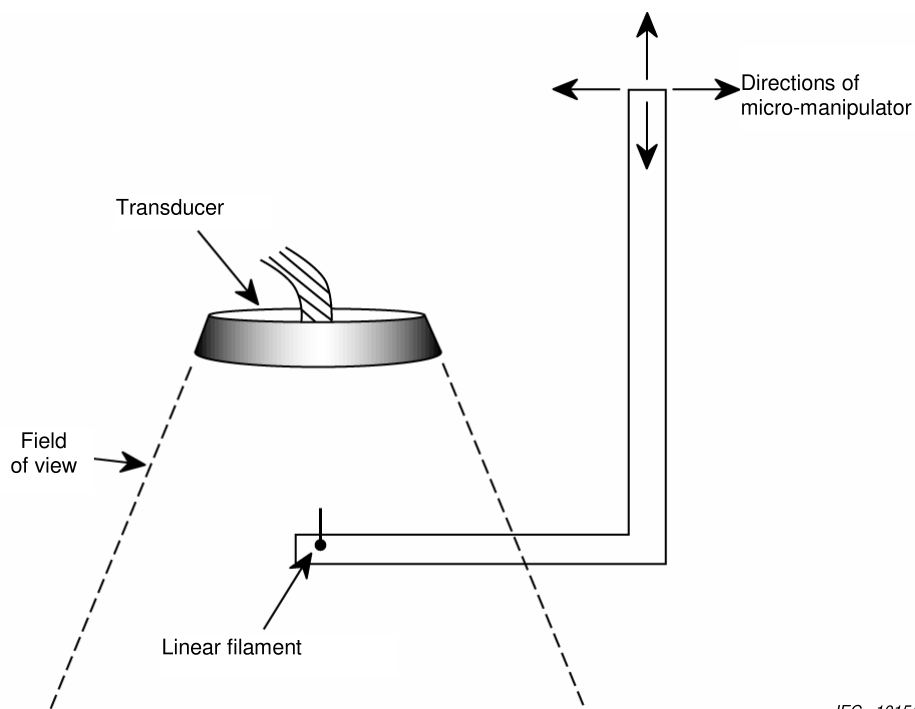
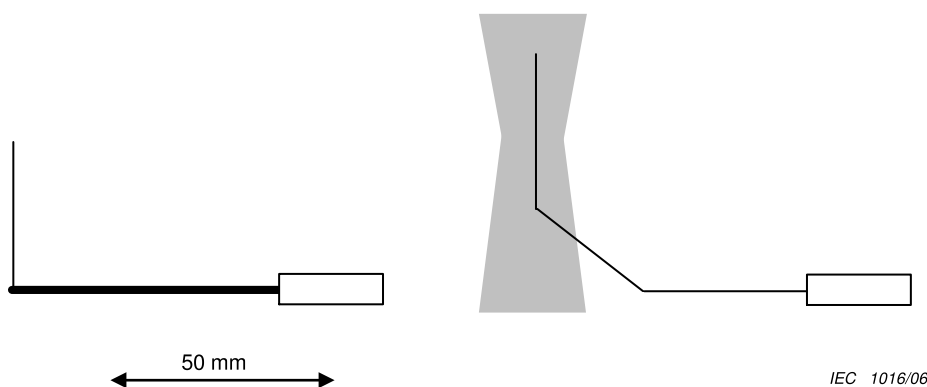


Figure C.2 – Axial resolution test object



IEC 1015/06

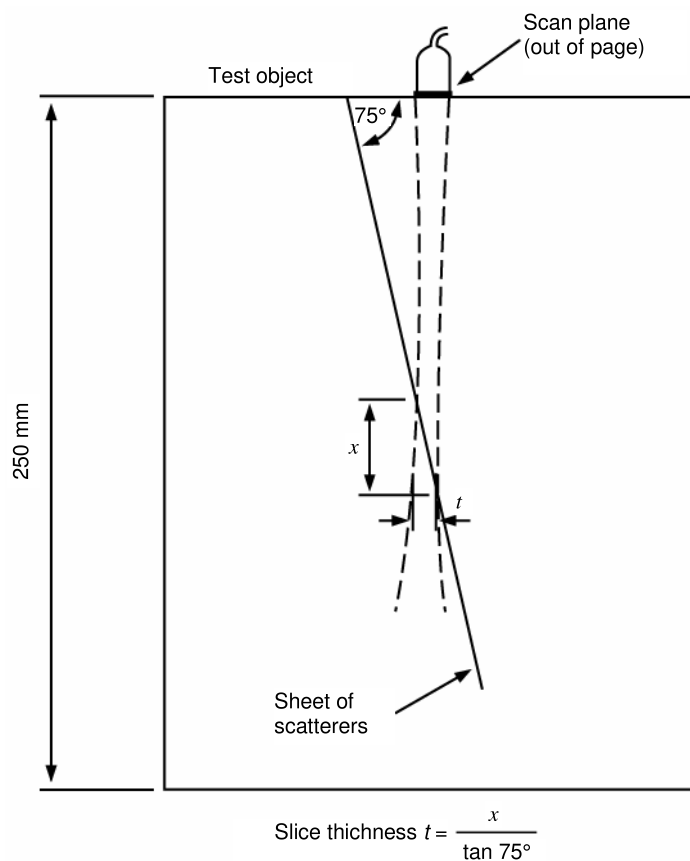
Figure C.3 – Movable single filament in water



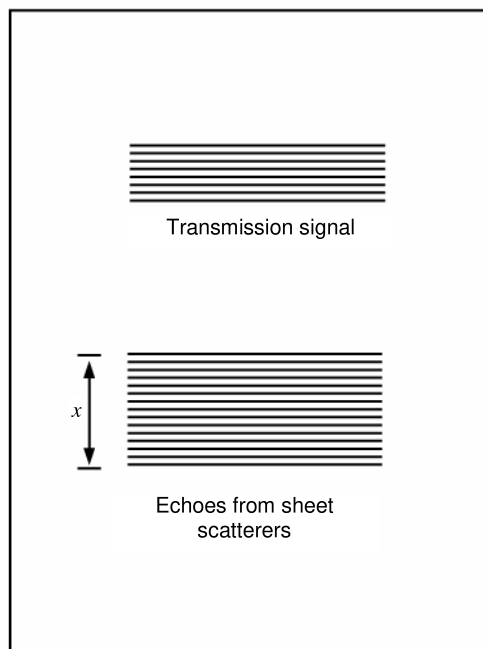
IEC 1016/06

At their right side the targets can be connected to a scanning table.

Figure C.4 – Sketch of two point targets made from dental wire (diameter 0,24 mm)



Image



IEC 1017/06

Figure C.5 – Slice thickness measurement and calculation

Annex D (informative)

Quality parameters derived by PSF-mapping analysis

D.1 General

A quality assessment system is vitally needed to provide an accurate and well-defined set of production-quality parameters for new or refurbished scanners or transducers in acceptance tests before their introduction to medical practice. It is important that products delivered by third-party sales groups, system-refurbishers and/or transducer manufacturers be carefully tested to be able to declare technical parameters of their products to be comparable to those of the new, originally manufactured systems. The methods used for quality assessment in medical applications are not certain and accurate enough to be used for such kinds of technical **performance evaluation**. **PSF-mapping analysis** gives reliable parameters suitable for this kind of tests. These parameters do not directly indicate the effectivity of a clinical diagnostic process, even though a close correlation between the assessed technical quality and success in the diagnostic process may be expected [40].

The ultrasound scanner used as a diagnostic system is composed of the system-control/user-interface unit and the ultrasonic-transducer assembly. Either unit can contain the transmitter- and the receiver- electronic systems and some of the beam-former electronics. The **ultrasonic transducer** converts electrical signals to ultrasound field and vice versa. Electrical and acoustic parameters of the transducer determine quality of the scanning **ultrasound beam**. The electronic system controls the transmitted and received ultrasound signal, conversion from mechanical to electrical signals, and the signal processing and conversion to video-signal inputted to the imaging unit. The imaging unit transfers the information to the human preceptors. The **PSF-distribution analysis** evaluates qualitative parameters of the whole ultrasound-scanner system, excluding the display unit. The analysed signal is affected by the quality of the whole imaging cycle, and the transmitting and receiving parts of the scanner. The analysed system function is affected by a complex set of control functions. Therefore, it is important that the combination of the control settings of the scanner be exactly specified and recorded as a part of the measurement.

D.2 Method

Annex D describes a method for precise and reliable measurement of several qualitative parameters of whole ultrasound scanning systems including both the transmitting and receiving parts of the systems, excluding the parameters of scanner display. The method is based on **PSF-distribution analysis** over a scanning area. In the case of **PSF-mapping**, the measured parameters are derived by analysis of sonograms generated by scanning a spherical target moving over a defined scanning volume on a specified trajectory.

The **PSF-mapping** system evaluates a set of parameters acquired over a user-defined area in one scanning plane of a B-mode grey-scale sonogram, scanned in a tank filled by degassed working liquid and using one measuring procedure. The whole target sonogram is not evaluated in the **PSF-mapping analysis**. The test signal is obtained by reflection of a transmitted ultrasound wave from a point-reflector surface and working-liquid boundary only. The point reflector used is a highly reflective sphere of diameter D [41].

The method is suitable for all kinds of echo(reflections)-evaluating sonographs using different types of beam-forming and plane-wave compounding of ultrasound signal in the frequency range 0,5 MHz to 50 MHz. The upper frequency limit is determined by a ball target of minimum diameter available to assure reflection effectivity and fulfil the condition $\lambda \leq D \leq 4\lambda$, where λ is the ultrasound wavelength in the working liquid [42]. Further limiting factors are a minimum size of step and precise mechanical construction of the positioning system to assure measurement reliability and adequate scan size.

The method is relevant for all the types of transducers used with these scanners, including

- mechanical probes including annular arrays,
- electronic phased arrays,
- linear arrays,
- curved arrays,
- two-dimensional arrays, and
- 3D-volume scanning probes based on a combination of the above types.

The **PSF**-measuring system is not a tissue-mimicking object. It is dedicated to performing accurate, stable and reliable measurements under conditions appropriate to achieving these measurements of parameters, some of which may be obtained by use of sophisticated electronic measurements of the scanner's electronic system and some by **PSF**-mapping analysis only [43].

The following data are acquired and are analysed using the method:

- a) the ROI digital image stored for the scanned-plane axis in each point of the **measuring grid**;
- b) the echo-signal amplitude distribution over the measured area;
- c) the distribution of the parameter $W_{F, HM}$ which is Full-Width-at-Half-Maximum (FWHM) of the **point-spread function (PSF)** in the azimuthal direction over the measured area;
- d) the distribution of the parameter $W_{H, HM}$ Half-Width-at-Half-Maximum (HWHM) of the **point-spread function (PSF)** in the axial direction over the measured area;
- e) the peak echo-amplitude received $a_{r, max}(x, y, z)$ at each y_k step of the target position in the elevation (transversal) direction;
- f) the (x, y, z) coordinates set for stored position of the point reflector generating $a_{r, max}(x, y, z)$ from MER in each point of the **measuring grid** (position in centre of ultrasound beam).

Data analysis derives the following ultrasound scanner parameters and functions:

- 1) focal areas in both the azimuth and the elevation directions;
- 2) visualization of the distribution of ultrasound scanning lines;
- 3) manufacturer's preloaded **TGC** function;
- 4) width (elevation) of the scanning plane over the depth of scan;
- 5) side-lobes signal-level distribution in the scan plane;
- 6) amplification uniformity in the azimuth direction;
- 7) scan geometry linearity and **accuracy**.

D.3 Environmental conditions

The most temperature-sensitive parameters are those assessing geometry of the sonogram and related calculations. The temperature-dependent deviations may be compensated mathematically from known working-liquid temperature and thermal coefficient of speed of sound.

Water condensation on electronic system components should be avoided.

D.4 General requirements of the method

Ultrasound waves produce a **PSF**-signal that is neither singular nor isotropic. Furthermore, the ultrasound **PSF** can be asymmetrical, having different axial and lateral dimensions, and it also

varies with distance from the transducer in both the axial and the azimuthal directions. Thus, it is important that many different measurements of the **PSF** at different positions and depths be performed to obtain representative values of the system's imaging performance at specific positions along the beam axis. It is also important that the measured area be covered by a grid of the measuring points, the density of which is determined by the expected parameters and the **accuracy** of the measurement [44].

NOTE For example, determination of a focal point's position may need an axial step $\Delta z = 5$ mm; visualization of scanning lines demands an azimuthal step $\Delta x = 0,1$ mm for a conventional linear transducer of nominal frequency 3 MHz.

The following features are necessary to apply the **PSF**-analyser to the sonograph:

a) The sonograph to be tested:

- video-signal output of live, dynamic scanning available in analog (composite) or digital (DVI-D, HDMI) form;
- operating instructions or skilled operator to assure proper manipulation and operating adjustment;
- record of proper evidence and registration of the measurement process, including identification of operator and all apparatus used, record of parameters preset in the measured system, record of environmental conditions, including a time stamp.

b) The basic configuration of the measurement tank:

- The **scanning window(s)** is(are) localized in the side wall(s) of the tank.
- The spherical-target positioning system is fixed on top of the tank, controlling movement of the target fixed in a holder.
- Filling the tank with degassed working liquid is recommended to prevent bubble generation in the working fluid. Bubbles may mimic the point reflector and/or produce spurious reflections from the point reflector after having accumulated on it, due to surface tension.
- Temperature should be kept in the specified range to eliminate measurement uncertainties generated by dependence of the speed of ultrasound propagation upon temperature.

c) The transducer:

- The transducer is acoustically coupled to the **scanning window** by standard coupling ultrasound gel. The **scanning window** is covered by tough, flexible foil made of material with similar acoustic properties to the working liquid to avoid ultrasound field reflections and absorption.
- The scanning plane is oriented in the horizontal direction and the transducer is fixed to keep the whole slice thickness below the water surface. It is important that the lateral orientation of the transducer be specified.

d) The positioning system:

- A computer-controlled micromanipulator is used to move the target or transducer to a determined position. It is important that mechanical construction, **accuracy** and stability of the positioning system correspond with ultrasound frequency. The shorter the wavelength, the more accurate and robust the system should be to avoid systematic measuring errors.

e) The control and analysing system:

- The software controls the video-signal acquisition, determines ROI, selects and saves the ROI frames to be used for analysis and finally maintains the complex analysis of the stored information.
- The basic parameters used for **PSF**-analysis are the $W_{F, HM}$ (Full Width at Half Maximum, i.e. at -6 dB down from $a_{r, max}$) and the **pixel** level of noise at an area without reflections.

- The $W_{F,HM}$ depends upon the intensity of the received signal. Therefore, it is important that the receiving gain and output power be properly adjusted to utilize the whole dynamic range of the analyser.

D.5 Measuring conditions

D.5.1 General

In Figure D.1 a principal schematic of the **PSF**-analyser is introduced. The **PSF**-analyser consists of the data-acquisition components (test-tank and point-reflector parts of the schema), a linear transducer, a personal computer (PC) with video-signal input, running acquisition and analysis software. The sonograph being measured is shown with a linear transducer but the results analysis is displayed for a sector-scan transducer to illustrate transducer-type independency of the system.

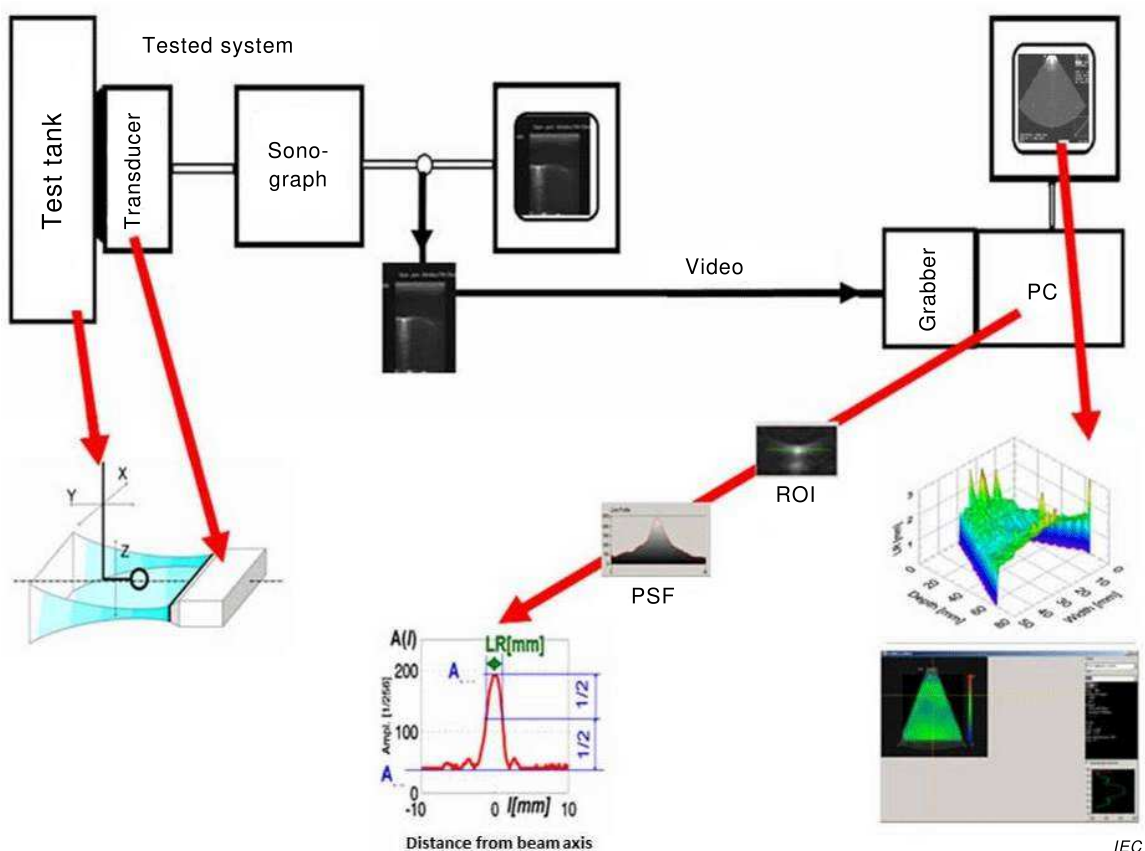


Figure D.1 – Principal schematic of the PSF-analyser function

D.5.2 Sonograph

D.5.2.1 General

Sonographs are equipped with a large set of different control functions to ensure optimal handling of received ultrasound signals to create the best image. These control functions affect remarkably the measurement results because the method evaluates the signal after it has passed through a whole imaging system. Therefore, it is important that all the adjustments and settings be carefully documented in the measuring protocol to be available for use for repeated measurements.

D.5.2.2 The amplitude transfer-characteristic adjustment

The sonograph receiver employs a logarithmic amplifier to compress a very wide range of received signal amplitudes (more than 100 dB). It is important that all the other additional, nonlinear, so-called pre-processing and post-processing functions be switched off or adjusted to linear working regions to avoid further nonlinear distortions of the signal. This requirement includes functions dedicated to eliminating different types of noise either by scan-sequences correlation or averaging or cutting off low-level signals.

It is important that the amplified signal-peak amplitude limitation due to low amplifier dynamic range be eliminated to assure correct measurement of the FWHM-function. Therefore, it is also important that the widest dynamic range accessible by the measuring system be selected, except in the case when the gain of the sonograph is not sufficient to achieve full dynamic range for the $A_{r,max}$.

D.5.2.3 The look-up table setting

The relation of the grey-scale level to the digitalized amplitude of the signal is derived from a look-up table (LUT) or Gamma curve. If possible to assure a pure logarithmic characteristic of the system, only a most linear LUT should be used. It is important that all the settings be documented in the measuring protocol, to be recorded for use during further periodical measurements. A method to find the most linear LUT should be part of the measuring software.

D.5.2.4 The measured gain adjustment

The sonograph's receiver gain is controlled in two parts: the **overall gain** (G_o) and the **time-gain compensation (TGC)** functions. Their adjustment prior to measurement should be as follows: The **TGC** is adjusted first. The basic condition is to adjust the **TGC** to be constant within the whole range of the scanned depth. The basic condition of the **TGC** adjustment is its independence from the depth parameter (elimination of the function). Then the G_o is increased to just obtain little visible thermal noise of the system in an echo-free area on the screen, e.g. when the transducer is energized when coupled to the air. It is important that the G_o be decreased or a smaller reflector used, if the $A_{r,max}$ amplitude limitation occurs.

Some sonographs utilize an **ATGC** (automatic time gain compensation) function. It is important that this function be disabled for gain (sensitivity) dependence from depth and focal-points localization from MER-profile assessments.

D.5.3 Measuring tank

D.5.3.1 General

The measuring tank serves as a holder of the volume of homogenous non-reflecting medium – the working liquid – in which the point reflector is moving by the precise, computer-controlled electromechanical system. An appropriate **scanning window** allows effective acoustic connection between the working liquid inside the tank and the active part of the measured transducer. The size of the measuring tank should be appropriate to the size of largest scanned area expected.

D.5.3.2 Working liquid

It is important that the working liquid used in the tank be degassed to avoid collection of bubbles at the point reflector [45]. It is also important that the temperature be recorded for the measurements to be based on accurate values of speed of sound, which significantly affects reading geometry of the scan. However, measurements based on reading the echo intensity are not remarkably affected by working liquid temperature because temperature dependence of both the reflectivity and the water absorption are negligible in this case.

The accuracy of the distance calculations is determined by the accuracy of knowledge of ultrasound speed of propagation in the working liquid. The coefficient of speed of ultrasound in

pure water as a function of temperature is approximately $4 \text{ ms}^{-1}\text{°C}^{-1}$ in the temperature range 10 °C to 30 °C . According to the distance calculation from the echo delay, a change of temperature of $\pm 1 \text{ °C}$ results in error of $\pm 0,15 \%$ of the measured distance value. A calculation giving the temperature error compensation is possible by use of an expression given by [46] or another by [47].

D.5.3.3 Reflections

Due to low ultrasound absorption of the working liquid and high reflectivity of the tank walls and relatively small size of the measuring tank, many multiple reflections are generated in the scanned area. These multiple reflections may be eliminated by a simple system of absorbers and reflector shielding because a pulsed signal is generated and only a small ROI surrounding and close to the reflector is analysed in the sonogram.

D.5.3.4 Scanning windows

The ultrasound signal is transferred between transducer and reflector via the **scanning window**. The window works as an acoustic coupler between the active surface of the transducer and the working liquid. The window is closed with flexible foil made of material with similar acoustic properties to the working liquid to assure effective transfer of the ultrasound signal and elimination of ultrasound-signal reflections and absorption. The transducer is attached to the foil with use of conventional coupling gel so that the whole active surface of the transducer is in acoustic contact with the working liquid. Proper shape of the window is needed to assure effective contact of different types of transducers. The measured transducer holder is part of the **scanning window**. Its design should ensure adjustment of the scanning plane to be parallel to the xz -plane of the target positioning system and stable transducer positions during measurement.

D.5.4 Target

D.5.4.1 General

The target system is composed of the sphere reflector, the reflector holder and the reflector positioning system.

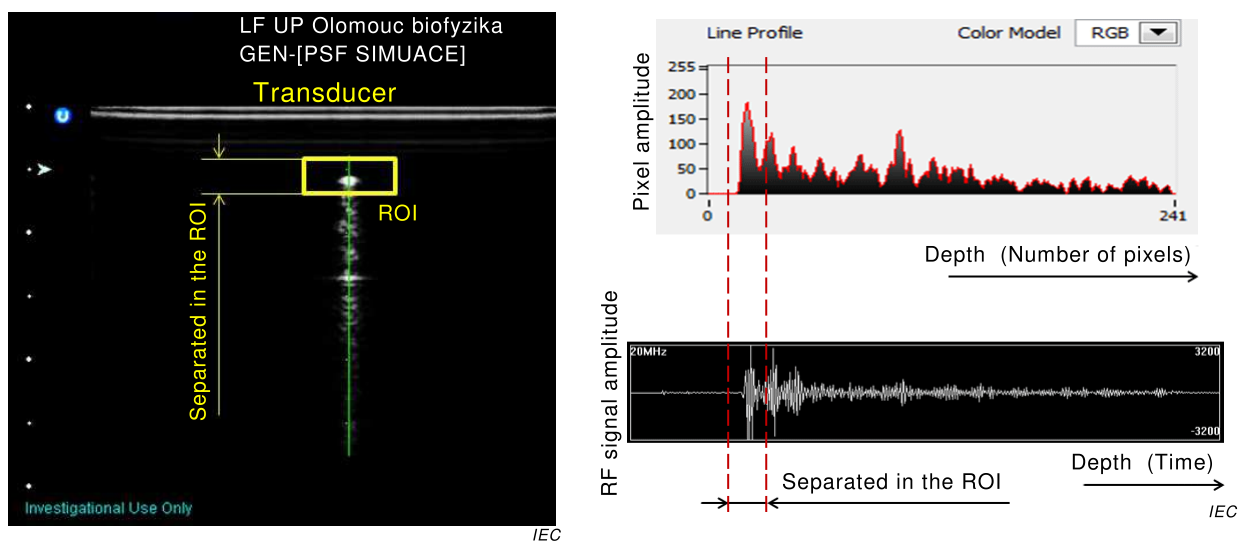
D.5.4.2 Reflector

In contrast to C.4 and Figure C.4, the point reflector is not the flat end of a wire but a sphere made of steel or similar highly reflective material. The reflector is fixed to a positioning system with use of a holder made of tiny, hard wire. The sphere reflection is independent of the incident angle of irradiation.

The diameter D of the rigid metallic reflector is critical for the method used but should be kept within a range $\lambda < D < 4\lambda$ according to [42]. Use of a reflector of $D < \lambda$ is possible, but the reflected signal amplitude will not be strong enough for low-sensitivity systems. Therefore decrease of the target diameter, D , may be employed when too-sensitive sonographs limit high amplitude of signals, even if the **overall gain** G_0 and output power are adjusted to minimum levels (see D.5.2.4).

If $D > 4\lambda$, values and variations of the $W_{F,HM}$ increase to an erroneous extent in both the lateral and the elevational directions.

The reflections from any inner structure of the sphere and multiple reflections in the sphere and/or sphere resonances [48] [49] are eliminated in the received-signal evaluation process. They do not affect the measurement results. The **PSF**-mapping method evaluates reflections of a transmitted ultrasound wave from a point-reflector surface and working-liquid boundary only (Figure D.2). It is important that the sphere reflector be made of a highly reflective material, i.e. the difference of characteristic acoustic impedance of the working liquid and sphere-target material should be as high as possible.



The yellow rectangle specifies ROI. The green line corresponds with the signal displayed in b).

A linear transducer, nominal frequency 10 MHz, was used.

In the top figure, RF signal amplitude is converted to pixel brightness, time domain to space domain. The ROI separated for analysis is marked.

In the bottom figure, the RF signal corresponds with the green line in a).

a) Scan of spherical target 0,6 mm diameter

b) Process of demodulation (top) and the target-scan received-RF signal (bottom)

NOTE The late reflections are not included in the analysed data.

Figure D.2 – Principle of elimination of internal multi-reflections in the spherical target using filtration in time domain

D.5.4.3 Sphere holder

The holder fixes the sphere to the positioning system. It is important that the holder construction avoid any influence upon the reflected signal. It is also important that the connection be rigid enough to eliminate any shift of the target during its displacement in the water, due to hydraulic resistance. A possible alternative for such a holder is a tiny, stiff wire bent to rectangular form of the letter L with the sphere fixed on one end (the bottom – horizontally oriented) of the wire while the second (the top) end is mounted to the positioning system. It is important that the wire diameter be smaller than D , but the ratio is not critical. Reflections from the holder are positioned outside the ROI; therefore they do not affect the measurement results.

The holder construction should allow easy replacement (exchange) of the sphere reflector to allow use of sphere reflectors of various diameters, according to the transducer's nominal frequency and the fundamental wavelength of ultrasound in the working liquid ($\lambda < D < 4\lambda$).

D.5.4.4 Positioning system

The positioning system (micro-manipulator) is dedicated to placing the sphere within the scanned area in the working liquid. The required system **accuracy** and stiffness is determined by the measuring ultrasound frequency and transducer-scanning parameters.

D.5.5 Signal acquisition

It is important that live, full-resolution video be captured from the sonograph for processing the measurements.

Two basic types of signals are available from the sonograph output. The old and low-end types are equipped with analog composite VHS video output in PAL standard only. The newer and higher technological level systems do not use the obsolete analog video but use digital video

DVI-D or HDMI outputs. Both types of video signal need special hardware (video-grabber) to be acquired by the computer for further analysis. Most higher end systems have direct digitally detected- and/or IQ-data output.

The following parameters of the acquired video signal are essential for recording and analysis:

- the ultrasound scanner frame rate;
- dynamic range and amplitude resolution (quantization-step size);
- spatial resolution of the sonograph, related to number of lines for analog video signals and/or size of **pixel** for digital video signals.

The digital resolution parameters are not too critical for the resolution of the measurement results, because the analysing system may interpolate between adjacent, discrete values.

The **pixel** value corresponds to echo amplitude received. The amplitude dynamic range of the analysed input signal is derived from the video monitor's grey-level range, which is 256 grey levels at maximum. Therefore, 8-bit (256 grey levels) digitalization is a minimum to fulfil the needs of inputted analog-signal conversion.

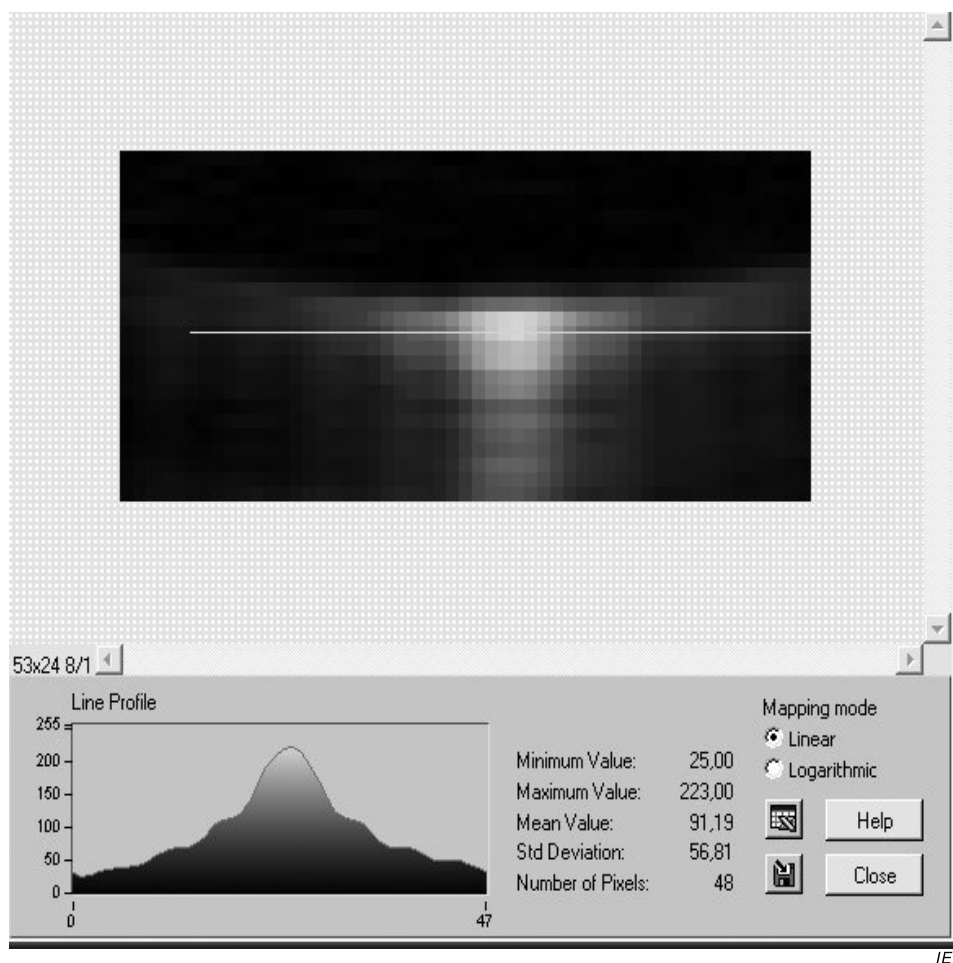
D.5.6 Signal analysis

The system grabs a continuously recorded video signal from the sonograph. Prior to running the acquisition, a proper region of interest (ROI) is chosen in the sonogram. The software controls ROI position according to the target placement in the scanning plane to keep the target inside the ROI. Only those ROI frames are used for further analysis, which coincide with the target position in the pre-programmed measuring grid at points (x,z) within the measured volume of the scan. Each of these ROIs is analysed to find the **pixel** containing $a_{r,\max}(x,z)$ the highest value of MER when the target moves in an elevation (y) direction. Not whole ROIs but only their highest **pixel** value $a_{r,\max}(x,y,z)$ is stored for each y , including the corresponding target position (x,y,z) for each elevation path. Only one ROI containing the highest of the $a_{r,\max}(x,y,z)$ from each elevation set (x,z) is stored as a picture $I(x,y,z)$. The stored $I(x,y,z)$ is a picture of the target on the central (axis) of the ultrasound beam and therefore may be analysed to find the **PSF** in both the azimuth and the axial directions. See Figure D.3 and Figure D.4.

The input video-signal level and the transfer characteristic of the video-grabber may affect the measurement results when the input data are obtained from a composite video signal. Thus, it is important that the video-grabber amplitude-transfer characteristic be well adjusted to assure use of the whole range of the AD-converter to prevent artificial limitation of the high level signal.

The digital video-grabber transfer function does not affect the measurement results when digital video-signal input is used.

An example of the **PSF** derived from the data recorded in one point of the **measuring-grid** is presented in Figure D.3.



NOTE The white line is the line of **PSF**-acquisition from the digital picture of the target at position (x,y,z) . The line passes through the pixel containing the maximum of the signal $a_{r,max}(x,y,z)$.

Figure D.3 – A pixel maximum level and PSF-trace estimation in ROI stored digital data

D.6 Results – measured parameters

D.6.1 Spatial resolutions in PSF-mapping

D.6.1.1 General

The **PSF**-mapping method demands modification of the spatial-resolution evaluation method defined earlier (See 3.46, 3.50 and 3.53). The assessment of target displacement complicates the mapping process; it has been changed to simply deriving the spatial resolution value in **PSF**-mapping from a single measurement in the **measuring-grid** point. The values of mapped parameters may differ slightly from the ones defined earlier (3.46, 3.50 and 3.53), but they are measured with higher **accuracy** and repeatability when using the mapping system [44].

The spatial resolution is defined as a distance. The lateral and axial resolutions are enumerated from FWHM- and/or HWHM-parameters, respectively. These parameters are derived from the **PSF** in number of **pixels** and recalculation using the **pixel** size in a scanned area is needed. The **elevational resolution** is derived directly from positions of the reflector generating half-maximum reflected signal during vertical displacement of the reflector.

The lateral- and axial-resolution conversion to units of distance should be performed by evaluating software using the following relations:

$$R_{L,PSF} = W_{F,HM,n} \times p_x \quad (D.1)$$

$$R_{A,PSF} = 2 \times W_{H,HM,n} \times p_z \quad (D.2)$$

where

$W_{F,HM,n}$ and $W_{H,HM,n}$ are normalized values according to Formula (D.3) in D.6.1.2;

$R_{L,PSF}$ is **lateral resolution in a PSF map**;

$R_{A,PSF}$ is **axial resolution in a PSF map**;

p_x is lateral size of one **pixel** in scanned area;

p_z is axial size of one **pixel** in the scanned area.

D.6.1.2 Lateral resolution in PSF-mapping – $R_{L,PSF}$

Resolution in the lateral direction can be approximated from the FWHM of the **PSF** as the $W_{F,HM}(x,z)$ which is a distance between **pixel** numbers where there is a dip of –6 dB in **pixel** values relative to the maximum value. Because the $W_{F,HM}$ value depends upon the value of the maximum amplitude of the **PSF**-profile, normalization according to the maximum value is necessary. A linear extrapolation is used by application of the formula

$$W_{F,HM,n}(x,z) = W_{F,HM}(x,z) \times \frac{M}{a_{r,max}(x,z) + 1} \quad (D.3)$$

where

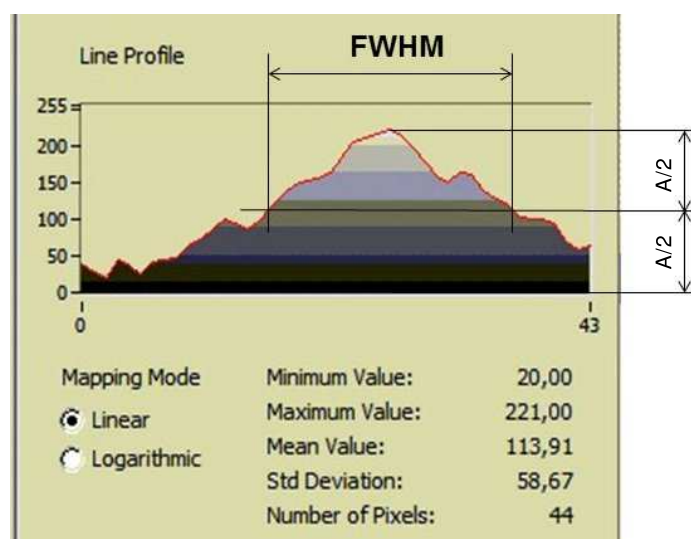
$W_{F,HM,n}(x,z)$ is the normalized value of FWHM at point (x,z) of the **measuring grid**;

$W_{F,HM}(x,z)$ is the measured value of FWHM at point (x,z) of the **measuring grid**;

x,z are coordinates of a point in the **measuring grid**;

$a_{r,max}(x,z)$ is the MER **pixel** value measured for the reflector at the point (x,z) ;

M is number of quantization levels defined by $M = 2^m$ where m is number of **pixel** bits.

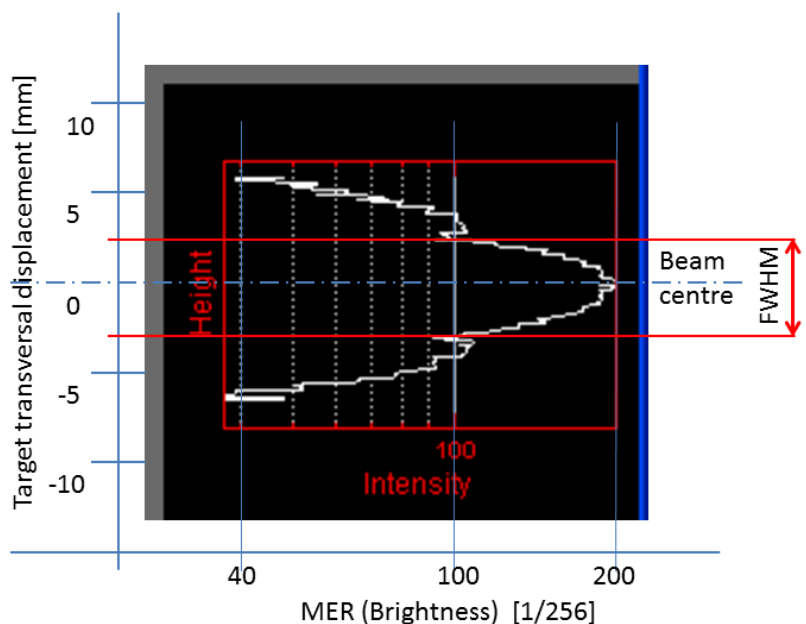


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Figure D.4 – The derivation of FWHM-value from the lateral-amplitude profile of PSF received-echo signal ($a_{r,max}(x_i, z_k) = 221$) with reflector positioned on axis of the ultrasound beam in point (x_i, z_k) of the measuring grid

D.6.1.3 Elevational resolution in PSF-mapping – $R_{E,PSF}$

Resolution in elevational directions can be approximated as the FWHM of a function MER versus point-reflector vertical position at the measured points, where there is a dip of –6 dB below the maximum value of MER; see Figure D.5 and Figure D.11. Because the $W_{F,HM}$ depends upon the value of MER, normalization according to the maximum value is necessary. The normalization procedure applies (see Formula (D.3)).



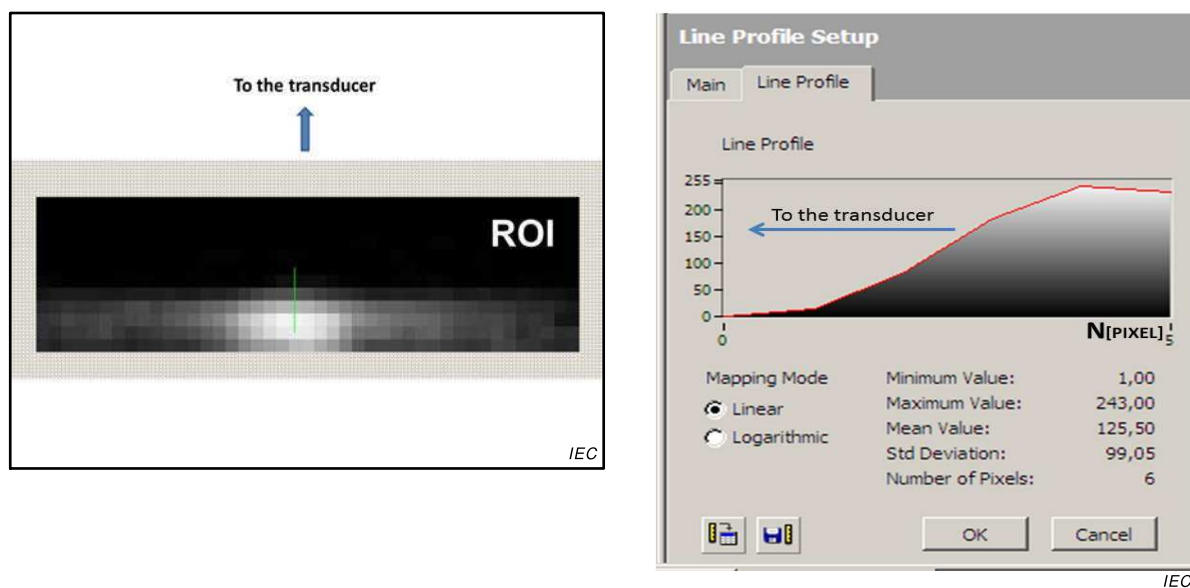
IEC

Figure D.5 – The FWHM-derivation from the elevational (transversal) profile of MER in one point of the measuring grid

The $R_{E,PSF}$ distribution over the scan plane specifies its thickness according to depth of scan and therefore elevational focus and focal-area position. A position of $R_{E,PSF}$ minimum localizes the elevational focal area.

D.6.1.4 Axial resolution in PSF-mapping – $R_{A,PSF}$

Because the echo generated at the front surface of a ball reflector is evaluated for axial direction only (Figure D.6), the **PSF**-mapping process needs a different method to access axial resolution as compared to the $R_{L,PSF}$ and the $R_{E,PSF}$. The Half-Width-at-Half-Maximum (HWHM) is used instead of the FWHM; see Figure D.7. The FWHM may be simply derived as two times HWHM to obtain the $R_{A,PSF}$ (see Formula (D.2)).



The pixel containing MER is crossed by the green line in the axial direction. The line determines a set of pixels to be evaluated.

- a) The ROI containing the target sonogram b) The axial profile line of the evaluated pixels, and basic statistical data

Figure D.6 – The derivation of the reflected-signal axial-profile from the ROI

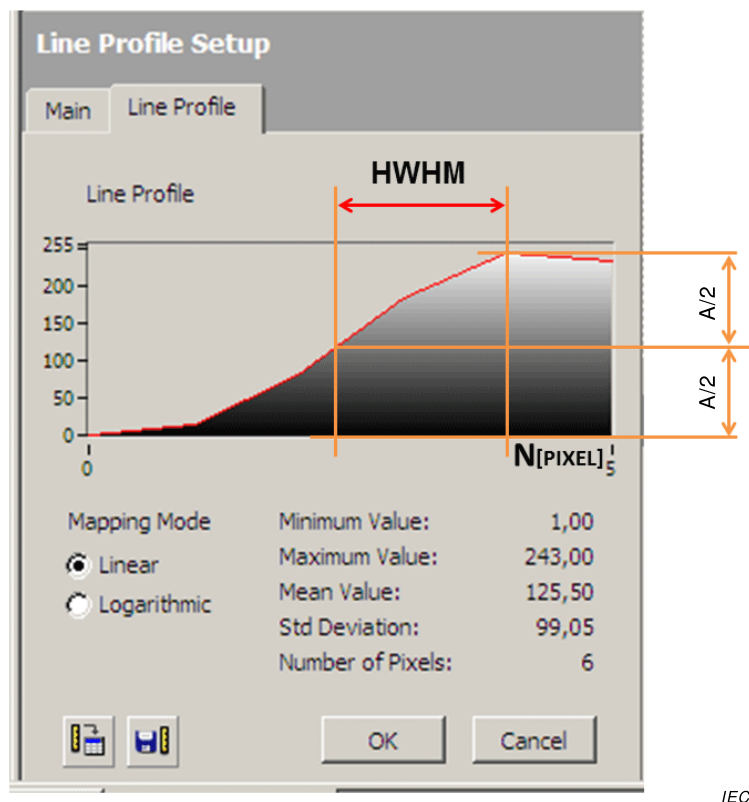
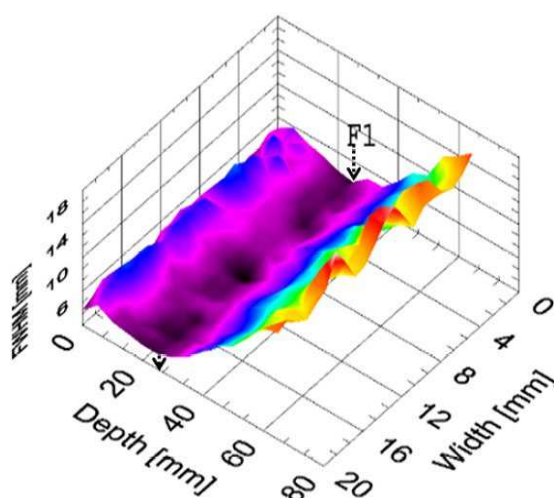


Figure D.7 – The enumeration of the $W_{H, HM}$ parameter from the axial-profile line: $a_{r, \max}(x, z) = 243$

Because the $W_{H, HM}$ value depends upon the value of the maximum of MER, normalization according to the maximum value is necessary. The normalization procedure applies (see Formula (D.3)).

D.6.2 Focal areas in both the azimuth and the elevation directions

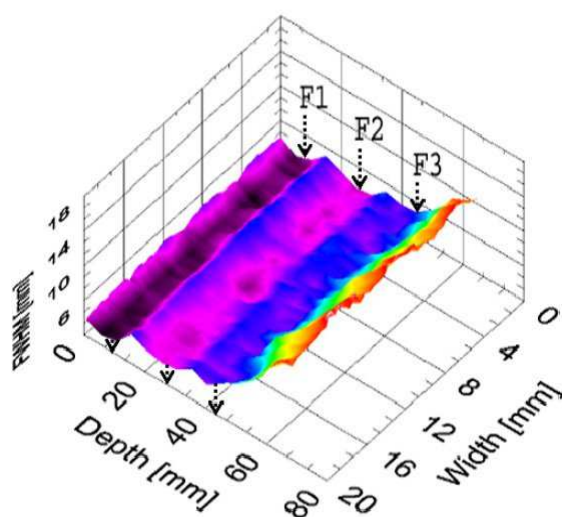
The focal-area position is characterized by local extremes of spatial resolution and sensitivity level. Therefore the focus position is derivable from both the FWHM of **PSF** and the echo-amplitude distribution over the corresponding direction. The focal area has lowest value of the $W_{F, HM}$ (the best spatial resolution) and highest value of the received echo amplitude. In the case of dynamic focusing, not only the position but the number of foci is available from the characteristic. See Figure D.8 and Figure D.9.



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NOTE The focal point F1 (local minimum of $W_{F, HM}$) is at depth approximately 25 mm in the measured area.

Figure D.8 – The distribution of FWHM over a scan area of width 20 mm to depth 80 mm, made with a monofocal scan using a linear 5 MHz transducer



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NOTE The focal-point positions (local minima of the $W_{F, HM}$ values) are observable at depths approximately 10 mm, 27 mm and 42 mm in the measured area.

Figure D.9 – The distribution of FWHM over a scan area of width 20 mm to depth 80 mm, made with the same system as for Figure D.7 but using three focal points (F1, F2 and F3) for the scan

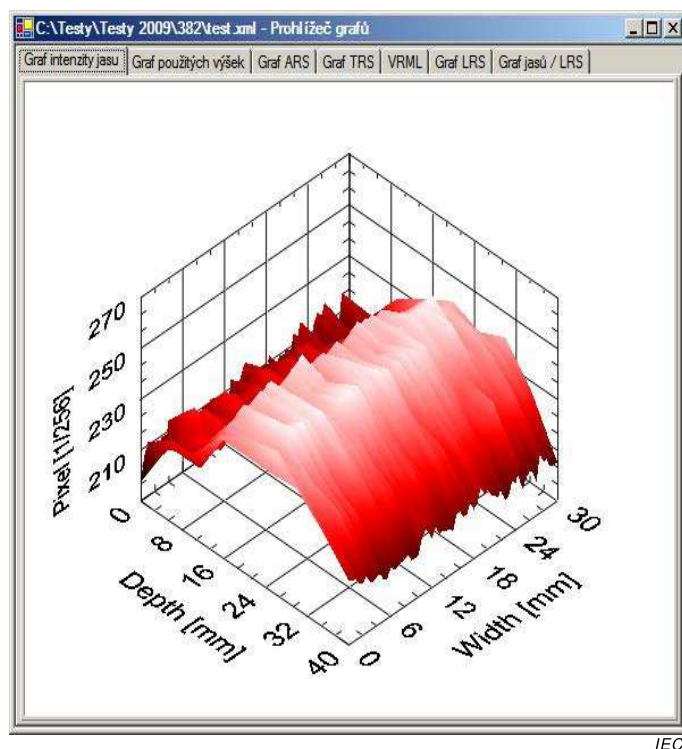
D.6.3 Amplification uniformity in both the axial and the azimuthal directions

The 2D-sonogram is composed of individual **scan lines** that fill the scanning plane in the azimuthal direction. (The 3D-sonogram is created from separate 2D-scan planes.) The MER amplitude varies in the scan plane due to **TGC**, density of the **scan lines**, localization and number of foci, non-uniformity of channel amplifiers and different sensitivity of transducer elements. Some of these effects are desirable, e.g. **TGC**, focusing; some spoil the picture, e.g. defective amplifiers or elements.

All the information can be obtained by analysing the distribution of MER over the scanned area. See Figure D.10.

D.6.4 Preloaded TGC-function

When the **TGC** manual control is adjusted to be depth independent, the manufacturer's preloaded **TGC**-function may be derived from the MER-level distribution along the axial profile. See Figure D.10. The preloaded **TGC**-curve is obviously pre-programmed in the sonograph software and is transducer type-dependent. An aim of this function is to reduce sensitivity for the image area close to the transducer or according to the individual element position in the array of elements in the transducer (e.g. phased array transducers). The echo-amplitude distribution over the scan area contains information about the gain distribution, because the scanned volume in the measuring bath is filled in by degassed working liquid with negligible ultrasound field absorption.



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NOTE A linear transducer with nominal frequency 10 MHz, scan width 38 mm, using one focus active in 20 mm depth of the measured area was used.

**Figure D.10 – Plot of the distribution of MER-intensity over the scanning area
30 mm wide and 40 mm deep**

D.6.5 Visualization of ultrasound scanning lines

When the elevation step of the ball-target placement is small enough, a fluctuation of MER-amplitude is observable along the scanning lines in the elevation direction. The maximum intensity is received when the reflector stays on the **ultrasound beam** axis. The same effect is observable on the plot of the FWHM-distribution, only the reflector position on the beam axis

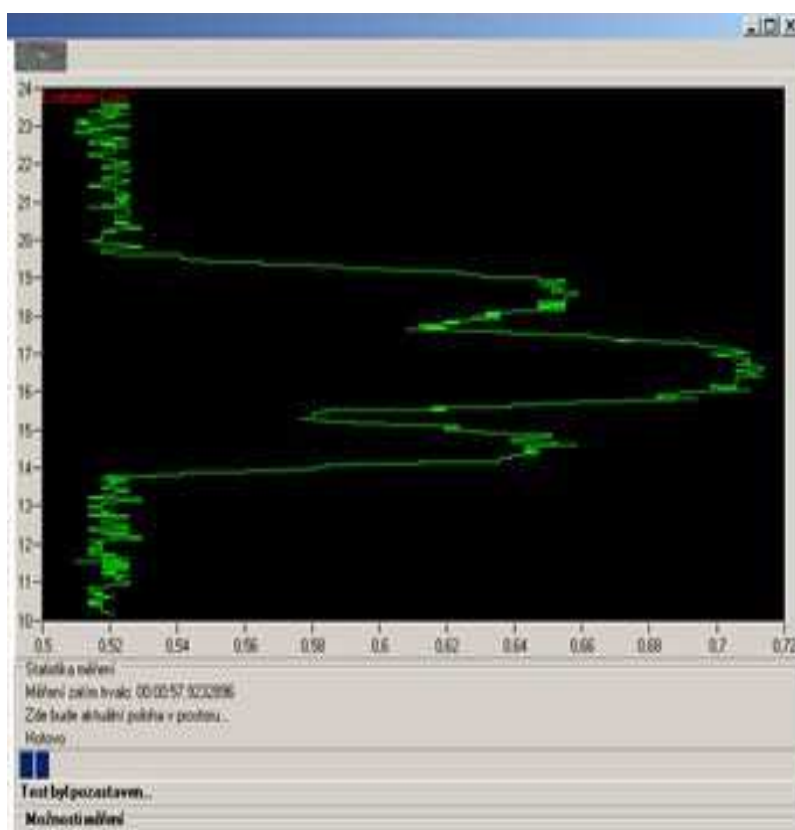
generates drop-down of the parameter value. This method may be used to visualize separate **scan lines**, their density and changes of profile. See Figure D.10.

D.6.6 Elevation width of a scanning plane

This is assessable from the elevational profile of MER-amplitude for the spherical target passing perpendicularly to the scanning plane in each point of the **measuring grid**. The data generates the FWHM of **PSF**-distribution in the azimuthal direction over the scan area.

D.6.7 Side-lobes level in elevational plane

The MER-profile recorded in the elevational direction can display level and beam direction of **side lobes**. See Figure D.11. The side-lobes level is an important parameter of sonograph-imaging quality. Side-lobes generate false reflections creating ghost structures in the sonogram.



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NOTE The abscissa scale is MER $a_{r,max}(x_i, y, z_k)$ enumerated as a ratio relative to value 256 (M for 8-bit pixel); the ordinate scale is elevational displacement y of the reflector in millimetres.

Figure D.11 – The elevational profile recorded from $a_{r,max}(x_i, y, z_k)$ values for a spherical target passing perpendicularly to the scanning plane in one point (x_i, z_k) of the measuring grid

In Figure D.11 the main peak value corresponds with target position in the centre of the beam. The two minor peaks are generated by side-lobes in the azimuthal plane. The relative value of MER is on the abscissa; the ordinate gives the target displacement in millimetres.

D.6.8 Accuracy of sonogram geometry and caliper measurement

The system moves the reflector to specified positions with defined **accuracy**. The reflecting point gives point reflections in the sonogram at geometric positions specified with **accuracy** defined by **accuracy** of mechanical positioning system, speed of sound in the water, the

precision of reflection-delay measurement and size of **pixel**. These data may be used to check the **accuracy** of the displayed sonogram geometry, including the dynamic process needed for TM-mode examination.

The method is suitable for an imaging-geometry test of the reflected-signal imaging modes like A-mode, B-mode 2D and 3D, TM, US-elastography. The detailed instructions for the evaluation process are available in 6.4 and Clause 7. The referenced fixed test-objects with reflecting structures may be substituted by a spherical reflector positioned at an exactly defined position. The spherical reflector is approach-angle independent. A partial geometric parameter will be evaluated using suitable software.

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FINAL VERSION



Ultrasonics – Pulse-echo scanners – Part 1: Techniques for calibrating spatial measurement systems and measurement of system point-spread function response

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INTERNATIONAL ELECTROTECHNICAL COMMISSION

ULTRASONICS – PULSE-ECHO SCANNERS –**Part 1: Techniques for calibrating spatial measurement systems
and measurement of system point-spread function response**

FOREWORD

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This Consolidated version of IEC 61391-1 bears the edition number 1.1. It consists of the first edition (2006-07) [documents 87/336/FDIS and 87/343/RVD] and its amendment 1 (2017-07) [documents 87/650/FDIS and 87/653/RVD]. The technical content is identical to the base edition and its amendment.

This Final version does not show where the technical content is modified by amendment 1. A separate Redline version with all changes highlighted is available in this publication.

International Standard IEC 61391-1 has been prepared by IEC technical committee 87: Ultrasonics.

This publication has been drafted in accordance with the ISO/IEC Directives, Part 2.

Terms in **bold** in the text are defined in clause 3.

This standard is intended to be published in two or more parts:

- Part 1 deals with techniques for calibrating spatial measurement systems and measurement of system point-spread function response;
- Part 2 will deal with measurement of system sensitivity, dynamic range, and low-contrast resolution.

The committee has decided that the contents of the base publication and its amendment will remain unchanged until the stability date indicated on the IEC web site under "<http://webstore.iec.ch>" in the data related to the specific publication. At this date, the publication will be

- reconfirmed,
- withdrawn,
- replaced by a revised edition, or
- amended.

A bilingual version of this publication may be issued at a later date.

IMPORTANT – The 'colour inside' logo on the cover page of this publication indicates that it contains colours which are considered to be useful for the correct understanding of its contents. Users should therefore print this document using a colour printer.

INTRODUCTION

An ultrasonic pulse-echo scanner produces images of tissue in an ultrasonic **scan plane** by sweeping a narrow pulsed beam of **ultrasound** through the section of interest and detecting the echoes generated at tissue boundaries. A variety of **ultrasonic transducer** types are employed to operate in a transmit/receive mode for the ultrasonic signals. Ultrasonic scanners are widely used in medical practice to produce images of many soft-tissue organs throughout the human body.

This standard describes test procedures that should be widely acceptable and valid for a wide range of types of equipment. Manufacturers should use the standard to prepare their specifications; the users should employ the standard to check specifications. The measurements can be carried out without interfering with the normal working conditions of the machine. Typical **test objects** are described in the annexes. The structures of the **test objects** have not been specified in detail, rather suitable types of overall and internal structures are described. The specific structure of a **test object** should be reported with the results obtained using it. Similar commercial versions of these **test objects** are available.

The performance parameters specified and the corresponding methods of measurement have been chosen to provide a basis for comparison with the manufacturer's specification and between similar types of apparatus of different makes, intended for the same kind of diagnostic application. The manufacturer's specification should allow comparison with the results obtained from the tests in this standard. Furthermore, it is intended that the sets of results and values obtained from the use of the recommended methods will provide useful criteria for predicting the performance of equipment in appropriate diagnostic applications. This standard concentrates on measurements of images by digital techniques. Methods suitable for inspection by eye are covered here as well. Discussion of other visual techniques can be found in IEC 61390 [1] ¹⁾.

Where a diagnostic system accommodates more than one option in respect of a particular system component, for example the **ultrasonic transducer**, it is intended that each option be regarded as a separate system. However, it is considered that the performance of a machine is adequately specified, if measurements are undertaken for the most significant combinations of machine control settings and accessories. Further evaluation of equipment is obviously possible but this should be considered as a special case rather than a routine requirement.

1) Figures in square brackets refer to the Bibliography.

ULTRASONICS – PULSE-ECHO SCANNERS –

Part 1: Techniques for calibrating spatial measurement systems and measurement of system point-spread function response

1 Scope

This International Standard describes methods of calibrating the spatial measurement facilities and **point-spread function** of ultrasonic imaging equipment in the ultrasonic frequency range 0,5 MHz to 15 MHz. This standard is relevant for ultrasonic scanners based on the pulse-echo principle of the types listed below:

- mechanical sector scanners;
- electronic phased-array sector scanners;
- electronic linear-array scanners;
- electronic curved-array sector scanners;
- water-bath scanners based on any of the above four scanning mechanisms;
- 3D-volume reconstruction systems.

2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60050-801:1994, *International Electrotechnical Vocabulary – Chapter 801: Acoustics and electroacoustics*

IEC 60050-802:2011, *International Electrotechnical Vocabulary – Part 802: Ultrasonics*

IEC 61685:2001, *Ultrasonics – Flow measurement systems – Flow test object*

IEC 62127-1:2007, *Ultrasonics – Hydrophones – Part 1: Measurement and characterization of medical ultrasonic fields up to 40 MHz*

3 Terms and definitions

For the purposes of this document, the terms and definitions given in IEC 60050-801:1994, IEC 60050-802:2011, IEC 62127-1:2007 and the following apply. See also related International Standards, Technical Specifications and Technical Reports for definitions and explanations [1] [2] [3] [4] [34] [35] [36] [37] [38] [39].

3.1

A-scan

class of data acquisition geometry in one dimension, in which echo strength information is acquired from points lying along a single **beam axis** and displayed as amplitude versus time of flight or distance

3.2

acoustic coupling agent (also, coupling agent)

a material, usually a gel or other fluid, that is used to ensure acoustic contact between the transducer and the patient's skin, or between the transducer and the surface of a sealed test object

3.3

acoustic working frequency

arithmetic mean of the frequencies f_1 and f_2 at which the amplitude of the acoustic pressure spectrum is 3 dB below the peak amplitude

(See 3.4.2 of IEC 61102)

3.4

automatic time-gain compensation

ATGC

automatic working time gain control based on the observed decrease in echo amplitudes due to the attenuation in ultrasonic pulse amplitude with depth

3.5

axial resolution

minimum separation along the beam axis of two equally scattering volumes or **targets** at a specified depth for which two distinct echo signals can be displayed

3.6

backscatter coefficient

mean acoustic power scattered in the 180° direction by a specified object with respect to the direction of the incident beam, per unit solid angle per unit volume, divided by the incident beam intensity. For a volume filled with many scatterers, the scatterers are considered to be randomly distributed. The mean power is obtained from different spatial realisations of the scattering volume

NOTE **Backscatter coefficient** is commonly referred to as the differential scattering cross-section per unit volume in the 180° direction

3.7

backscatter contrast (normalized)

difference between the **backscatter coefficients** from two defined regions divided by the square root of the product of the two **backscatter coefficients**

3.8

beam axis

the longitudinal axis of the pulse-echo response pattern of a given **B-mode scan line**, a pulse-echo equivalent to the transmitted beam axis of IEC 61828 [2]

3.9

B-scan

class of data acquisition geometry in which echo information is acquired from points lying in an ultrasonic **scan plane** containing interrogating ultrasonic beams. See **B-mode** below.

NOTE B-scan is a colloquial term for **B-mode** scan or image. (See 3.10)

3.10

Brightness-modulated display

B-mode

method of presentation of **B-scan** information in which a particular section through an imaged object is represented in a conformal way by the scan plane of the display and echo amplitude is represented by local brightness or optical density of the display

[IEC 60854: definition 3.18, modified]

3.11**displayed dynamic range**

ratio, expressed in decibels, of the amplitude of the maximum echo that does not saturate the display to the minimum echo that can be distinguished in the display under the scanner test settings

3.12**elevational resolution**

minimum separation perpendicular to the ultrasonic **scan plane** of two equally scattering **targets** at a specified depth for which two distinct echo signals can be displayed. Often used here informally for slice thickness for purposes of 3D-scanning

3.13**field-of-view**

area in the ultrasonic **scan plane** which is insonated by the **ultrasound beam** during the acquisition of echo data to produce one image frame

3.14**frame rate**

number of sweeps comprising the full-frame refresh rate that the ultrasonic beam makes per second through the **field-of-view**

3.15**gain**

ratio of the output to the input of a system, generally an amplifying system, usually expressed in decibels

3.16**grey scale**

range of values of image brightness, being either continuous between two extreme values or, if discontinuous, including at least three discrete values

[IEC 60854: definition 3.14]

3.17**lateral resolution**

minimum separation of two **line targets** at a specified depth in a **test object** made of **tissue-mimicking material** for which two distinct echo signals can be displayed. The **line targets** should be perpendicular to the scanned plane; the separation between the **targets** should be perpendicular to the beam-alignment axis

3.18**line-spread function****LSF**

characteristic response in three dimensions of an imaging system to a high-contrast line target

3.19**line target**

cylindrical reflector whose diameter is so small that the reflector cannot be distinguished by the imaging system from a cylindrical reflector with diameter an order of magnitude smaller, except by signal amplitude. The backscatter from a standard **line target** should be a simple function of frequency over the range of frequencies studied

3.20

M-mode time-motion mode

method of presentation of **M-scan** information in which the motion of structures along a fixed **beam axis** is depicted by presenting their positions on a line which moves across a display to show the variation with time of the echo

3.21

M-scan time-motion scan

class of acquisition geometry in which echo information from moving structures is acquired from points lying along a single beam axis. The echo strength information is presented using an **M-mode** display

3.22

nominal frequency (of a transducer)

intended **acoustic working frequency** of a transducer as quoted by the designer or manufacturer

[adapted from definition 3.7 of IEC 60854]

3.23

pixel

picture element

smallest spatial unit or cell size of a digitized 2-dimensional array representation of an image. Each **pixel** has an address (x-and y-coordinates corresponding to its position in the array) and a specific brightness level

NOTE **Pixel** is a contraction of 'picture element'.

3.24

point target

reflector whose scattering surface dimensions are so small that it cannot be distinguished (except by signal amplitude) by the imaging system from a similar **target** whose scattering surface is an order of magnitude smaller. The backscatter cross section of a standard point **target** should be a simple function of frequency over the range of frequencies studied.

3.25

point-spread function PSF

characteristic response in three dimensions of an imaging system to a high-contrast **point target**.

NOTE For most ultrasound systems, an individual ultrasound **PSF** cannot be used as the overall system impulse response, due to changes in the **PSF** with depth, with other positions in the region of use and with system focal and frequency settings. The problem is solved by PSF mapping – see Annex D.

3.26

scan line

one of the component lines which form a **B-mode** image on an ultrasound monitor. Each line is the envelope-detected **A-scan** line in which the echo amplitudes are converted to brightness values

3.27

scan plane

a plane containing the **ultrasonic scan lines**

[IEC 61102: definition 3.38, modified]

3.28

side lobe

secondary beam, generated by an **ultrasonic transducer**, that deviates from the direction of the main beam. Usually, the intensity of the **side lobes** is significantly less than that of the central axis beam

NOTE The presence of **side lobes** may be responsible for introducing artifactual echoes into the **ultrasound** image.

3.29

slice thickness

thickness, perpendicular to the **ultrasonic scan plane** and at a stated depth in the **test object**, of that region of the **test object** from which acoustic information is displayed

3.30

speckle pattern

image pattern or texture, produced by the interference of echoes from the scattering centres in tissue or **tissue-mimicking material**

3.31

spot size

the –6 dB width or otherwise specified width of the **PSF** or **LSF**

3.32

target

an object to be interrogated by an **ultrasound** beam

NOTE Examples of **targets** are:

- a) a device specifically designed to be inserted into the ultrasonic field to serve as the object on which the radiation force is to be measured;
- b) a scatterer or ensemble of scatterers giving rise to a signal within the effective ultrasonic beam;
- c) a wire or a filament in a **test object**.

3.33 test object

device containing one or more groups of object configurations embedded in a **tissue-mimicking material** or another medium

3.34 test object scanning surface

surface on the tissue-mimicking **test object** recommended for transducer location during a test procedure

3.35

time-gain compensation

TGC

change in amplifier gain with time, introduced to compensate for loss in echo amplitude with increasing depth due to attenuation in tissue

3.36

tissue-mimicking material

material in which the propagation velocity (speed of sound), reflecting, scattering and attenuating properties are similar to those of soft tissue for **ultrasound** in the frequency range 0,5 MHz to 15 MHz.

[See 6.4 and Annex D of IEC 61685]

3.37

transmitted ultrasound field

three-dimensional distribution of ultrasound energy emanating from the ultrasonic transducer

3.38

ultrasonic scan line

for automatic scanning systems, the beam-alignment axis either for a particular **ultrasonic transducer** element or for a single or multiple excitation of an **ultrasonic transducer** or of an **ultrasonic transducer element group**

[IEC 61157: definition 3.27, modified]

3.39

ultrasonic transducer

device capable of converting electrical energy to mechanical energy within the ultrasonic frequency range and/or reciprocally capable of converting mechanical energy to electrical energy

[IEC 61102: definition 3.58]

NOTE For the purposes of this standard, **ultrasonic transducer** is taken to refer to a complete assembly that includes the transducer element or elements and mechanical and electrical damping and matching provisions.

3.40

ultrasonic transducer element group

group of elements of an **ultrasonic transducer** which are excited together in order to produce a single acoustic pulse

[IEC 61102: definition 3.60]

3.41

ultrasound

acoustic oscillation whose frequency is above the high-frequency limit of audible sound (conventionally 20 kHz)

[IEV 801 21-04, modified]

3.42

ultrasound beam (pulse-echo response pattern)

region adjacent to the transducer face from which an echo signal from a specified **target** may be detected for the test settings of the scanner and with the scanner operating in a non-scanning mode. This term should be distinguished from the **transmitted ultrasound field**

3.43

voxel

smallest spatial unit or cell size of a digitized 3-dimensional array representation of an image. Each **voxel** has an address (x, y, and z-coordinates) corresponding to its position in the array, and a specific brightness and/or color value

3.44

working liquid

a mixture of water and other solvent that adjusts the speed of sound to 1 540 m/s

[See also 6.4 and Annex D of IEC 61685:2001]

3.45

accuracy

closeness of agreement between a test result and the accepted reference value

[SOURCE: ISO 5725-1:1994, 3.6]

3.46

axial resolution in a PSF-map

twice the Half-Width-at-Half-Maximum (HWHM) of a function's trace created from a set of increasing **pixel** values, commencing near zero and terminating at the first maximum value

(centre of the **PSF**) and representing the leading edge of the echo signal from a point reflector located on the main beam axis

Note 1 to entry: The **axial resolution in a PSF map** differs from the **axial resolution** specified by 3.5. It is used for the **PSF**-mapping only to simplify the data acquisition.

Note 2 to entry: A detailed explanation of the **axial resolution in the PSF-map** measuring method is in D.6.1.4.

Note 3 to entry: The axial resolution mainly depends on the ultrasound frequency used, not on sonograph construction.

Note 4 to entry: **Axial resolution in a PSF-map** is expressed in metres.

3.47

brightness

luminance as perceived by the human visual system

[SOURCE: IEC 62563-1:2009, 3.1.2]

3.48

contrast

C

ratio of the difference of the luminance of two image areas, $L_1 - L_2$, divided by the average of the two luminance values:

$$C = 2 (L_1 - L_2) / (L_1 + L_2)$$

[SOURCE: IEC 62563-1:2009, 3.1.6]

3.49

dynamic imaging

real-time imaging

imaging with a frame rate that is high enough to observe moving structures in apparently continuous motion

3.50

elevational resolution in a PSF-map

difference of point-reflector displacements in passing through the scanning plane in an elevational direction, which result in decreases of MER of –6 dB compared to the MER-value in the beam centre

Note 1 to entry: The **elevational resolution in a PSF-map** differs from the **elevational resolution** specified by 3.12. It is used for the **PSF**-mapping only to simplify the data acquisition.

Note 2 to entry: Detailed explanation of the method is in D.6.1.3.

Note 3 to entry: **Elevational resolution in a PSF-map** is expressed in metres.

3.51

overall gain

G_0

basic level of gain that is uniform for the whole scan area but modified by **TGC** relative to the depth of the scan

3.52

profile line

set of **pixel** values ordered along an abscissa according to the sequence during their acquisition

3.53

lateral resolution in a PSF-map

Full-Width at Half-Maximum (FWHM) of the **PSF**, measured in a lateral direction

Note 1 to entry: The **lateral resolution in a PSF-map** differs from the **lateral resolution** specified by 3.17. It is used for the **PSF**-mapping only to simplify the data acquisition.

Note 2 to entry: Detailed explanation of the method is in D.6.1.2.

Note 3 to entry: **Lateral resolution in a PSF-map** is expressed in metres.

3.54

measuring grid

matrix of points specified by Cartesian coordinates x_i and z_j defined in a plane parallel to the scanning plane

Note 1 to entry: Each point determines the position (x_i, z_j) in which individual measurement of **PSF** is performed.

Note 2 to entry: The step Δx is defined as an increment $x_{i+1} - x_i$ in the lateral direction. The step Δz is defined as an increment $z_{j+1} - z_j$ in the axial direction.

3.55

performance evaluation

tests performed to assess specific absolute performance of the object tested

Note 1 to entry: Typical times for ultrasound-system **performance evaluation** are at pre-purchase evaluation, new- and repaired-system acceptance testing, at time of performance difficulties, and at end-of-useful-life evaluations.

[SOURCE: IEC TS 62736:2016, 3.5]

3.56

precision

closeness of agreement between independent test results obtained under stipulated conditions

[SOURCE: ISO 5725-1:1994, 3.12]

3.57

scanning window

area on the surface of the **test tank** dedicated for transducer application to obtain a suitable sonogram of the target

Note 1 to entry: It is important that the **scanning window** be covered by flexible foil made of material with similar acoustic properties to the working liquid to avoid ultrasound field reflections and absorption.

Note 2 to entry: The foil flexibility should assure proper acoustical contact of any type of curved transducer.

Note 3 to entry: It is important that the foil covering the scanning-window be tough enough to prevent its damage during coupling the measured transducer to the **scanning window**, to prevent resultant leakage of working liquid from the measuring tank.

Note 4 to entry: The **scanning window** has the identical function as the **test object scanning surface** in the case of tissue-mimicking test objects (see 3.34).

3.58

side-lobe signal

echo signal generated by ultrasound signal transmitted/received in a direction different from the central axis of the transducer

3.59

test tank

tank designed to be suitable for providing specified kind of tests, which is filled with a **working liquid** and equipped with **scanning window(s)**

4 Symbols

A	surface area
A_c	cross-sectional area
a_i	length of the semi-major axes for a given half ($i = 1$ or 2) of the ellipsoid of an ovoid object
$A_{r,max}$	greatest $a_{r,max}$ evaluated for whole measured volume
$a_{r,max}$	MER pixel value evaluated from ROI
$a_{r,max}(x,y,z)$	MER pixel value evaluated from ROI scanned for reflector in position (x,y,z)
b	mean of the lengths of the minor axes of the ellipsoid of an ovoid object
C	contrast
D	diameter of the reflector sphere
f	acoustic working frequency
G_o	overall gain
$I(x,y,z)$	ROI specified in a digital picture of scan stored with reflector in position (x,y,z)
k	circular wave number; ($= 2\pi / \lambda$ in which λ is the wavelength)
M	number of quantization levels defined by $M = 2^m$ where m is number of pixel bits
P	perimeter of cross-section of ovoid object
p_x	pixel size in lateral (azimuthal) direction
p_z	pixel size in axial direction
R	ratio of mean of measured spacings to known spacings (see 7.3.1)
$R_{A,PSF}$	axial resolution in a PSF-map
$R_{E,PSF}$	elevational resolution in a PSF-map
$R_{L,PSF}$	lateral resolution in a PSF-map
R_x	lateral dimension calibration factor (see 7.4.2); ratio of mean filament spacings to known spacings for the horizontal direction
R_y	ratio of mean filament spacings to known spacings for the vertical direction
r	radius of a wire or filament target
V	volume of an ovoid object
$W_{F,HM}$	value of FWHM (full width at half of maximum)
$W_{H,HM}$	value of HWHM (half width at half of maximum)
$W_{F,HM,n}$	normalized $W_{F,HM}$ according to Formula (D.3) in D.6.1.2
$W_{H,HM,n}$	normalized $W_{H,HM}$ according to Formula (D.3) in D.6.1.2
Z_m	characteristic acoustic impedance of a wire or filament material
Z_w	characteristic acoustic impedance of the surrounding medium (working liquid or tissue-mimicking material)
λ	ultrasound wavelength in the working liquid, calculated from the nominal frequency of the transducer used
ε	$1-(b/(2a))^2$ eccentricity of an ellipsoid or an ovoid object
σ	backscattering cross-section for a point-like target
ATGC	automatic time-gain compensation
FWHM	full width at half of maximum
HFHM	half width at half of maximum

LUT	look-up table
MER	maximum echo received
PSF	point-spread function
RF	radio frequency
ROI	region of interest
TGC	time-gain compensation
US	ultrasound

5 General conditions

The tests should be performed within the following ambient conditions:

- temperature $23\text{ °C} \pm 3\text{ °C}$;
- relative humidity 45 % to 75 %;
- atmospheric pressure 86 kPa to 106 kPa.

This standard permits the use of test objects of various constructions. Therefore it is essential that the following data of the test object be reported. The following standard choices are recommended:

- a) medium: either **working liquid** or **tissue-mimicking material** [6]
- b) use of coupling gel: thin layer or gel with adapted sound velocity
- c) geometry (one of the models given in Annex A, B or C, where needed with a different spacing between **targets**).

For the medium **working liquid**, the following properties are required:

- speed of sound = $(1\,540 \pm 15)\text{ m/s}$;
- low attenuation ($< 0,1\text{ f dB cm}^{-1}\text{ MHz}^{-1}$);
- negligible scattering (see IEC 61685).

For adjusting the speed of sound in working liquid, see [7, 8].

For the medium **tissue-mimicking material** [9], the following properties are required:

- speed of sound = $(1\,540 \pm 15)\text{ m/s}$;
- attenuation $(0,5 \pm 0,05)\text{ f dB cm}^{-1}\text{ MHz}^{-1}$ in the frequency range used in the tests;
- scattering (moderate, no value imposed).

Note: Where an ultrasound system is designed for particular applications where the mean speed of sound is different from 1 540 m/s, a medium with that design speed of sound should be employed and that change reported with the results.

For tissue-mimicking properties, see also 6.4 and Annex D of IEC 61685:2001 .

Tissue-mimicking material is usually protected by a thin cover. Its thickness and acoustic properties (attenuation and sound velocity) should be reported if these influence the measurement.

The transducer is usually coupled to the cover of **tissue-mimicking material** by an **acoustic coupling agent** (ultrasound gel). If the layer is thin (compared to the wavelength) its influence can be ignored. For a thick layer, for example as needed for a curved-array transducer, the sound velocity of the **acoustic coupling agent** shall be equal to $(1\,540 \pm 15)\text{ m/s}$.

Sound velocity of a medium has two different effects: if it is larger then 1 540 m/s, the axial distances in the medium are rendered proportionally shorter and the focus of the transducer

moves away from the transducer. If the sound velocity is lower, the opposite occurs. The effect on the focus becomes more important for transducers with a high numeric aperture. Therefore the use of the correct sound velocity ($1\,540 \pm 15$ m/s, to which ultrasonic systems are standardized) is essential in Clauses 6 and 7, dealing with geometrical distortions. In Clause 8, dealing with the **PSF**, a deviation can be tolerated for not too high numeric apertures.

In describing scanning procedures with “horizontal” and “vertical”, it is assumed that a **test object** is insonated from above, and that the image on the scanner is oriented correspondingly.

6 Techniques for calibrating 2D-measurement systems

6.1 Test methods

To carry out the test procedures, the following items are required:

- a) tissue-mimicking **test objects** containing **targets** at accurately specified positions;
- b) tissue-mimicking **test object** containing a 3D-object of accurately specified dimensions;
- c) a tank equipped with target holder to position the target at accurately specified positions and containing degassed **working liquid**.

The specifications of these devices are given in Annexes A, B, C and D.

6.2 Instruments

6.2.1 General

The equipment specified in this subclause has been selected to permit testing of ultrasonic scanners in clinical usage. The devices described will ensure that the data collection and analysis will be objective and reproducible.

6.2.2 Digitizers

While some spatial measurements can be made with long-existing digital callipers, for more generally applicable, objective, reproducible data, the **ultrasound** images obtained for testing should be digitally encoded. Many modern **ultrasound** imaging devices produce digital images from the scan converter that can be used for these measurements and are most closely representative of the displayed images. Such measurements can be employed well by hospital-based users with some digital measurement expertise. For spatial measurements this procedure is directly applicable. For **PSF** and **LSF** measures it is necessary, however, to have a characteristic curve of linear echo amplitude at the transducer as a function of the digital image values or to create a sparse representation of that curve by use of calibrated reflectors as described in 7.2.1.2 of [19]. In some systems, rf scan-line data are available. Such data are more accurate for precision measurements in which the linear signal amplitude is important. Measurements made with rf data should be clearly indicated as such and the level at which they came from the system documented. For those machines that do not produce digital images, a frame grabber may be used to acquire and digitize **ultrasound** images. This digitizer requires adequate spatial resolution (at least 512×512 **pixels**) and sufficient **grey scale** (at least 256 grey shades). Also, adequate image analysis software should be used to perform the simple measurements described below on the digital **ultrasound** images of the **test objects**. The digitizer shall exhibit a linearity producing spatial uncertainty of <1 % over 75 % of the image dimension measured, signal level (grey scale) linearity <3 % of full range and signal level stability over a year of <5 % of full range.

The digital imaging software should allow the user to be able to place the cursor at any location on the screen and obtain the pixel address (i.e. row, column coordinates). This will allow the user to calibrate the digital image to actual distances recorded in the **ultrasound** images. Once the digitizer is calibrated, digitized ultrasound images can be subjected to more sophisticated

software analysis than that which is possible directly on the ultrasound display. The digital imaging software should allow the reading of the grey value at any pixel address.

To calibrate the digital image pixel distance (i.e. calibrate the digitizer relative to the **ultrasound** imaging system):

- a) Scan an image of a **test object** containing appropriate **working liquid**. Make a note of the magnification level for this image and make all subsequent measurements and comparisons with the same level of magnification.
- b) Measure the known distances between the positions of two wires or filaments at a distance of about 75 % of the screen size with the electronic callipers to confirm that the calliper measured distance corresponds to the actual distance. The measurement should be performed for a pair of wires or filaments connected by a vertical line and for a pair of wires or filaments connected by a horizontal line. In case deviations are found the scanner should be adjusted before proceeding further. If adjustment is not possible the actual distances shall be used in d).
- c) Digitize the scanned image and use the imaging software to measure the distances in pixels between pairs of wires or filaments by obtaining the pixel address of each wire or filament location, and subtracting to obtain the distances in pixels. Repeat for several different locations, checking both vertical and horizontal distances.
- d) Measure the distance in pixels for various positions of wires, using various directions of the connecting line with respect to the vertical. Divide each distance by the actual distance in mm. Average these ratios; this average ratio is the pixels per millimetre calibration for your digitizer. Once this calibration has been performed, this ratio can be used to compute relative distances in all subsequent digitized images for the particular scanner and magnification used.

See [10].

6.2.3 Tissue-mimicking test objects

Tissue-mimicking **test objects** shall contain structures that allow the following types of measurement to be made:

- a) linear;
- b) curvilinear;
- c) circumferential;
- d) area;
- e) volume;
- f) image distortion;
- g) **M-mode** calibration.

Examples of tissue-mimicking **test objects** are given in Annex B.

6.3 Test settings

6.3.1 General

The many combinations of scanner settings and transducers make it impracticable to carry out tests for all of them. Tests are therefore carried out for each **ultrasonic transducer**, with two settings, one which provides a complete image and one which provides the highest resolution of the **test objects**. The focusing of the ultrasonic beam should be extended over as large a range as possible, to achieve the best resolution over all visible **targets**.

The **test object**, containing an array of filaments as in Figure A.1, is used for the procedures described below.

6.3.2 Display settings (focus, brilliance, contrast)

The focus is made sharp and the brilliance and contrast controls are turned to their lowest positions. The brilliance is now increased until the echo-free zone at the side of the image becomes the minimum perceptible shade of grey. The contrast control is then increased to make the image contain the greatest range of grey shades possible. The focus is then checked for sharpness. If it requires further adjustment, the whole procedure is repeated.

6.3.3 Sensitivity settings (frequency, suppression, output power, gain, TGC, ATGC)

- a) The **nominal frequency** of the **ultrasonic transducer** is noted.
- b) If there is a suppression- or reject-control, it is adjusted to allow the smallest possible signals to be displayed.
- c) The output power and **gain** are adjusted to present the images of the **target** filaments as the smallest visible points on the display.
- d) The **time-gain compensation (TGC)** controls are arranged to present the images of the **target** with equal brilliance over the image. For scanning in **working liquid** the TGC slope should be close to zero

6.3.4 Final optimisation

A final optimisation of the image may be carried out by a small change in the suppression level, **gain** or output power.

When **automatic time-gain compensation (ATGC)** is an option in a scanner, tests should be carried out in this mode of operation. The **test object** is imaged with **ATGC** enabled, the image being optimised using any control which still functions manually, e.g. the overall **gain** or output power.

6.3.5 Recording system

The digital acquisition of **ultrasound** images allows for objective measurements and also allows the images to be saved for comparison at a later date. A major advantage of digital recording is that images are not subjected to the degradation that occurs with either photographic or video recording systems.

6.4 Test parameters

6.4.1 General

Techniques are described in this standard for the following types of measurement:

- linear;
- curvilinear;
- circumferential;
- area;
- volume;
- display and recording distortion;
- **M-mode** calibration.

Transmitted intensity should be low enough to avoid pulse distortion due to non-linear propagation (see IEC 61102). A list of all factors influencing the operation of the scanner for example transducer, frequency, sensitivity control settings, focusing, image processing option shall be made. These data are to be recorded in sufficient detail so as to allow the test to be repeated exactly at a later date by another operator and shall accompany the measuring results.

6.4.2 Measurement accuracy (linear, curvilinear, circumferential, area)

To assess the accuracy of the measurement system of a scanner, the wires or filaments in the **test object** shown in Figure A.1 or Figure A.2 are imaged with the sensitivity adjusted to make the displayed echoes as sharp as possible. If the test object is sealed, a coupling agent shall be used. An **ultrasound** image is obtained and digitized of the set of filament **targets** situated at the middle of the typical working range for the **ultrasonic transducer** assembly being used. Other factors that may affect the value of the resolution are also noted, for example the image processing options of the scan converter or focusing. The procedure is repeated for the other **ultrasonic transducers** of the scanner.

Measurements are made in straight lines on the screen of lengths approximately equal to 75 % of the displayed range. Using image analysis software, a linear brightness profile is obtained along each dimension. Distances are measured “from peak to peak” of the wire or filament brightness profiles. (In case the measuring results are noisy, the position of a peak value is replaced by the midpoint between the –3 dB points and that action noted.) These measurements are carried out along at least a vertical and a horizontal line in Figures A.1, and A.2 and, when possible, along near-vertical directions in the **field-of-view**. The average percentage error is tabulated for each length in each direction. The process is repeated for the available display scales.

To evaluate the accuracy of measurements of curved lines and cross-sectional areas, closed figures having an area approximately 0,75 of the **field-of-view** are traced centrally on the display. The circumferences and areas are measured and the percentage errors calculated. The tracing is done point-to-point, so that a polygon-shaped region is traced. The circumference and area of the polygon are measured. Additional measurements are made with two smaller figures (areas 0,1 and 0,25 of the **field-of-view**) located at the top and bottom of the display. This process is repeated for the available scales of the display.

For rigorous testing, sources of variability should be evaluated. This holds true for both short-term and long-term variation in measurement and analysis procedure. Short-term testing (same-day) should be repeated multiple times from setup to final analysis. In manual testing, an operator should repeat the tests over a short period of time and then the tests should be repeated by several operators.

6.4.3 Display and recording of image distortion

Scan the two-dimensional, regular array of filaments of the **test object** shown in Figure A.2, so that their echoes are seen with equal brilliance throughout the **field-of-view**. Select wires or filaments located horizontally and vertically from the centre throughout the **field-of-view**. Measure on the digitized image the distance from the centre of each wire or filament to the centre of a reference wire or filament located at approximately the centre of the **field-of-view**. Calculate and tabulate the percentage errors.

Observe the directly viewed image of the array of filaments to check that any distortion (i.e. failing orthogonality) of dimensions in the display is less than 3 %.

6.4.4 M-mode calibration

6.4.4.1 General

An **M-mode** facility exists on most real-time scanners. A partial assessment of its performance can be carried out using the **test objects** described in Annex A.

6.4.4.2 Spatial measurements (scrolling A-scan line)

Performing an **M-mode** scan with the **ultrasound beam** directed at wires or filaments in a resolution **test object**, as described earlier for the **B-mode**, enables the measurement errors of a system to be ascertained.

Distortions of the display or record are checked and recorded using the array of **target** filaments in the **test object** as is done for a **B-mode** image.

The accuracy of the time-axis calibration of the **M-mode** trace can be checked by injecting bursts of **ultrasound** into the **ultrasonic transducer** using an external pulse generator and transducer at accurately known intervals, for example 1 ms bursts at 200 ms intervals.

Measurement checks should be carried out for the digitized image of the **M-mode** trace on the display screen. The errors in measurement should be less than 3 %.

6.4.4.3 Tissue-thickness M-mode

For tissue-thickness **M-mode**, the system measures changes in relative thickness of moving tissue. Thus, evaluation of the accuracy of these measurements requires a tissue-like phantom that can be compressed and relaxed at pre-determined positions, and the ability to compare the compressed and relaxed thicknesses with the **M-mode** readings. Also, it should be possible to compress and relax the phantom at different rates. A deformable sponge phantom might be useful for this measurement. It is important to have the ability to test the synthetic **M-mode** at various depths, so the phantom should allow for re-positioning the transducer for different **target-to-transducer** distances. Also, the test should be repeated for each of the **M-mode** sweep speeds.

7 Methods for calibrating 3D-measurement systems

7.1 General

Three-dimensional (3D) imaging systems exist which are only used for visualization, while others include measurement capabilities. Since 3D-reconstruction of volumes is achieved in different ways, it is important to examine the volume reconstruction method and its associated problems, and evaluate the accuracy of the reconstructed images. This discussion is limited to measurement of dimensional accuracy of reconstruction. Measurement of 3D-system resolution will be discussed in document IEC 61391-2 dealing with system resolution and sensitivity.

7.2 Types of 3D-reconstruction methods

7.2.1 General

True 3D-imaging requires that the imaging system assemble data in a 3-dimensional **voxel** matrix. This matrix is usually composed of data from a stack of ultrasonic **scan planes** that contain the **target** volume. The 3D-imaging system stores the information as a 3-dimensional matrix. The spatial density of data-matrix points depends on the number of **ultrasonic scan lines** within each ultrasonic **scan plane**, the pulse length, and the number and spacing of ultrasonic **scan planes** that make up the elevational (depth) dimension. The manner in which the volume of interest is scanned is important, since reconstruction accuracy will depend on how well physical distances are preserved. The distance between successive ultrasonic **scan planes** should be constant, but it should be less than the elevational resolution (**slice thickness**) of the **ultrasonic transducer**. Otherwise, interpolation of adjacent ultrasonic **scan planes** will be included in the reconstructed volume. This interpolation can lead to errors in the dimensions of the reconstructed volumes.

Once the 3D-matrix is constructed, data may be obtained along any dimension within that volume. For example, if multiple images of xy-**ultrasonic scan-plane** data are collected to form a 3D-volume, the resulting 3D-matrix can be “sliced” normal to the y-axis in the xz plane, producing C-scan slices from the resulting data. Also, the resulting 3D-reconstruction can be rotated in space, so as to be viewed and measured from angles that were not available in the original ultrasonic **scan planes**.

[See [11, 12] for reviews of 3D-scanning techniques.]

3D-volume-matrix acquisition and reconstruction are performed in two basic ways:

- a) reconstruction by external positioning;
- b) sequential reconstruction.

Each method has its own characteristics, strengths, and problems.

7.2.2 3D-volume reconstruction methods

7.2.2.1 External positioning methods

Reconstruction methods for 3D-volume by external positioning methods use a reference point and a coordinate reference frame, and all dimensions and positions within the 3D-volume matrix are recorded with respect to the reference coordinates. In these types of systems there is usually a scanning framework, into which the scanning volume is inserted, and the transducer is usually motorized and rides on a rail at a constant speed. Other forms of rigid support may be used that restrain the **ultrasonic transducer** to maintain its 3D-coordinates accurately. This type of reconstruction method is the most accurate and reliable, but is subject to some problems due to initial positioning of the **ultrasonic transducer's** support frame, motor speed deviations, or changes in the positioning system during the data-collection phase. A variant of external positioning systems exists, in which the transducer is guided manually and its position and direction are sensed with respect to a reference coordinate system.

[See [13].]

7.2.2.2 Sequential positioning methods

Sequential positioning and reconstruction methods use different techniques but are usually based on attachment of subsequent **scan planes** to the 3D-matrix based on the position of the previous plane. One such method uses the rate of change of image speckle in at least one dimension [14, 15]. This encoding involves several assumptions that are not always valid. One assumption is that the motion is either a purely linear sweep or a purely angular sweep. In some commercial implementations, a simple, uniform, linear sweep speed is assumed. Tests are therefore important to demonstrate the capabilities and limitations of the measurements under laboratory conditions. Problems occur if the transducer is not moved at a uniform speed, or if the transducer angle shifts from its previous orientation. The reconstruction scheme, in most cases, cannot compensate for these shifts in reference plane orientation, and the reconstructed volume will contain inaccuracies.

7.3 Test parameters associated with reconstruction problems

7.3.1 Reconstruction by external positioning

For reconstruction by external positioning-system testing, either water-based or tissue-mimicking-based **test objects** may be used.

Test parameters: From the reconstructed volume, measure the reconstructed lengths in all three Cartesian coordinate directions and compare with the physical object dimensions along the same coordinate directions. Check the **ultrasonic transducer** orientation with respect to frame and point of reference; speed of motor; distance between ultrasonic **scan planes**; and Cartesian dimensions of reconstructed volume. Procedures for measuring the following parameters are described below.

- a) linear dimensions;
- b) areas;
- c) perimeters of areas;
- d) volumes.

7.3.2 Sequential plane reconstruction systems

Systems in which 3D-spatial encoding is based on the image speckle require a **test object** with relatively uniform speckle backscatter and structures to allow tests of the accuracy and precision of the registration. With certain **ultrasound** systems not only should the accuracy of measurements over long distances be tested, but also the uniformity of the distance scale. Many sequential encoding systems require additional tests due to imprecise position encoding. When such imprecision exists, the direction in an image can be distorted by local jumps or retardation of recorded position in the direction of scanning.

Test parameters: from the reconstructed volume, measure the reconstructed lengths in all three Cartesian coordinates and compare with the physical object dimensions along these same coordinates. Procedures for measuring the following parameters are described below.

- a) linear dimensions (axes);
- b) areas;
- c) perimeter of areas;
- d) volumes.

7.3.3 Test instruments (phantoms) for evaluation of 3D-reconstruction accuracy

7.3.3.1 Filament phantom (water-filled)

For the reconstruction method by external positioning, a filament **test object**, filled with **working liquid** as described in Annex A, Figure A.1 or Figure A.2, may be used. Since the system does not depend on speckle correlation to place the ultrasonic **scan planes** into the 3D-matrix, the filament matrix is a useful structure for testing reconstruction accuracy. For each of the rows of filaments, the distance r from the image location of a filament to a reference filament is measured and plotted against the known distance r' in the phantom. The maximum and r.m.s. deviations of the measured filament positions from the linear regression lines of r on r' are measured, as well as the slopes of the fitted lines.

If the volume measurement algorithm of the system under test can work with sharp corners and flat surfaces, the accuracy of measurement of well-defined volumes are best tested with a filament **test object** as in Figure A.1 or A.2.

7.3.3.2 Tissue-mimicking phantom

A second **test object** that can be used to test both types of systems is shown in Annex B. Figures B.1 to B.4 show different views of ovoid-shaped **tissue-mimicking material** structures with minimal specular boundaries, set in a **tissue-mimicking material** matrix. Such a **test object** is volumetric and the **targets** are defined in the image by differences in **backscatter contrast**. Only the **grey scale** texture and average signal level define the borders. This reliance on volumetric backscatter rather than specular reflectors or **point targets** allows evaluation of image formation, display and measurement aspects of the ultrasound system's reconstruction of volumetric **targets** imaged in the body.

7.4 Test methods for measurement of 3D-reconstruction accuracy

7.4.1 General

These methods are for 3D-measurements with conventional 2D-scanners as well as 3D-scanners. For high quality 3D-imaging and testing thereof, the separation of the recorded image planes should be less than the width of the ultrasonic **scan plane** thickness at its focus, ideally, less than one half the elevational (ultrasonic **scan plane**) focal width. If controls for scan-plane separation are provided, such settings should be employed.

7.4.2 Measurement methods and accuracy using the filament test object

Volumetric measurements from two orthogonal 2D-images: 3D-measurements are often made with simple 2D-imaging systems by acquisition of two orthogonal views of a roughly spherical object, measurement of the three major axes of the object, and calculation of the spherical or ellipsoidal volume by appropriate equations therefore or by the ultrasound system. Either volumetric calculation method can be tested on one of the images from the stack of the previous paragraph by measuring the diameters of the assumed sphere passing through four or more filaments in the phantom and then repeating the measurements on the same or adjacent image, assuming that the two were acquired at 90-degree angles to each other. The calculation of the sphere with cross-section equal to the circle measured is then calculated, compared with the volume of the assumed sphere, and the error computed.

Tests for correction of angulation of scan plane in the elevational direction and for verification of the measurement algorithm from a 3D-sweep parallel to filaments: Perform a 3D-scan with the central acquired image planes normal to the filaments and the scan direction parallel to the filaments, i.e., with the transducer in View B of Figure A.1. If the system allows rotation of the transducer in an arc in the elevational direction, perform the scan in that way and, if possible, display the reconstructed volume with all the reconstructed images normal to the filaments. If that reformatting is not an option in a sector scanner, correct the separation of the filaments for the known viewing angle. If not already so done in the same way in 6.4.2, perform the measurements of 6.4.2 on the first, middle and last image of the 3D-set and document their errors and variances. The ratios of mean filament spacings to known spacings for the horizontal and vertical directions are referred to as the lateral- and longitudinal-dimension calibration factors, R_x and R_y respectively. Check that the mean spacings of groups of filaments that should have the same spacings are the same in each of the image planes within 1 %.

See [16].

For calibration of the **ultrasonic scan-plane** separation in 3D-imaging and for assessing distortion of images reconstructed (from a volume data set) with one axis in the **ultrasonic scan-plane's** thickness direction, the transducer is moved or rotated slowly in the elevational (ultrasonic **scan-plane's** thickness) direction normal to the filaments. That is, the transducer is moved from left to right (Transducer View A) on Figure A.1. This motion is performed in accordance with directions for this type of 3D-scanning as provided by the **ultrasound** system manufacturer. Often only a linear translation or a sector sweep is allowed but it is instructive to deviate from the instructions to see the amount of error generated.

For each of the rows of filaments from this second scan (View A), display reconstructed images that are perpendicular to the filaments. Calculate and report the maximum and r.m.s. deviations of:

- a) the measured filament spacings from their known values;
- b) the measured filament positions for the fitted lines;
- c) the slopes of the fitted lines from the expected values.

To evaluate the accuracy of curved lines and cross-sectional areas, closed figures are traced centrally on the display of curves and areas covering approximately 0,75 of the field-of-view. The lengths, circumferences and areas are measured and the percentages of measured-to-known areas calculated. Volume measurements are tested in these data with the sweep direction orthogonal to the filaments. Mark a known area, A , in the reconstructed image that is perpendicular to the filaments. Mark an **ultrasound** system-indicated length L' along the filaments for the third dimension of a 3D-volume enclosed by the filaments. For Figure A.1 this volume would be a cylindrical rod. Compare the measured volume $A' \times L'$ with the known

volume $A \times L/R_x$, where R_x is the lateral-dimension calibration factor from the second paragraph of this subclause. Further examples of measurements with filament **test objects** are given in [16].

7.4.3 Measurement accuracy using volumetric targets in a backscattering object phantom (Figure B.1) with a 2D-scanner

7.4.3.1 General

In these measurements, the transducer is rotated and tilted to find a circular cross-section of each of the **targets** in the 3D-**test object** that can be fully imaged in one view. For each **target**, move and adjust the image plane to find the minor axis, the largest diameter where the object still appears circular. The calliper markers are placed on the ends of the largest vertical (axial) diameter, b_y , obtainable through that circular cross-section and the value is noted on the calliper read-out. A horizontal (lateral) diameter, b_h , is measured in a similar way. The average of those two diameter measures is labelled b .

Rotate the transducer through 90° and find and measure the longest dimension of the ellipsoid, referred to as a , where a is the sum of the lengths of each half of the egg, $a_1 + a_2$. These procedures are repeated for the smaller 3D-object, if possible. The results obtained are tabulated in Table 2. Compare the measured values with the known values of the diameters given in Table 1.

7.4.3.2 Perimeters

Using an image of a cross-section of an ovoid **target**, begin to measure at a desired point on the perimeter of the **target** to be measured. Calliper markers are placed all along the selected image of the 3D-object until the starting point is reached. On average, the separation of the calliper markers should be no more than 1/20 of the estimated length of the perimeter except when an ellipse, or at least a curved line, fit is performed by the **ultrasound** system. The equation for the perimeter [17] of each of the two half-ellipses is approximately:

$$P = \frac{\pi}{\sqrt{2}} \left[a_i^2 + \left(\frac{b}{2} \right)^2 \right]^{1/2} \quad (1)$$

where

a_i is either a_1 or a_2 , the semi-major axes for a given half of the ellipsoid;

b is the mean minor axis of the ellipsoid.

The perimeter of the entire egg-shaped object is the sum of the perimeters of the two halves. The perimeter of the circular cross-section is $2\pi b$. See Table 1 for expected values for the two objects in Figures B.1 and B.2.

7.4.3.3 Areas

On virtually all machines, a value for the enclosed (cross-sectional) area is calculated from the same measurement points defined in the perimeter measurements (see 7.4.3.2).

The measured area values should be compared against the known areas for the 3D-object cross-sections. In the largest elliptical and circular cross-sections, respectively, the areas are [17]

$$A_c = 0,79 b (a_1 + a_2) = \pi b (a_1 + a_2) / 4 \text{ and } A_c = 0,79 b^2 = \pi b^2 / 4$$

The surface area of the ellipsoids is given by:

$$A = 2 \pi \left(\frac{b}{2} \right)^2 + \pi a_1 \left(\frac{b}{2 \varepsilon_1} \right) \arcsin \varepsilon_1 + \pi a_2 \left(\frac{b}{2 \varepsilon_2} \right) \arcsin \varepsilon_2 \quad (2)$$

where

ε_1 is the eccentricity $(1 - (b/(2a_1))^2)$;

ε_2 is the eccentricity $(1 - (b/(2a_2))^2)$.

See Table 1 for expected values for the two objects in Figure B.1.

7.4.3.4 Volumes

Volume measurements or relative measurements of objects of consistent shape can be made by measuring maximum linear dimensions on three orthogonal axes. In the case of 3D-ellipsoidal objects [17], the actual volume is given as:

$$V = 0,52 (a_1 + a_2) b^2 = \frac{4}{3} \pi \frac{(a_1 + a_2)}{2} \left(\frac{b}{2}\right)^2 \quad (3)$$

The volume can be determined by measuring individually the two lengths, a and b , where $a = a_1 + a_2$. Averaging two perpendicular measures of b is appropriate. These measurements can be made from two image planes, with the transducer rotated 90° between them. Note this calculation method is essentially correct only for volumes approximating two half-ellipsoids with circular cross-sections. (For estimating the volume of any mass that resembles an ellipsoid with orthogonal axes a , b , and c , equation 3 would have $V = 0,52abc$). See Table 1 for expected values for the two objects in Figure B.3.

For applicability to other shapes and probably increased accuracy, a series of images is obtained from equally spaced ultrasonic **scan planes**. For high accuracy, the separation of the ultrasonic **scan planes** should be less than the width of the ultrasonic **scan-plane** thickness at its focus, ideally, less than half that focal width. In the conceptually simplest calculation, the volume is estimated by considering it to be composed of a number of cylinders of base area equal to that measured in a ultrasonic **scan plane** and height equal to the separation of the planes, i.e., multiply the cross-sectional area of the 3D-object in each plane by the scan separation and add these volumes for each slice. This calculation can be tedious by hand. More sophisticated volume-measurement algorithms are implemented on most 3D cross-section imaging systems and should be tested.

Table 1 – Expected values for the two ellipsoidal objects in Figure B.3

	Half the perimeter ^a cm	Perimeter cm	Area cm ²	Surface area cm ²	Volume cm ³
<i>Small object</i>					
Long half	3,01	6,02	3,82	14,51	4,58
Short half	1,65	3,31	1,70	9,94	2,04
Total	4,66	9,32	5,51	24,45	6,61
Cross-section	2,83	5,65	2,54		
<i>Large object</i>					
Long half	5,35	10,7	16,5	63,6	55,0
Short half	4,16	8,3	11,0	69,9	36,6
Total	9,5	19,0	27,5	133,6	91,6
Cross-section	7,9	15,7	19,6		
The perimeters and half-perimeters here are calculated from the exact elliptical integrals, rather than from Equation 1.					
^a Half the perimeter is a test of curved path length					

Table 2 – Suggested table of reported values

	Half the perimeter		Perimeter		Area		Surface Area		Volume	
	cm	%	cm	%	cm ²	%	cm ²	%	cm ³	%
<i>Small object</i>										
Long half										
Short half										
Total										
Cross-section										
<i>Large object</i>										
Long half										
Short half										
Total										
Cross-section										
Enter the measurement then that measurement as a percent of the expected value from Table 1. This form should be filled out for each of the measurement algorithms under study. It is instructive to fill out all or parts of the table from values calculated from the linear dimensions, a_1 , a_2 , and b , using equations 1, 2 and 3, as well as from any other measurement method provided on the imaging system. Similar measurements are performed in 3D-fetal biometry [18]. This table may also be used for curved line, area and volume measurements from the filament test objects.										

8 Measurement of point-spread and line-spread functions (high-contrast spot size)

8.1 General

High-contrast resolution characteristics of most imaging systems can be characterized by measurement of the **point-spread function (PSF)**, which is the characteristic response of the imaging system to a high-contrast **point target** or the similar **line-spread function (LSF)**. For most optical systems, the **PSF** is singular, symmetric and isotropic. Thus, a measurement of the **PSF** usually is sufficient to characterize the system's impulse response.

Unlike optical imaging systems, **ultrasound** produces a **PSF** and **LSF** that is neither singular nor isotropic. Further, the **ultrasound PSF** and **LSF** are asymmetrical, having different axial and lateral dimensions, and they also vary with distance from the transducer (i.e. depth in the

image). Thus, many different measurements of the **PSF** and **LSF** at different positions and depths must be performed to obtain representative values of the system's imaging performance at specific positions along the beam axis.

An individual **PSF** or **LSF** cannot be used as the system's impulse response. Due to changes in the **PSF** and **LSF** with depth, there are different levels of resolution, depending on the position within the **ultrasound** scan.

Another complication arises from the way **ultrasound** behaves with high-contrast **targets** versus low-contrast **targets**. At high contrast, **ultrasound** can image very small structures, such as thin wires, filaments or the tip of a needle, or an air microbubble. Each of these will produce a characteristic smear that is associated with the **PSF** or **LSF**. For low-contrast **targets**, such as tissue, however, **ultrasound** images do not produce images of individual point **targets**, but rather, form a grey **speckle pattern** that appears as a grainy background, whose characteristics arise from the coherent nature of the **ultrasound beam**. A measurement of the dimensions of the PSF for an ultrasound system at a given depth describes the high-contrast resolution of the system at that depth. For low-contrast resolution measurements, a tissue-mimicking phantom is used, and the contrast-detail performance is sometimes measured. In this document, only the high-contrast behaviour of the **ultrasound PSF** or **LSF** will be measured. Low-contrast measurements will be described in another standard (See future IEC 61391-2).

In this document **PSF** and **LSF** are measured at two levels: FWHM (full width at half maximum), i.e. at –6 dB from the maximum, and at –20 dB. This requires that the relation between grey level and signal intensity be known (see 8.4.5).

NOTE Resolution on a two-target phantom (axial, lateral, or elevational) can be approximated from the PSF or LSF by adding two PSF- or LSF-curves as functions of distance between them and identifying the separation where there is a dip in the combined curves of –6 dB. Rigorously, the rf signal at each point in the PSF or LSF, and for each assumed target, must be summed to more accurately simulate the two-target response.

8.2 Test methods

To carry out the test procedures, the following items are required:

- a) **test objects**, as described in Annex A and Annex C, containing **targets** at accurately specified positions;
- b) a tank containing degassed liquid and, optionally, movable targets as described in Clause C.4 and D.5.4.2;
- c) digitized images, as described in 6.2.2 and 6.3.5.

The specifications of these devices are given in Annexes A and C.

8.3 Instruments

8.3.1 General

The instruments specified in this subclause have been selected to permit testing of real-time ultrasonic scanners without requiring electronic signals to be output from or input to the scanners.

8.3.2 Test object and tank

A **test object** should contain structures that allow the following machine features to be measured:

- a) **axial resolution**;
- b) **lateral resolution**;
- c) scan **slice thickness**;

Examples of **test objects** are given in Annexes A and C.

A tank of degassed **working liquid** is required. Under circumstances that allow a deviation of the speed of sound (see 5) the **working liquid** can be replaced by water to facilitate handling.

8.3.3 Image digitizer

An image digitizer as described in 6.2.2 is required.

8.4 Test settings

8.4.1 General

The many combinations of scanner settings and transducers make it impossible to carry out tests for all of them. Tests are therefore carried out for specified settings and for each transducer. The scanner is set up using the procedures outlined below. The focusing of the ultrasonic beam should be extended over as large a range as possible i.e. to achieve the best average resolution over all visible **targets**. A setting should be specified by a test instruction for each test, if it differs from the general recommendations. See D.5.2.

8.4.2 Display settings (focus, brilliance, contrast)

The focus is made sharp and the brilliance and contrast controls are turned to their lowest positions. The brilliance is now increased until the echo-free zone at the side of the image becomes the minimum perceptible shade of grey by observing the grey-shade bar on the display. The contrast control is then increased to make the image contain the greatest range of grey shades possible. The focus is then checked for sharpness; if it requires further adjustment, the whole procedure is repeated. If the system allows multiple focal zones, activate all of them for the desired range of measurements.

8.4.3 Sensitivity settings (frequency, suppression, output power, gain, TGC, ATGC)

- a) The **nominal frequency** of the scanner's transducer is noted.
- b) If there is a suppression or reject control, it is adjusted to allow the smallest possible signals to be displayed.
- c) The output power and **gain** are set to the minimum values consistent with displaying an adequate image.
- d) The **ATGC** controls are adjusted to remove the **ATGC**.
- e) The overall **gain** control is increased until the signal from the **target** is clearly visualized on the display.

8.4.4 Recording system

Digitized images will be used, as described in 6.2.2 and 6.3.5.

8.4.5 Calibration of system characteristic curve

A well-established method for calibrating the system characteristic curve (output signal as a function of acoustic-pressure signal at the transducer) is with planar boundaries of graded reflectivity [19, 20]. A simple, calibration phantom provides specular reflectors located parallel to each other and normal to the intended path of the **ultrasound beam** at different ranges from the transducer position in a **tissue-mimicking material**. At each range at least two reflectors are provided, with different reflectivities.

In its simplest form one plate has a 10 dB lower reflection coefficient than the other. This is achieved by constructing one plate of 342 stainless steel and the other of plexiglas. The steel is located on each step of one staircase (A), and the plexiglas on each step of another, adjacent staircase (B). At usual diagnostic frequencies, the first step is 1 cm below the surface

followed by two additional steps at 2 cm and 5 cm depth. These steps are followed by two steps, one at 10 cm depth and another at 18 cm depth. The space above the steps is filled with **tissue-mimicking material**. The speed of sound, c , the density, ρ , and the attenuation coefficient slope, α/f (where α is the attenuation coefficient at the working frequency, f) of this **tissue-mimicking material** shall be as follows: $c = (1\,540 \pm 6)$ m/s; $\rho = (1,05 \pm 0,05)$ g/cm³; $\alpha/f = (0,5 \pm 0,05)$ dB cm⁻¹MHz⁻¹.

Echo signals due to backscatter in the **tissue-mimicking material** should be at least 20 dB weaker than the echo signal from the weakest reflecting interface at all frequencies in the tested waveforms.

In use to calibrate a system's sensitivity control, gain or output, this phantom is imaged at a range at or beyond the transducer's focal zone and at a gain where the weaker reflector, on staircase B, is approximately 6 dB above the electronic noise level. If signal levels are too high, choose a deeper step and reduce the TGC as much as possible. Signal levels (10 dB apart) are recorded from the steps at that range on staircases A and B, respectively. Then the system's sensitivity control is increased until the signal from the step on staircase B (plate B) is equal to the signal level from the step on staircase A (plate A) at the previous control setting. Thus, the signal from plate A is now 10 dB above that for plate B at the first setting. By repeating this process, the entire signal dynamic range can be calibrated in 10-dB steps. By recording settings of the system's sensitivity control, this control is thus calibrated relatively as well. Interpolation of those control settings will allow calibration of signal changes at finer increments.

Another method for constructing the phantom is used to produce weaker signals lying more within the linear range of most diagnostic **ultrasound** systems. This method should be compared against the primary method for accuracy. The secondary method is to fabricate the steps on one staircase from high-density polyethylene (LB-861) and the steps on the other staircase from low-density polyethylene (NA-117)²⁾. The high-density material has an impedance of $2,33 \times 10^6$ kgm⁻²s⁻¹, while the impedance of the low-density material is $1,79 \times 10^6$ kgm⁻²s⁻¹ [21]. The reflection coefficients of these interfaces should differ by 10 dB in a tissue-mimicking gel. More staircases with steps having other reflection coefficients can be employed to produce even weaker echoes and finer gradations, e.g., [22]. However, these also shall be calibrated against the primary method. With low echogenicities, the temperature of the medium becomes critical and shall be controlled sufficiently precisely to maintain the relative echogenicity within $\pm 0,5$ dB.

Even more convenient, when available, is electrical signal injection, either directly, or by acoustically coupling another transducer, with a calibrated transmitter and receiver to the transducer under test [23]. Connectors for direct injection are expensive and system-specific, while acoustic coupling must be performed carefully. However, if referenced to the primary method above, signal injection can be used in practice.

Point targets for **PSF** measurements: A second acoustic approach is based on **targets** of different sizes and should also be referenced to the primary method above. As scattering cross-section increases with frequency, this type of **target** emphasizes the high-frequency part of the signal from the scanner. However, the same emphasis occurs for signals from distributions of small scatterers, so it is not unrealistic to test a scanner in this way.

Theory indicates that, for **targets** much smaller than the wavelength, **targets** of all diameters have the same frequency dependence. Hence, it is possible to make a series of reflectors with well-defined scattering ratios. A series of sphere diameters increasing by 26 %, such as, in microns, 10 – 12,6 – 15,87 – 20 – 25,2 – 31,75 – 40 – 50,4 – 63,5 and 80 would give reflectivities in steps of 6 dB, spanning a range of 54 dB. This approach has been implemented

²⁾ Both materials are available from USI Corporation, Marlboro, MA, among others. They are examples of suitable products available commercially. This information is given for the convenience of users of this document and does not constitute an endorsement by IEC of these products.

with dense distributions of small scattering particles, but practicalities of implementation are not well documented for individual particles.

Similarly, a series of flat-ended wires can provide calibrated reflectivities. Here the ratio of diameters should be $\sqrt{2} = 1,41$, to obtain a step of 6 dB. For 5 MHz, diameters in the range 200 μ to 1 600 μ could span a range of 36 dB with a precision of circa 1 dB.

8.5 Test parameters

8.5.1 General

To measure **point-** or **line-spread functions**, an appropriate **target** is scanned in a test tank. A standard point **target** is the tip of a flat-ended wire viewed on end; a standard line **target** is a filament or wire, typically lying normal to the direction of ultrasound propagation. The dimensions of the **point-spread function (PSF)** in each of the three directions are measured for complete, detailed system-performance evaluation, usually in a developmental-, production, or research setting. **Line-spread functions (LSFs)** are employed in the field and should be used in specifications to be tested in the field. **LSFs** are recommended over **resolution**, as measured with multiple **targets**, because fewer targets are required and spatial spread-function measurements are required at many locations in the **field-of-view** and are dependent on a system's focal- and frequency-settings. Actual **resolution** with target pairs is sometimes preferred for visual measurements and when non-linear signal processing, such as signal differentiation may be employed. **PSF** and **LSF** measurement on a line normal to and through the **ultrasound beam axis** are closely related to the imaging system's high-contrast **resolution** and to each other when scanned in a direction along the same line at the same depth in a phantom. **PSF**, **LSF** and **resolution** each can be derived from a full set of measurements of the other, if the characteristic curve of the ultrasound system is calibrated. See the definition of **spot size**, used in this standard to relate to the width of a **PSF** or **LSF** in a way more self-explanatory to those not familiar with the spread-function terminology.

PSFs and **LSFs** currently yield relative measurements. Quantitative measurements indicating the sensitivity of the imaging system to a specified or standard **point target** or point pressure source can be obtained [24-28] but experience with these techniques is too limited for standardization at this time. These measurements are closely related to the high-contrast **axial resolution** and axial **LSF** of the imaging system for a specific position along the beam (i.e. depth in a phantom).

For all such measures, the sensitivity and angulation of the scanner is adjusted so that the maximum signal from the **target** is clearly visualized during each test.

If a **line target** is used, oriented perpendicularly to the ultrasonic **scan plane**, the cross-section of the line with the ultrasonic **scan plane** constitutes the **lateral line-spread function**, but this configuration has two limitations:

- a) the **line-spread function** can only be measured in the ultrasonic **scan plane**;
- b) the intensity is sensitive to the orientation of the ultrasonic beam perpendicularly to the **line target**.

To overcome this limitation a spherical **target** has been used. However, now interference between ultrasonic paths inside the sphere leads to complicated patterns of angular scattering [24] unless the sphere is extremely small and a relatively weak reflector. The complications generated by interference and multiple reflections inside the spherical target may be solved by time-domain analysis of the received echo when a larger and/or highly reflective sphere is used. See D.5.4.2.

An alternative **test object** for the measurement of resolution is a single filament (or wire) mounted vertically (in the direction of the ultrasound beam) on a micromanipulator which allows the filament to be moved through the **field-of-view** (Figure C.3). This permits the **spot size** to be measured at any point in the **field-of-view**. It also removes the problem, encountered with

an array of **targets**, of aligning all filaments perpendicularly to the beam at the same time to record a maximum signal from each.

When the point is much smaller than the wavelength (or than the diameter of the ultrasonic beam at the position of the **target** for measured transducers of large f-numbers), the **point-spread function** is measured independently of the properties of the **target**. Note that the response width from the **PSF** (from a **point target**) and the **LSF** (from a line **target**) are approximately equal. It has been shown that scattering strength depends smoothly on angle and frequency, and means of constructing **targets** have been proposed [25]. Figure C.4 gives two simpler alternatives.

From the theory of the scattering by the front plane (equation 3 of [25].), the value for the back-scattering cross section, σ , is found:

$$\sigma = \left[\frac{k r^2}{2} \left(\frac{Z_m - Z_w}{Z_m + Z_w} \right) \right]^2 \quad (4)$$

where

k is the circular wave number; ($= 2 \pi / \lambda$ in which λ is the wavelength in the medium)

r is the radius of the wire;

Z_m is the characteristic acoustic impedance of the wire material;

Z_w is the characteristic acoustic impedance of the surrounding medium (water).

Scattering is proportional to the square of the frequency and the fourth power of diameter (as long as the diameter of the **target** is so small that no phase differences between various parts of the **target** occur). At small angles the influence of the angle between front plane and the direction of the beam is small [25]. At larger angles observed scattering is diminished. Making the **target** small decreases the influence of the angle.

At oblique incidence of the **ultrasound** on the front end of the wire the scattering strength decreases. The consequence is that for a sector scanner, reorientation of transducer or **target** is necessary, in case quantitative measurements of reflected intensities in various parts of the field are to be compared. Quantitative measurements of beam width remain possible without reorientation.

Figure C.4 shows a sketch of two **point targets** made from dental wire (diameter 0,24 mm). The left **target** is intended for pulsed **ultrasound**; echoes from the wire tip and the supporting tube are received separated in time. The supporting tube (outer diameter 1,0 mm) provides mechanical stability. The right **target** is intended for CW **ultrasound**; the tip of the wire is the only feature inside the **ultrasound beam** (grey) which gives backscattering.

8.5.2 Axial PSF and LSF dimensions and axial resolution

8.5.2.1 General

The dimensions of the **PSF** or **LSF** in the axial direction should be measured at different depths. An appropriate phantom for **PSF** is shown in Figure C.3 or C.4, the latter being required when reverberations from the support wire are obtrusive. **LSF** is measured with a phantom as depicted in Figure A.1, A.2 or C.1 and axial resolution with the phantom of Figure C.2. Measurements can be made in increments of less than 1/3 of the axial **field-of-view** for a total depth depending on the transducer's imaging range. Sources of variability should be identified and documented, as indicated in 6.4.2. Axial resolution at a specific depth can be obtained as the minimum separation 1) of the two filaments in Figure C.2 that are distinguishable before the two filaments first blur together; and 2) quantitatively, at which there is at least a 6 dB drop in the signal between the two filaments.

8.5.2.2 Procedure – axial dimensions

Align the transducer for maximum signal from the **target(s)**. Alignment with the beam normal to any filament **target** is particularly critical. Set the system sensitivity controls so that all the **targets** are clearly visualized, but are not saturated. Make several **B-mode** images or image volumes of the **target** or phantom and digitize the best image. On some phantoms there is a scan window parallel to contained filament **targets**. Make sure that the **ultrasonic transducer** is well seated on the surface. When the transducer's face is curved, the central **scan line** should be normal to the scan window and the **ultrasound beams** should be normal to the filaments.

For **PSF** measurements, a 2D-scan is required for the **point target** in each measured location, unless alignment with the central axis or central plane of the beam or image plane has been accomplished. For **LSF** or **PSF** an image of the **test object** is obtained which contains the imaged targets for the entire depth range of the **ultrasonic transducer**. Use image-processing software to plot a brightness profile in the vertical direction for many successive **scan lines** covering an imaged **target**, moving laterally from left to right, until the maximum amplitude profile of the **target** is obtained. For this maximum amplitude of the **target**, at least, the full-width-at-half-maximum (FWHM) and the –20 dB width are measured and recorded for the given filament depth. Now, measure the axial response width for all the imaged-**target** positions (depths) in the phantom image. Once all the axial response widths are obtained, plot the FWHM or –20 dB width versus depth in the phantom image. This plot shows the variation in axial **PSF** or **LSF** for the imaging system with depth. For most systems, there will be only a small variation in axial **PSF** or **LSF** with depth.

Profile length (**PSF** or **LSF**) at a relative signal level near the expected axial clutter level or clutter level at a certain number of wavelengths is also most illuminating, if artefacts in the phantom allow that measurement.

8.5.3 Lateral PSF- and LSF-width and lateral resolution

8.5.3.1 General

The dimensions of the **PSF** or **LSF** in the lateral direction correspond to the high-contrast **lateral resolution** of the imaging system for a specific position along the beam (i.e. several depths in the phantom). An appropriate phantom for **PSF** is shown in Figure C.3 or C.4, the latter being required when reverberations from the support wire are obtrusive. **LSF** is measured with a phantom as depicted in Figure A.1, A.2 or C.1. Multi-**target** lateral resolution measurements are rarely required and are not addressed here. Measurements can be made over the total axial imaging range of the **ultrasonic transducer** in increments of less than 1/7 of the axial **field-of-view** along a generally central axis and at a distance from a lateral edge of the image of 1/8 of the image width. For volumetric imaging systems in which **PSF** might be dependent on elevational position in the volume (as for a linear array scanned in an enclosed housing), make a measurement at a distance from an elevational edge of the image of 1/8 of the volume thickness and in a corner of the volume at a distance from a lateral edge and an elevational edge of the image of 1/8 of the volume width and thickness, respectively. Sources of variability should be identified and documented, as indicated in 6.4.2.

8.5.3.2 Procedure – lateral dimensions

Align the transducer for maximum signal from the **target(s)**. Alignment with the beam normal to any filament **target** is particularly critical. Set the system sensitivity controls as described in 8.5.2.2. Make several **B-mode** images or image volumes of the **target** or phantom and digitize the best image. Make sure that the **ultrasonic transducer** is well seated on any scanning surface of the **test object**, so that the transducer face is perpendicular to the **test object's** surface.

For **PSF** measurements, a 2D-scan is required for the point **target** in each measured location, unless alignment with the central axis or central plane of the beam or image plane has been accomplished. For **LSF** or **PSF**, an image of the **test object** is obtained which contains the

imaged **targets** for the entire depth range of the **ultrasonic transducer**. Use the image processing software to plot a brightness profile in the horizontal direction for many successive **scan lines** covering an imaged **point target**, moving vertically from top to bottom (above and below the filament cross-section image), until the maximum amplitude profile of the **target** is obtained. For this maximum amplitude of the **target**, as a minimum, the full-width-at-half-maximum (FWHM) and the –20 dB width are measured and recorded for the given filament depth. Profile width at a relative signal level near the expected transducer clutter level or clutter level at a certain linear or angular beam width, is also most illuminating, if artefacts in the phantom allow that measurement.

Now measure the maximum lateral-response width versus depth for all the imaged **PSF** positions (depths) in the phantom image. Once all the lateral-response widths are obtained, plot the FWHM or –20 dB width versus depth in the phantom image. This plot shows the variation in lateral **PSF or LSF** (lateral **spot size**) for the imaging system with depth. For most systems, the **lateral resolution** varies much more with depth than the **axial resolution**.

NOTE The use of the **test object** described in Figure A.2 may be inadequate for some **ultrasonic transducers** which produce such large **side lobes** that they interfere with those of neighbouring filament images. For this case, a **test object** such as that described in Figure A.1 may be used. For that **test object**, the lines of **targets** are angled, so that individual **targets** are rarely positioned along the same horizontal line. A better **test object** is described in Figure C.1, where all filament **targets** are positioned individually, so that there are no laterally neighbouring **targets**. Finally, a single filament (or wire) on a micromanipulator, similar to that described in Figure C.3 but normal to the ultrasound beam, may be used.

8.5.4 Scan slice thickness (elevational PSF and LSF) or elevational resolution

For simple measurements in the field, the **slice thickness** is determined by scanning the thin sheet of scattering **targets** shown in Figure C.5. The **slice thickness** (the same as the elevational spot-size resolution) at each depth should be measured with the sensitivity and display controls in calibrated positions. Consider the maximum displayed range as being divided into five consecutive equal-sized sections. Place the plane of scan to intersect the sheet of **targets** along a line parallel to the top surface of the **test object**. The ultrasonic **scan plane** should also be perpendicular to the **test-object scanning surface**. To measure the **slice thickness** in the middle of the first section, arrange for the line of intersection of the ultrasonic **scan plane** and the **target** sheet to be at that depth. Using visual measurements or image-analysis software, measure the vertical thickness of the image of the sheet of **targets** and calculate the **slice thickness** at the centre of this first section, as illustrated in Figure C.5. Repeat the procedure for the four other sections. With image-analysis software, the width of the image can be measured at any desired level, e.g., FWHM, FWTM, (–6 dB, –20 dB) or clutter below that of the peak of the sheet signal.

For elevational **PSF** with a **point target** as described in Figures C.3 or C.4 or **LSF** using a horizontal wire similar to Figure C.3 or the filament phantom in Figures A.1, A.2 or C.1, a **B-mode** image is made. The **target** is moved perpendicularly to the ultrasonic **scan plane** (or *vice versa*) until the maximum amplitude of the **target** is obtained. The movement is then executed sequentially in two opposite directions along a line perpendicular to the ultrasonic **scan plane**, until two times a certain decrease in intensity is obtained repeatably twice. For example, at FWHM and FWTM (–6 dB and –20 dB), the clutter level and other desired levels may be obtained. The distance between the points on both sides of the ultrasonic **scan plane** is the **elevational resolution** at the measured level of, e.g. –6 dB, or –20 dB.

Measurements can be made over the total axial imaging range of the **ultrasonic transducer** in increments of less than 1/7 of the axial **field-of-view** along a generally central axis and at a distance from a lateral edge of the image of 1/8 of the image width. For volumetric imaging systems in which **PSF** might be dependent on elevational position in the volume (as for a linear array scanned in an enclosed housing), make a measurement at a distance, from an elevational edge of the image, of 1/8 of the volume thickness and in a corner of the volume, at a distance from a lateral edge and an elevational edge of the image, of 1/8 of the volume width and thickness, respectively. Sources of variability should be identified and documented, as indicated in 6.4.2.

Optionally the measurement of **elevational resolution** can be facilitated by using an array of equal **point targets**, placed on a line that makes an oblique angle with the **scan plane**. The **targets** should be placed at such a distance that their images are spaced with a greater distance than the **lateral resolution** of the scanner at that depth. When the **target** placed in the centre has a backscattering cross-section at the required level below that of the other **targets**, the measurement can be executed by obtaining images of equal intensity.

The most accurate and flexible method to derive the complex set of parameters based on the PSF mapping analysis is described in Annex D.

Annex A (normative)

Test objects – Calibration of 2D-spatial measurement systems

NOTE The **test objects** and materials described in this annex are derived from those described in several national reports and draft standards. The **test objects** have been restricted to those that have been demonstrated in practice.

A.1 Test object structures

Figures A.1 and A.2 contain sets of parallel filaments serving as or approximating **line targets**. The filaments are also parallel to the top surface, which serves usually as the scan window. The figures show two possible arrangements of the cross section perpendicular to the filaments. Figure A.2 has the advantage over A.1 of a more regular structure, but more risk of hiding a **target** by **targets** on top of it.

The scales of the test objects shown are for general abdominal imaging at 2 MHz – 5 MHz centre frequencies. The distances between filaments in a row are 2 cm. Accuracy of filament placement should be $\pm 0,2$ mm. For higher-frequency imaging, a finer, 1 cm, filament spacing, overall dimensions equal to 1/3 of the those shown in Figures A.1 and A.2, and $\pm 0,1$ mm placement accuracy are required.

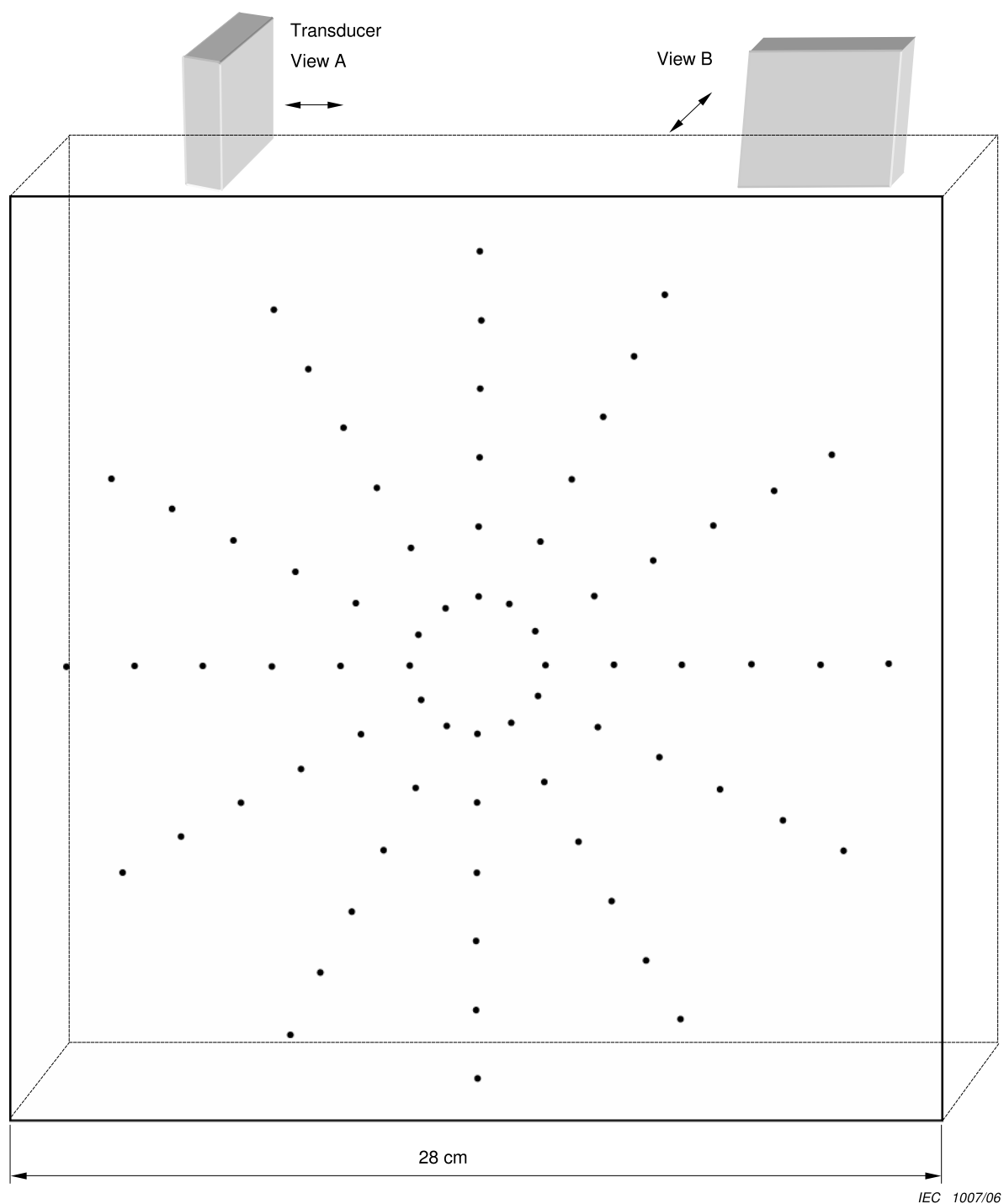
For this standard each **test object** (Figures A.1 and A.2) is designed to be scanned directly in **working liquid**. Nylon filaments in the objects are used to reduce the effect of shadowing. They are typically 0,078 mm in diameter (8X fly-fishing tippet). The tank should be sufficiently large to allow all transducers to be used in all required orientations to the internal structures.

Alternatively, the structures in Figures A.1 and A.2 may be immersed in **tissue-mimicking material** and scanned through an acoustic window of suitable material, such as polythene. A suitable window material should prevent the loss of water from the **tissue-mimicking material**. With **tissue-mimicking material** of typical backscatter, wires rather than nylon filaments should be used for the structures to give stronger echoes. Suitable wires for the **test object** are of stainless steel (type 316) with a diameter of 0,15 mm. These wires are sufficiently thin so that echoes from them are not elongated by internal reverberations at all but the highest diagnostic frequencies. They should be located in each **test object** to within $\pm 0,4$ mm.

These structures are designed with parallel filaments, as in most commercially available phantoms originally designed for calibrating 2D-imagers. They may be employed for 3D-imaging system calibration, two dimensions at a time, by rotating the scanhead 90 degrees about an axis normal to the filaments and lying centrally in the phantom and the imaged volume. A more general use of phantom motion has been published recently [29]. Numerous papers have been published since 1990 on filament phantoms for single view calibration of 3D-imaging and image-guided therapy systems [30].

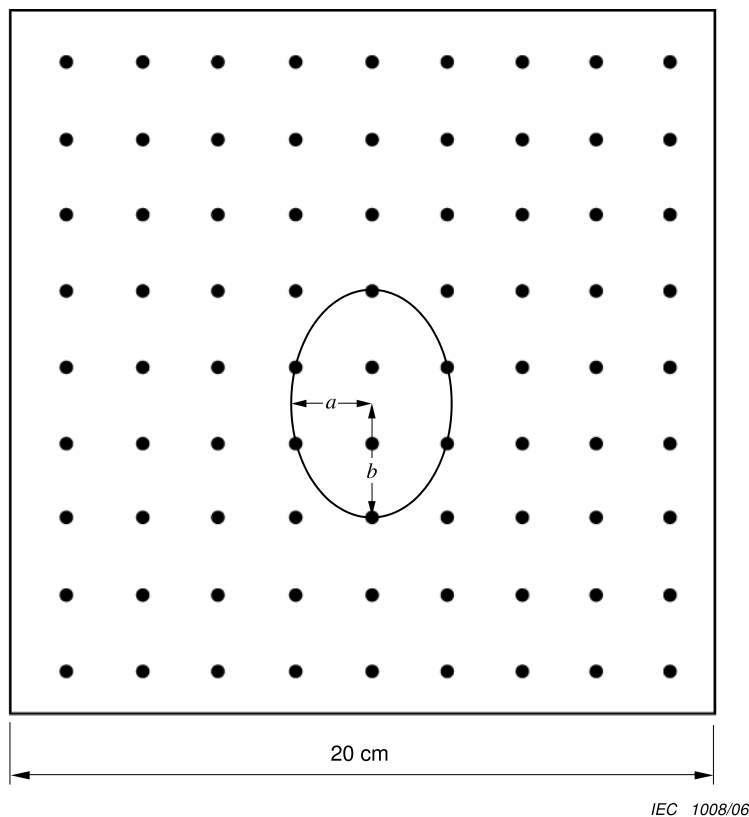
The calibration of **test objects** depends on the sound velocity of the medium (**working liquid** or **tissue mimicking material**). Sound velocity depends on temperature. With non- enclosed liquids, the change of composition by evaporation (e.g. of alcohol) should be considered. The stated margin in sound velocity ($1\,540 \pm 15$) m/s contributes ± 1 % uncertainty to measuring results. In view of variations in sound velocity in the body, this uncertainty is acceptable.

A.2 Test object diagrams



IEC 1007/06

Figure A.1 – Concentric circular arrays of nylon filaments



An imaginary ellipse is shown for curved-line length measurements, such as a perimeter.

Figure A.2 – Regular 2D-array of nylon filaments

Annex B (normative)

Test objects – Measurement and calibration of 3D-image reconstruction accuracy

B.1 Working-liquid-filled filament phantom

The phantoms described in Annex A (Figures A.1 and A.2) are appropriate for 3D as well as 2D measurements.

B.2 Tissue-mimicking filament phantom

A suitable phantom for a filament **test object** containing a backscattering background is shown (Figure A.1) for 2D- and 3D-distance measurement and image-distortion assessment. For 3D-imaging, the scanhead is moved in the elevational (ultrasonic **scan- plane** thickness) direction from left to right on this figure to assess position registration in the **slice thickness** direction. The relative positions of all filaments are maintained within $\pm 0,2$ mm at all locations. For arc length, perimeter, area and volume measurements the imaginary ellipse (or imaginary elliptical rod), $a = \pm 2,7$ cm, $b = \pm 3$ cm, is shown in Figure A.2. The area and circumference (perimeter) of the ellipse are $\pi a b = 25,4$ cm² and approximately $\sqrt{2} \pi(a^2 + b^2)^{1/2} = 17,9$ cm, respectively. Circular area and perimeter measurements and cylindrical rod volume and surface area measurements may be made easily on Figure A.1 and rectangular ones on Figure A.2. Volumes and surface areas of rods are, respectively, the cross-sectional area times the length and two times the cross-sectional area plus the perimeter times the length.

B.3 Tissue-mimicking ovoid target test object

Specification for 3D-, speckle-defined (egg-shaped) objects:

- Attenuation coefficient equal to that of the background material $\pm 0,2$ dB cm⁻¹ MHz⁻¹.
- Speed of sound equal to that of the background material ± 6 m s⁻¹ [9, 31].
- The egg-shaped objects have a **backscatter coefficient** -10 dB ± 2 dB relative to that of the standard background material, whose backscatter coefficient at 3 MHz is 3×10^{-4} cm⁻¹ sr⁻¹ ± 6 dB (with an approximately f^4 dependence from 1 MHz to 15 MHz).

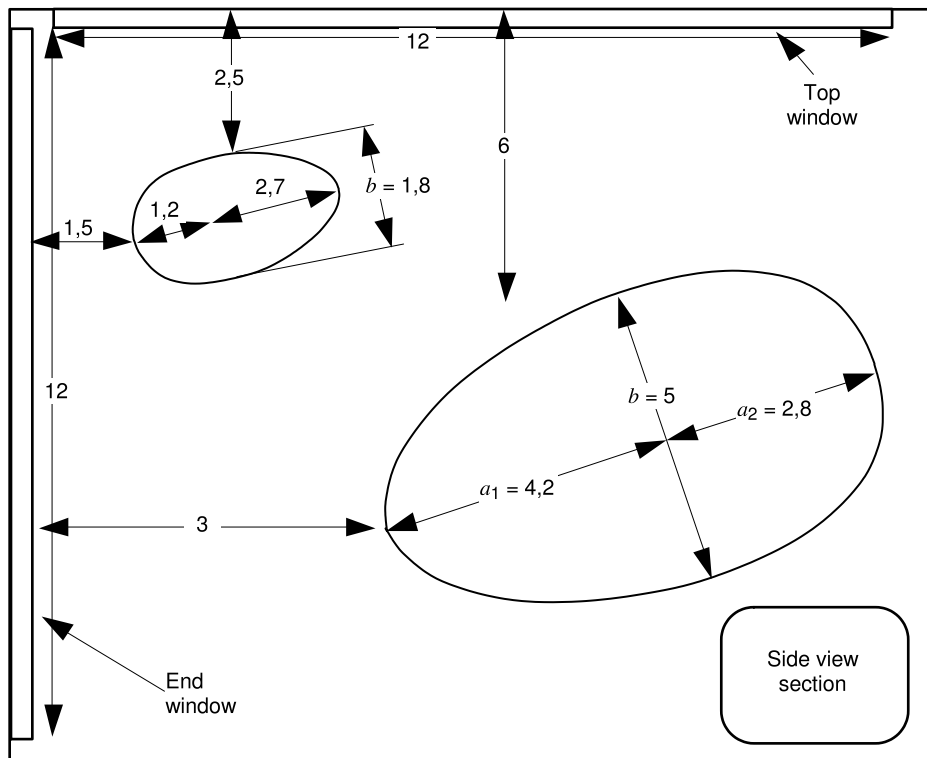
B.4 Spatial measurements from backscatter-defined objects

A suitable **test object** for spatial measurements of backscatter-defined objects is shown in Figures B.1 to B.4. This **test object** is referred to as the 3D-egg **test object** because the measured geometries consist of two half-ellipsoids of revolution (oblate spheroids) joined end to end.

The side view (Figure B.1) shows the long-axis views of the two ellipsoids. A composite of two cross-sectional views is shown in Figure B.2 and projections are illustrated in Figures B.3 and B.4.

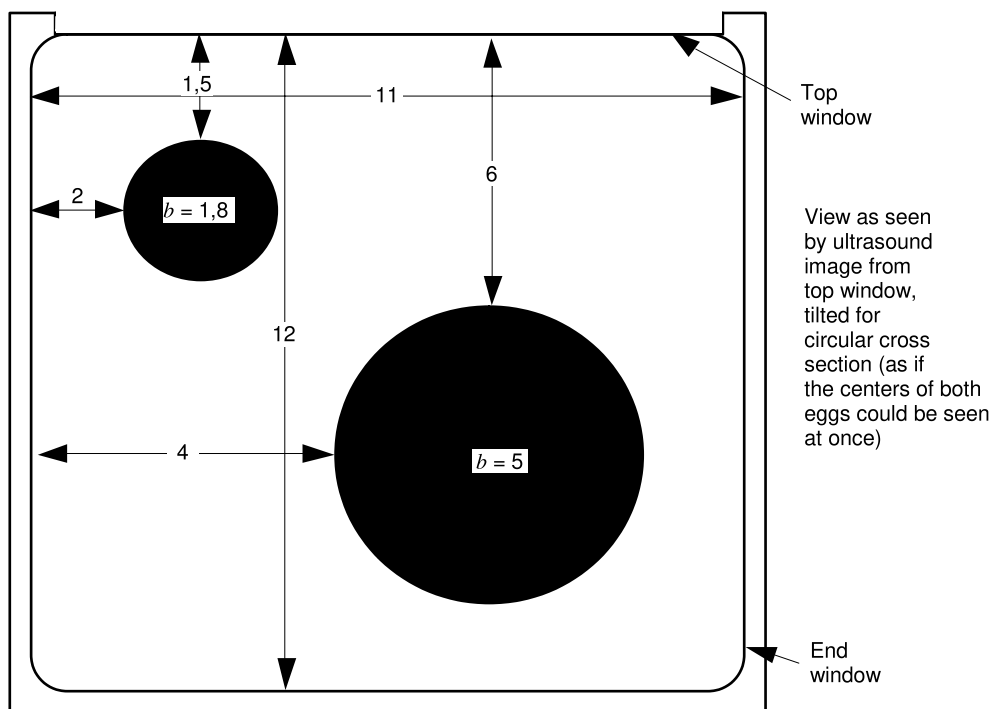
NOTE Alternatively, for practical purposes in the hospital, in which a standard phantom is not available, the 3D-object shapes may be constructed of a low -attenuation material, such as open pore reticulated foam [6, 32] and immersed in saline or other fluid, such that the 3D-objects and the fluid meet the above speed-of-sound criteria, the differential attenuation of $0,5$ dB cm⁻¹ MHz⁻¹, and the backscatter coefficient of the 3D-objects (egg) at 3 MHz of 3×10^{-5} cm⁻¹ sr⁻¹ ± 16 dB (with an f^2 to f^4 dependence from 1 MHz to 7 MHz).

B.5 Test object diagrams



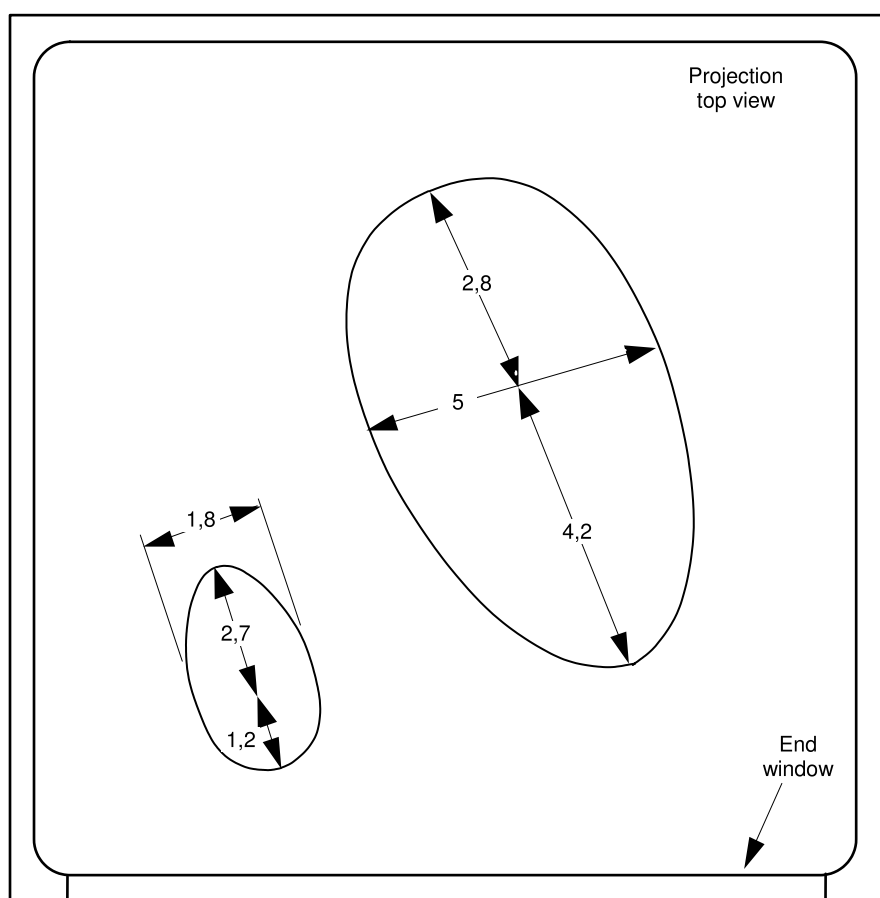
IEC 1009/06

Figure B.1 – Tissue mimicking ovoid target phantom



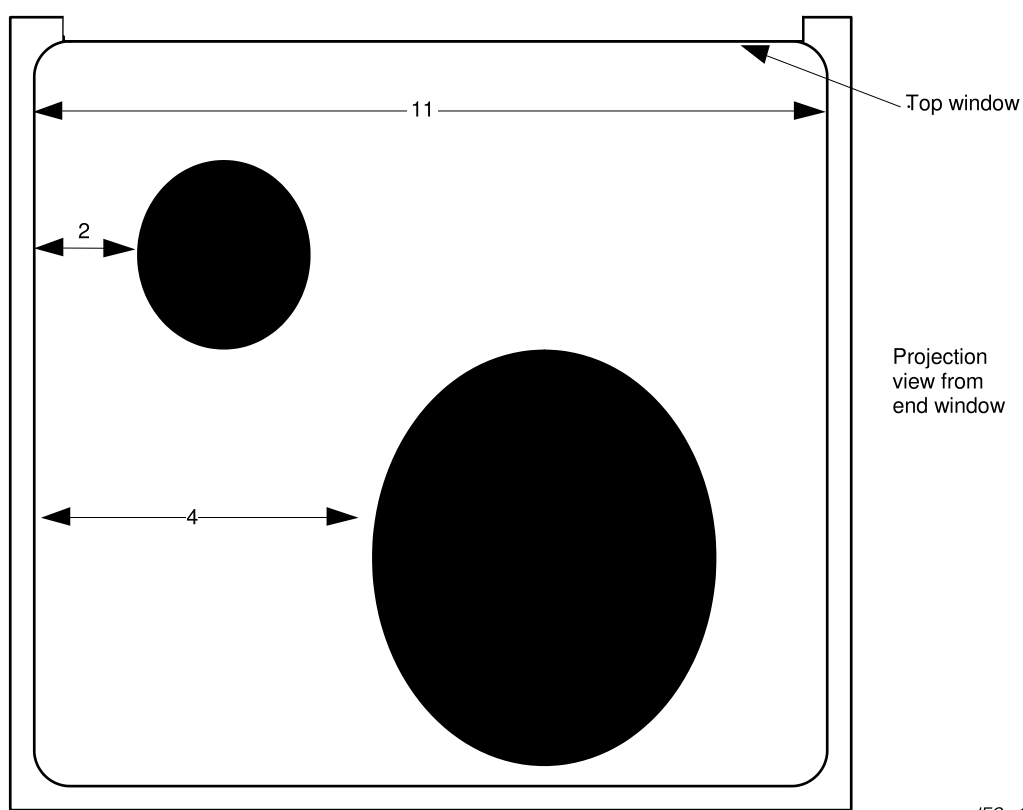
IEC 1010/06

Figure B.2 – Composite of two cross-sectional views of test object shown in Figure B.1



IEC 1011/06

Figure B.3 – Projection view from top of test object shown in Figure B.1



IEC 1012/06

Figure B.4 – Projection view from end window of test object shown in Figure B.1

Annex C (normative)

Test objects – Measurement of point-spread function response

NOTE The **test objects** and materials described in this annex are derived from those described in several national reports and draft standards. The **test objects** have been restricted to those which have been demonstrated in practice.

C.1 Test object structures

For this standard each **test object** (Figures A.1, A.2 and C.1 to C.5) is designed to be scanned directly in **working liquid**. Nylon filaments in the objects are used to reduce the effect of shadowing. They are typically 0,1 mm in diameter. The tank should be sufficiently large to allow all transducers to be used in all required orientations to the internal structures. Under some conditions (see 5) the **working liquid** may be replaced by water.

Alternatively, any of the mentioned structures may be immersed in **tissue-mimicking material** and scanned through an acoustic window of suitable material, such as polythene. A suitable window material should prevent the loss of water from **tissue-mimicking material**. Wires rather than nylon filaments should be used for the structures in the **tissue-mimicking material** to give stronger echoes. Suitable wires for the **test object** are of stainless steel (type 316) with a diameter of 0,15 mm. These wires are sufficiently thin so that echoes from them are not elongated by internal reverberations. They should be located in each **test object** to within $\pm 0,2$ mm.

C.2 Array of filaments in water (Figure C.1)

The two-dimensional array of filaments is supported in a frame. The wires or filaments are as specified in A.1.

C.3 Axial resolution test object (Figure C.2)

Two nylon filaments of typical diameter 0,1 mm are arranged to lie in the same vertical plane (Figure C.2) and at an angle to each other. The filaments can be supported over a range of depths. The volume of the liquid-filled tank should typically be a cube of side length equal to 20 cm.

C.4 Movable single filament or wire in water (Figures C.3, C.4)

The filament is mounted on a micromanipulator (see Figure C.3) which allows it to be moved throughout the **field-of-view** of the scanner. The filament is as specified in A.1. An alternative is to use a fine metal wire mounted as shown in Figure C.4. The diameter of the metal wire shall be much smaller than the beamwidth of the investigated transducer and smaller than $\lambda/2$ for low- f -number transducers.

The use of a movable spherical target for assessing quality parameters derived by **PSF**-mapping analysis is described in Annex D.

C.5 Scanner slice thickness test object (Figure C.5)

- A sheet of scatterers makes an angle of 75° to the top surface [33]. Suitable material for the sheet of scatterers has a **backscatter coefficient** more than 50 dB higher than that of the background material and the sheet is less than 0,4 mm thick.
- The method of calculating the scan **slice thickness** from the image is discussed in 8.5.4.

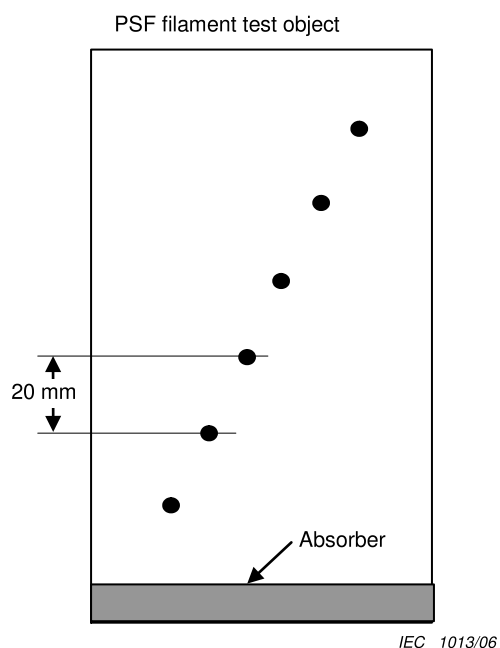


Figure C.1 – Filament test object for measuring the LSF

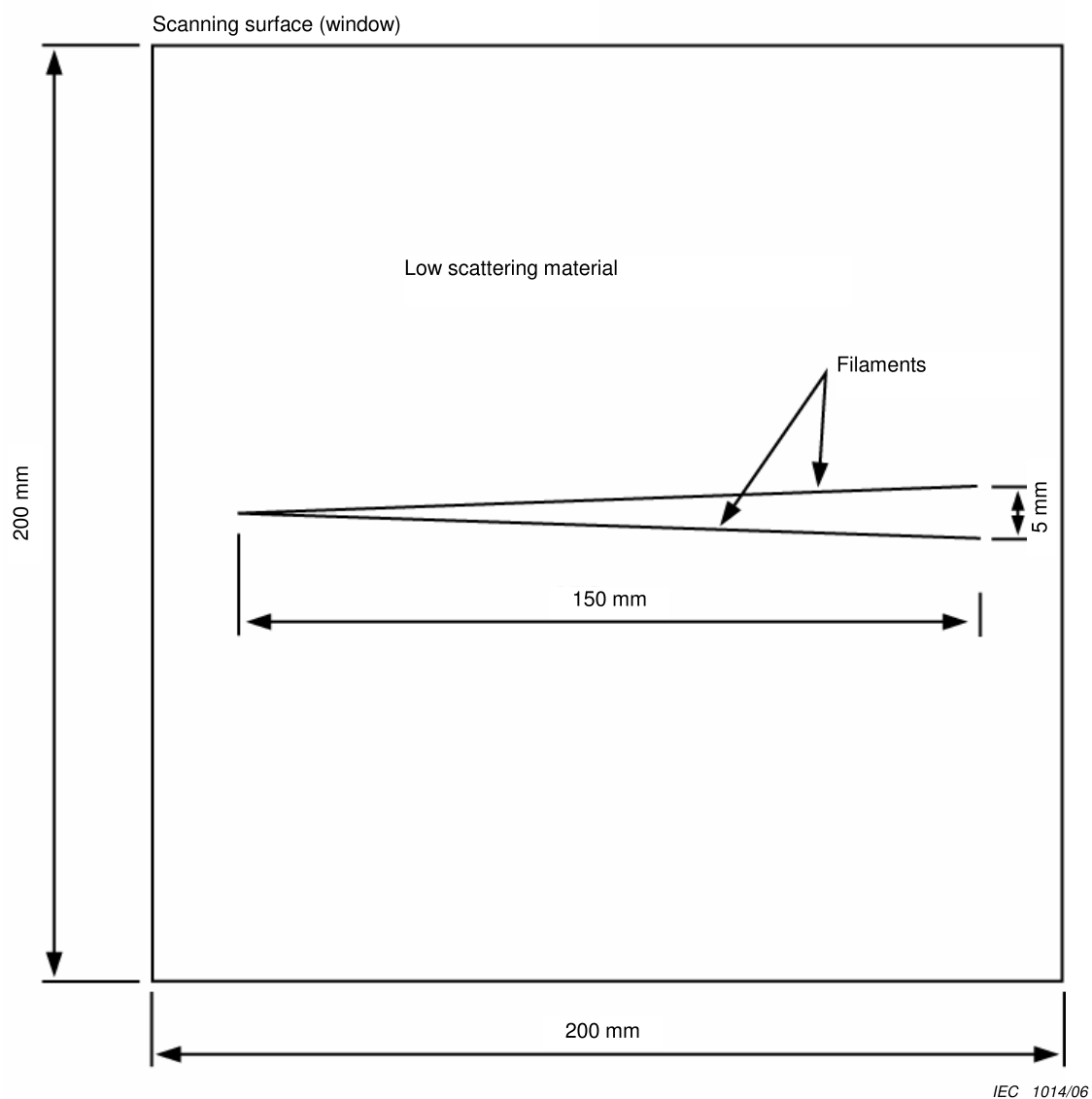
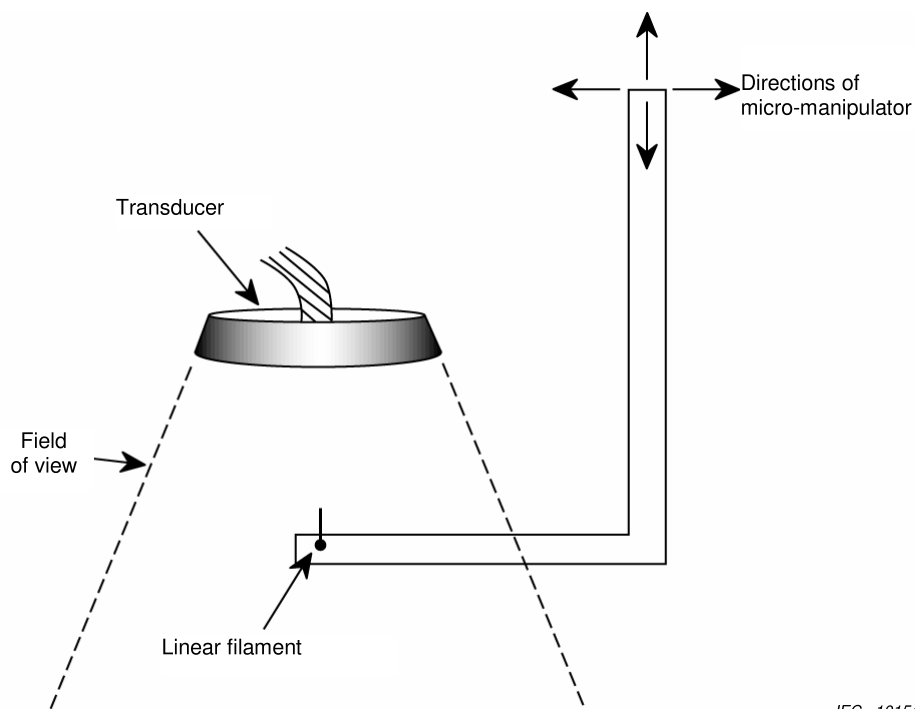
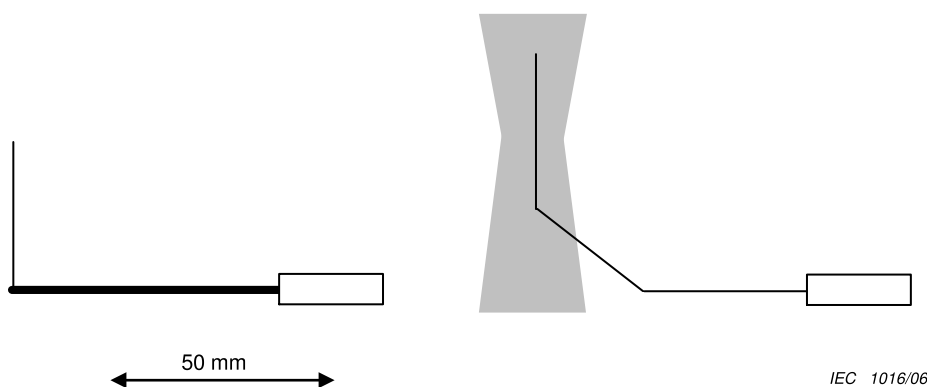


Figure C.2 – Axial resolution test object



IEC 1015/06

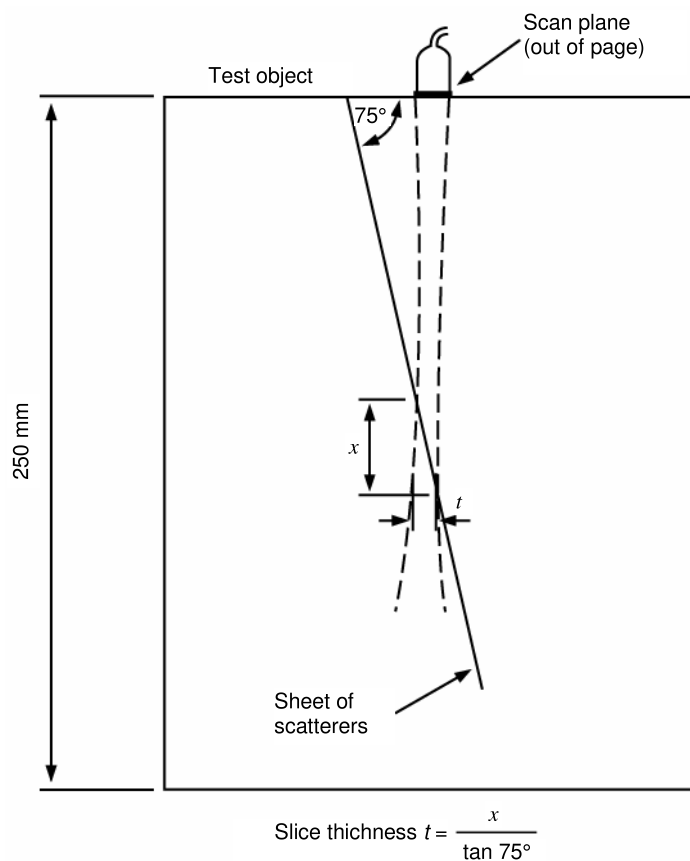
Figure C.3 – Movable single filament in water



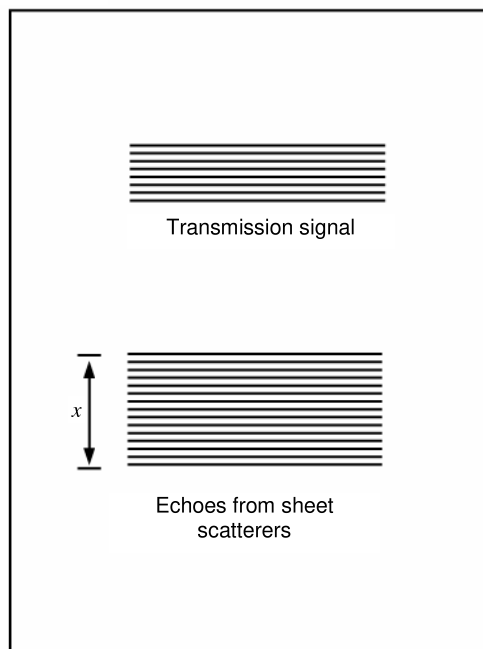
IEC 1016/06

At their right side the targets can be connected to a scanning table.

Figure C.4 – Sketch of two point targets made from dental wire (diameter 0,24 mm)



Image



IEC 1017/06

Figure C.5 – Slice thickness measurement and calculation

Annex D (informative)

Quality parameters derived by PSF-mapping analysis

D.1 General

A quality assessment system is vitally needed to provide an accurate and well-defined set of production-quality parameters for new or refurbished scanners or transducers in acceptance tests before their introduction to medical practice. It is important that products delivered by third-party sales groups, system-refurbishers and/or transducer manufacturers be carefully tested to be able to declare technical parameters of their products to be comparable to those of the new, originally manufactured systems. The methods used for quality assessment in medical applications are not certain and accurate enough to be used for such kinds of technical **performance evaluation**. **PSF-mapping analysis** gives reliable parameters suitable for this kind of tests. These parameters do not directly indicate the effectivity of a clinical diagnostic process, even though a close correlation between the assessed technical quality and success in the diagnostic process may be expected [40].

The ultrasound scanner used as a diagnostic system is composed of the system-control/user-interface unit and the ultrasonic-transducer assembly. Either unit can contain the transmitter- and the receiver- electronic systems and some of the beam-former electronics. The **ultrasonic transducer** converts electrical signals to ultrasound field and vice versa. Electrical and acoustic parameters of the transducer determine quality of the scanning **ultrasound beam**. The electronic system controls the transmitted and received ultrasound signal, conversion from mechanical to electrical signals, and the signal processing and conversion to video-signal inputted to the imaging unit. The imaging unit transfers the information to the human preceptors. The **PSF-distribution analysis** evaluates qualitative parameters of the whole ultrasound-scanner system, excluding the display unit. The analysed signal is affected by the quality of the whole imaging cycle, and the transmitting and receiving parts of the scanner. The analysed system function is affected by a complex set of control functions. Therefore, it is important that the combination of the control settings of the scanner be exactly specified and recorded as a part of the measurement.

D.2 Method

Annex D describes a method for precise and reliable measurement of several qualitative parameters of whole ultrasound scanning systems including both the transmitting and receiving parts of the systems, excluding the parameters of scanner display. The method is based on **PSF-distribution analysis** over a scanning area. In the case of **PSF-mapping**, the measured parameters are derived by analysis of sonograms generated by scanning a spherical target moving over a defined scanning volume on a specified trajectory.

The **PSF-mapping** system evaluates a set of parameters acquired over a user-defined area in one scanning plane of a B-mode grey-scale sonogram, scanned in a tank filled by degassed working liquid and using one measuring procedure. The whole target sonogram is not evaluated in the **PSF-mapping analysis**. The test signal is obtained by reflection of a transmitted ultrasound wave from a point-reflector surface and working-liquid boundary only. The point reflector used is a highly reflective sphere of diameter D [41].

The method is suitable for all kinds of echo(reflections)-evaluating sonographs using different types of beam-forming and plane-wave compounding of ultrasound signal in the frequency range 0,5 MHz to 50 MHz. The upper frequency limit is determined by a ball target of minimum diameter available to assure reflection effectivity and fulfil the condition $\lambda \leq D \leq 4\lambda$, where λ is the ultrasound wavelength in the working liquid [42]. Further limiting factors are a minimum size of step and precise mechanical construction of the positioning system to assure measurement reliability and adequate scan size.

The method is relevant for all the types of transducers used with these scanners, including

- mechanical probes including annular arrays,
- electronic phased arrays,
- linear arrays,
- curved arrays,
- two-dimensional arrays, and
- 3D-volume scanning probes based on a combination of the above types.

The **PSF**-measuring system is not a tissue-mimicking object. It is dedicated to performing accurate, stable and reliable measurements under conditions appropriate to achieving these measurements of parameters, some of which may be obtained by use of sophisticated electronic measurements of the scanner's electronic system and some by **PSF**-mapping analysis only [43].

The following data are acquired and are analysed using the method:

- a) the ROI digital image stored for the scanned-plane axis in each point of the **measuring grid**;
- b) the echo-signal amplitude distribution over the measured area;
- c) the distribution of the parameter $W_{F, HM}$ which is Full-Width-at-Half-Maximum (FWHM) of the **point-spread function (PSF)** in the azimuthal direction over the measured area;
- d) the distribution of the parameter $W_{H, HM}$ Half-Width-at-Half-Maximum (HWHM) of the **point-spread function (PSF)** in the axial direction over the measured area;
- e) the peak echo-amplitude received $a_{r, max}(x, y, z)$ at each y_k step of the target position in the elevation (transversal) direction;
- f) the (x, y, z) coordinates set for stored position of the point reflector generating $a_{r, max}(x, y, z)$ from MER in each point of the **measuring grid** (position in centre of ultrasound beam).

Data analysis derives the following ultrasound scanner parameters and functions:

- 1) focal areas in both the azimuth and the elevation directions;
- 2) visualization of the distribution of ultrasound scanning lines;
- 3) manufacturer's preloaded **TGC** function;
- 4) width (elevation) of the scanning plane over the depth of scan;
- 5) side-lobes signal-level distribution in the scan plane;
- 6) amplification uniformity in the azimuth direction;
- 7) scan geometry linearity and **accuracy**.

D.3 Environmental conditions

The most temperature-sensitive parameters are those assessing geometry of the sonogram and related calculations. The temperature-dependent deviations may be compensated mathematically from known working-liquid temperature and thermal coefficient of speed of sound.

Water condensation on electronic system components should be avoided.

D.4 General requirements of the method

Ultrasound waves produce a **PSF**-signal that is neither singular nor isotropic. Furthermore, the ultrasound **PSF** can be asymmetrical, having different axial and lateral dimensions, and it also

varies with distance from the transducer in both the axial and the azimuthal directions. Thus, it is important that many different measurements of the **PSF** at different positions and depths be performed to obtain representative values of the system's imaging performance at specific positions along the beam axis. It is also important that the measured area be covered by a grid of the measuring points, the density of which is determined by the expected parameters and the **accuracy** of the measurement [44].

NOTE For example, determination of a focal point's position may need an axial step $\Delta z = 5$ mm; visualization of scanning lines demands an azimuthal step $\Delta x = 0,1$ mm for a conventional linear transducer of nominal frequency 3 MHz.

The following features are necessary to apply the **PSF**-analyser to the sonograph:

- a) The sonograph to be tested:
 - video-signal output of live, dynamic scanning available in analog (composite) or digital (DVI-D, HDMI) form;
 - operating instructions or skilled operator to assure proper manipulation and operating adjustment;
 - record of proper evidence and registration of the measurement process, including identification of operator and all apparatus used, record of parameters preset in the measured system, record of environmental conditions, including a time stamp.
- b) The basic configuration of the measurement tank:
 - The **scanning window**(s) is(are) localized in the side wall(s) of the tank.
 - The spherical-target positioning system is fixed on top of the tank, controlling movement of the target fixed in a holder.
 - Filling the tank with degassed working liquid is recommended to prevent bubble generation in the working fluid. Bubbles may mimic the point reflector and/or produce spurious reflections from the point reflector after having accumulated on it, due to surface tension.
 - Temperature should be kept in the specified range to eliminate measurement uncertainties generated by dependence of the speed of ultrasound propagation upon temperature.
- c) The transducer:
 - The transducer is acoustically coupled to the **scanning window** by standard coupling ultrasound gel. The **scanning window** is covered by tough, flexible foil made of material with similar acoustic properties to the working liquid to avoid ultrasound field reflections and absorption.
 - The scanning plane is oriented in the horizontal direction and the transducer is fixed to keep the whole slice thickness below the water surface. It is important that the lateral orientation of the transducer be specified.
- d) The positioning system:
 - A computer-controlled micromanipulator is used to move the target or transducer to a determined position. It is important that mechanical construction, **accuracy** and stability of the positioning system correspond with ultrasound frequency. The shorter the wavelength, the more accurate and robust the system should be to avoid systematic measuring errors.
- e) The control and analysing system:
 - The software controls the video-signal acquisition, determines ROI, selects and saves the ROI frames to be used for analysis and finally maintains the complex analysis of the stored information.
 - The basic parameters used for **PSF**-analysis are the $W_{F, HM}$ (Full Width at Half Maximum, i.e. at -6 dB down from $a_{r, max}$) and the **pixel** level of noise at an area without reflections.

- The $W_{F,HM}$ depends upon the intensity of the received signal. Therefore, it is important that the receiving gain and output power be properly adjusted to utilize the whole dynamic range of the analyser.

D.5 Measuring conditions

D.5.1 General

In Figure D.1 a principal schematic of the **PSF**-analyser is introduced. The **PSF**-analyser consists of the data-acquisition components (test-tank and point-reflector parts of the schema), a linear transducer, a personal computer (PC) with video-signal input, running acquisition and analysis software. The sonograph being measured is shown with a linear transducer but the results analysis is displayed for a sector-scan transducer to illustrate transducer-type independency of the system.

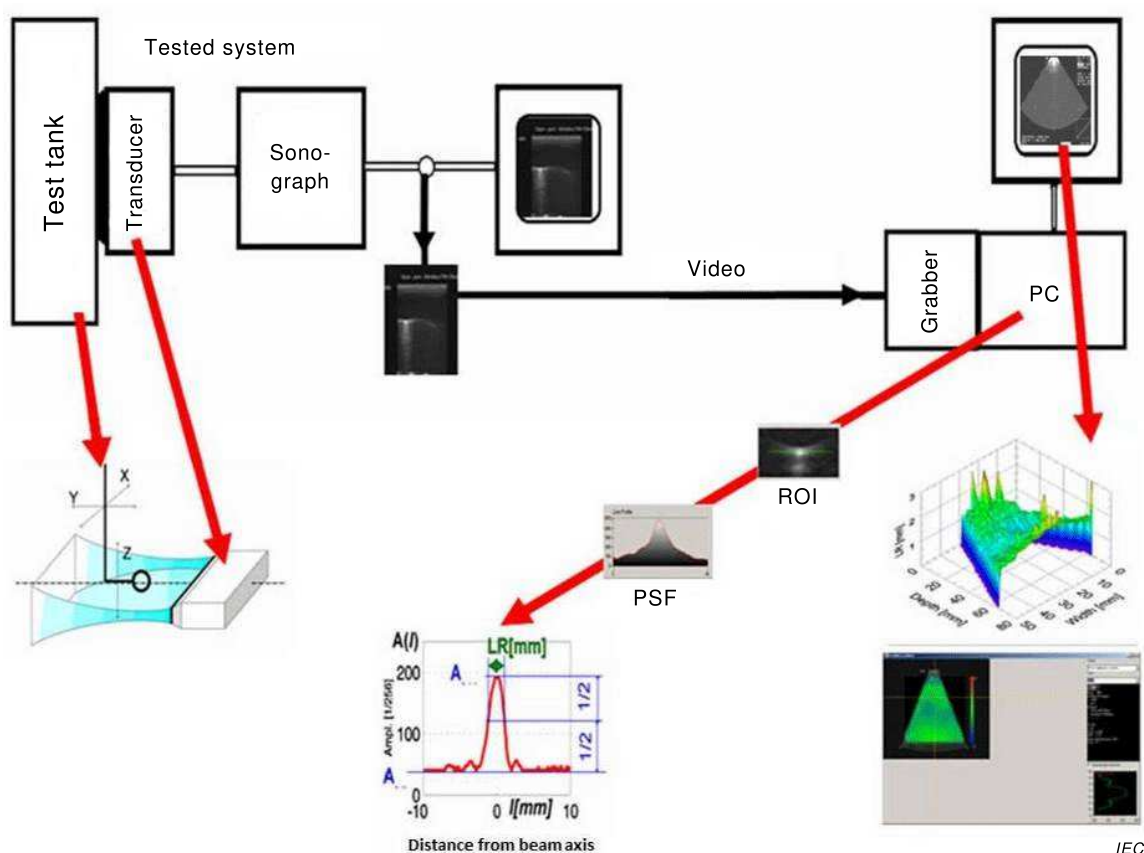


Figure D.1 – Principal schematic of the PSF-analyser function

D.5.2 Sonograph

D.5.2.1 General

Sonographs are equipped with a large set of different control functions to ensure optimal handling of received ultrasound signals to create the best image. These control functions affect remarkably the measurement results because the method evaluates the signal after it has passed through a whole imaging system. Therefore, it is important that all the adjustments and settings be carefully documented in the measuring protocol to be available for use for repeated measurements.

D.5.2.2 The amplitude transfer-characteristic adjustment

The sonograph receiver employs a logarithmic amplifier to compress a very wide range of received signal amplitudes (more than 100 dB). It is important that all the other additional, nonlinear, so-called pre-processing and post-processing functions be switched off or adjusted to linear working regions to avoid further nonlinear distortions of the signal. This requirement includes functions dedicated to eliminating different types of noise either by scan-sequences correlation or averaging or cutting off low-level signals.

It is important that the amplified signal-peak amplitude limitation due to low amplifier dynamic range be eliminated to assure correct measurement of the FWHM-function. Therefore, it is also important that the widest dynamic range accessible by the measuring system be selected, except in the case when the gain of the sonograph is not sufficient to achieve full dynamic range for the $A_{r,max}$.

D.5.2.3 The look-up table setting

The relation of the grey-scale level to the digitalized amplitude of the signal is derived from a look-up table (LUT) or Gamma curve. If possible to assure a pure logarithmic characteristic of the system, only a most linear LUT should be used. It is important that all the settings be documented in the measuring protocol, to be recorded for use during further periodical measurements. A method to find the most linear LUT should be part of the measuring software.

D.5.2.4 The measured gain adjustment

The sonograph's receiver gain is controlled in two parts: the **overall gain** (G_o) and the **time-gain compensation (TGC)** functions. Their adjustment prior to measurement should be as follows: The **TGC** is adjusted first. The basic condition is to adjust the **TGC** to be constant within the whole range of the scanned depth. The basic condition of the **TGC** adjustment is its independence from the depth parameter (elimination of the function). Then the G_o is increased to just obtain little visible thermal noise of the system in an echo-free area on the screen, e.g. when the transducer is energized when coupled to the air. It is important that the G_o be decreased or a smaller reflector used, if the $A_{r,max}$ amplitude limitation occurs.

Some sonographs utilize an **ATGC** (automatic time gain compensation) function. It is important that this function be disabled for gain (sensitivity) dependence from depth and focal-points localization from MER-profile assessments.

D.5.3 Measuring tank

D.5.3.1 General

The measuring tank serves as a holder of the volume of homogenous non-reflecting medium – the working liquid – in which the point reflector is moving by the precise, computer-controlled electromechanical system. An appropriate **scanning window** allows effective acoustic connection between the working liquid inside the tank and the active part of the measured transducer. The size of the measuring tank should be appropriate to the size of largest scanned area expected.

D.5.3.2 Working liquid

It is important that the working liquid used in the tank be degassed to avoid collection of bubbles at the point reflector [45]. It is also important that the temperature be recorded for the measurements to be based on accurate values of speed of sound, which significantly affects reading geometry of the scan. However, measurements based on reading the echo intensity are not remarkably affected by working liquid temperature because temperature dependence of both the reflectivity and the water absorption are negligible in this case.

The accuracy of the distance calculations is determined by the accuracy of knowledge of ultrasound speed of propagation in the working liquid. The coefficient of speed of ultrasound in

pure water as a function of temperature is approximately $4 \text{ ms}^{-1}\text{°C}^{-1}$ in the temperature range 10 °C to 30 °C . According to the distance calculation from the echo delay, a change of temperature of $\pm 1 \text{ °C}$ results in error of $\pm 0,15 \%$ of the measured distance value. A calculation giving the temperature error compensation is possible by use of an expression given by [46] or another by [47].

D.5.3.3 Reflections

Due to low ultrasound absorption of the working liquid and high reflectivity of the tank walls and relatively small size of the measuring tank, many multiple reflections are generated in the scanned area. These multiple reflections may be eliminated by a simple system of absorbers and reflector shielding because a pulsed signal is generated and only a small ROI surrounding and close to the reflector is analysed in the sonogram.

D.5.3.4 Scanning windows

The ultrasound signal is transferred between transducer and reflector via the **scanning window**. The window works as an acoustic coupler between the active surface of the transducer and the working liquid. The window is closed with flexible foil made of material with similar acoustic properties to the working liquid to assure effective transfer of the ultrasound signal and elimination of ultrasound-signal reflections and absorption. The transducer is attached to the foil with use of conventional coupling gel so that the whole active surface of the transducer is in acoustic contact with the working liquid. Proper shape of the window is needed to assure effective contact of different types of transducers. The measured transducer holder is part of the **scanning window**. Its design should ensure adjustment of the scanning plane to be parallel to the xz -plane of the target positioning system and stable transducer positions during measurement.

D.5.4 Target

D.5.4.1 General

The target system is composed of the sphere reflector, the reflector holder and the reflector positioning system.

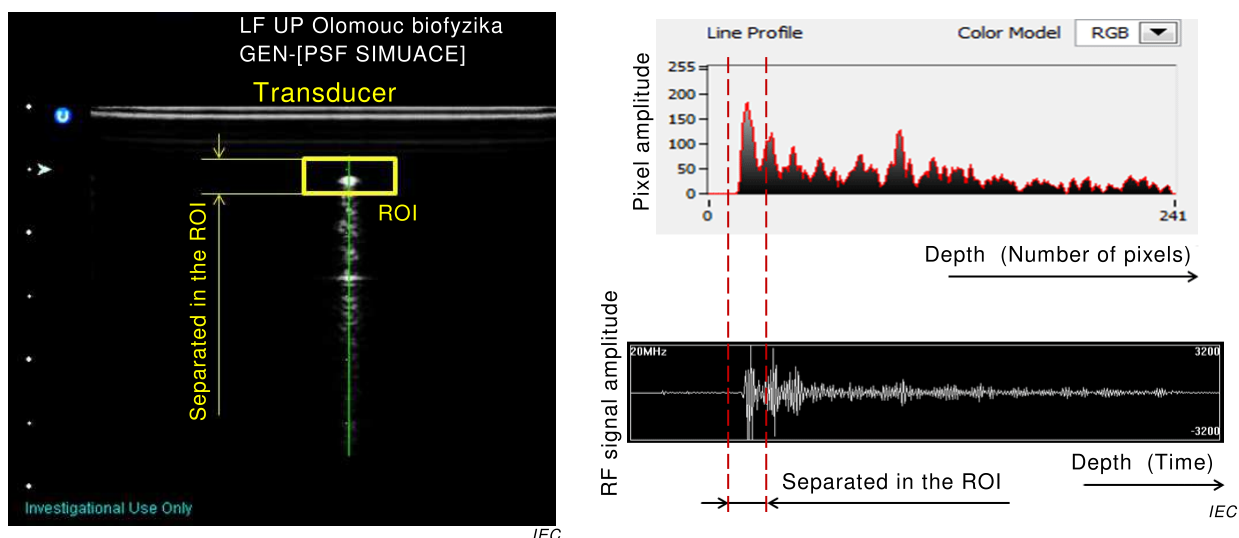
D.5.4.2 Reflector

In contrast to C.4 and Figure C.4, the point reflector is not the flat end of a wire but a sphere made of steel or similar highly reflective material. The reflector is fixed to a positioning system with use of a holder made of tiny, hard wire. The sphere reflection is independent of the incident angle of irradiation.

The diameter D of the rigid metallic reflector is critical for the method used but should be kept within a range $\lambda < D < 4\lambda$ according to [42]. Use of a reflector of $D < \lambda$ is possible, but the reflected signal amplitude will not be strong enough for low-sensitivity systems. Therefore decrease of the target diameter, D , may be employed when too-sensitive sonographs limit high amplitude of signals, even if the **overall gain** G_0 and output power are adjusted to minimum levels (see D.5.2.4).

If $D > 4\lambda$, values and variations of the $W_{F,HM}$ increase to an erroneous extent in both the lateral and the elevational directions.

The reflections from any inner structure of the sphere and multiple reflections in the sphere and/or sphere resonances [48] [49] are eliminated in the received-signal evaluation process. They do not affect the measurement results. The **PSF**-mapping method evaluates reflections of a transmitted ultrasound wave from a point-reflector surface and working-liquid boundary only (Figure D.2). It is important that the sphere reflector be made of a highly reflective material, i.e. the difference of characteristic acoustic impedance of the working liquid and sphere-target material should be as high as possible.



The yellow rectangle specifies ROI. The green line corresponds with the signal displayed in b).

A linear transducer, nominal frequency 10 MHz, was used.

a) Scan of spherical target 0,6 mm diameter

In the top figure, RF signal amplitude is converted to pixel brightness, time domain to space domain. The ROI separated for analysis is marked.

In the bottom figure, the RF signal corresponds with the green line in a).

b) Process of demodulation (top) and the target-scan received-RF signal (bottom)

NOTE The late reflections are not included in the analysed data.

Figure D.2 – Principle of elimination of internal multi-reflections in the spherical target using filtration in time domain

D.5.4.3 Sphere holder

The holder fixes the sphere to the positioning system. It is important that the holder construction avoid any influence upon the reflected signal. It is also important that the connection be rigid enough to eliminate any shift of the target during its displacement in the water, due to hydraulic resistance. A possible alternative for such a holder is a tiny, stiff wire bent to rectangular form of the letter L with the sphere fixed on one end (the bottom – horizontally oriented) of the wire while the second (the top) end is mounted to the positioning system. It is important that the wire diameter be smaller than D , but the ratio is not critical. Reflections from the holder are positioned outside the ROI; therefore they do not affect the measurement results.

The holder construction should allow easy replacement (exchange) of the sphere reflector to allow use of sphere reflectors of various diameters, according to the transducer's nominal frequency and the fundamental wavelength of ultrasound in the working liquid ($\lambda < D < 4\lambda$).

D.5.4.4 Positioning system

The positioning system (micro-manipulator) is dedicated to placing the sphere within the scanned area in the working liquid. The required system **accuracy** and stiffness is determined by the measuring ultrasound frequency and transducer-scanning parameters.

D.5.5 Signal acquisition

It is important that live, full-resolution video be captured from the sonograph for processing the measurements.

Two basic types of signals are available from the sonograph output. The old and low-end types are equipped with analog composite VHS video output in PAL standard only. The newer and higher technological level systems do not use the obsolete analog video but use digital video

DVI-D or HDMI outputs. Both types of video signal need special hardware (video-grabber) to be acquired by the computer for further analysis. Most higher end systems have direct digitally detected- and/or IQ-data output.

The following parameters of the acquired video signal are essential for recording and analysis:

- the ultrasound scanner frame rate;
- dynamic range and amplitude resolution (quantization-step size);
- spatial resolution of the sonograph, related to number of lines for analog video signals and/or size of **pixel** for digital video signals.

The digital resolution parameters are not too critical for the resolution of the measurement results, because the analysing system may interpolate between adjacent, discrete values.

The **pixel** value corresponds to echo amplitude received. The amplitude dynamic range of the analysed input signal is derived from the video monitor's grey-level range, which is 256 grey levels at maximum. Therefore, 8-bit (256 grey levels) digitalization is a minimum to fulfil the needs of inputted analog-signal conversion.

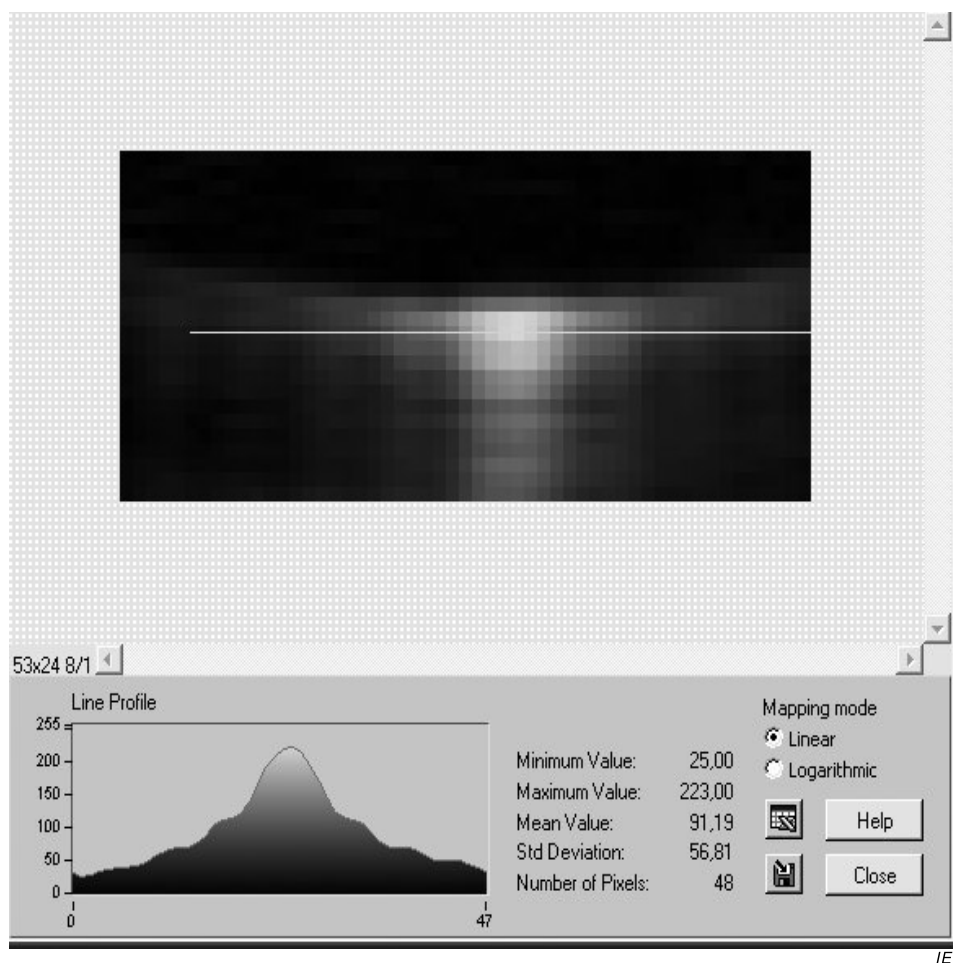
D.5.6 Signal analysis

The system grabs a continuously recorded video signal from the sonograph. Prior to running the acquisition, a proper region of interest (ROI) is chosen in the sonogram. The software controls ROI position according to the target placement in the scanning plane to keep the target inside the ROI. Only those ROI frames are used for further analysis, which coincide with the target position in the pre-programmed measuring grid at points (x,z) within the measured volume of the scan. Each of these ROIs is analysed to find the **pixel** containing $a_{r,\max}(x,z)$ the highest value of MER when the target moves in an elevation (y) direction. Not whole ROIs but only their highest **pixel** value $a_{r,\max}(x,y,z)$ is stored for each y , including the corresponding target position (x,y,z) for each elevation path. Only one ROI containing the highest of the $a_{r,\max}(x,y,z)$ from each elevation set (x,z) is stored as a picture $I(x,y,z)$. The stored $I(x,y,z)$ is a picture of the target on the central (axis) of the ultrasound beam and therefore may be analysed to find the **PSF** in both the azimuth and the axial directions. See Figure D.3 and Figure D.4.

The input video-signal level and the transfer characteristic of the video-grabber may affect the measurement results when the input data are obtained from a composite video signal. Thus, it is important that the video-grabber amplitude-transfer characteristic be well adjusted to assure use of the whole range of the AD-converter to prevent artificial limitation of the high level signal.

The digital video-grabber transfer function does not affect the measurement results when digital video-signal input is used.

An example of the **PSF** derived from the data recorded in one point of the **measuring-grid** is presented in Figure D.3.



NOTE The white line is the line of **PSF**-acquisition from the digital picture of the target at position (x,y,z) . The line passes through the pixel containing the maximum of the signal $a_{r,max}(x,y,z)$.

Figure D.3 – A pixel maximum level and PSF-trace estimation in ROI stored digital data

D.6 Results – measured parameters

D.6.1 Spatial resolutions in PSF-mapping

D.6.1.1 General

The **PSF**-mapping method demands modification of the spatial-resolution evaluation method defined earlier (See 3.46, 3.50 and 3.53). The assessment of target displacement complicates the mapping process; it has been changed to simply deriving the spatial resolution value in **PSF**-mapping from a single measurement in the **measuring-grid** point. The values of mapped parameters may differ slightly from the ones defined earlier (3.46, 3.50 and 3.53), but they are measured with higher **accuracy** and repeatability when using the mapping system [44].

The spatial resolution is defined as a distance. The lateral and axial resolutions are enumerated from FWHM- and/or HWHM-parameters, respectively. These parameters are derived from the **PSF** in number of **pixels** and recalculation using the **pixel** size in a scanned area is needed. The **elevational resolution** is derived directly from positions of the reflector generating half-maximum reflected signal during vertical displacement of the reflector.

The lateral- and axial-resolution conversion to units of distance should be performed by evaluating software using the following relations:

$$R_{L,PSF} = W_{F,HM,n} \times p_x \quad (D.1)$$

$$R_{A,PSF} = 2 \times W_{H,HM,n} \times p_z \quad (D.2)$$

where

$W_{F,HM,n}$ and $W_{H,HM,n}$ are normalized values according to Formula (D.3) in D.6.1.2;

$R_{L,PSF}$ is **lateral resolution in a PSF map**;

$R_{A,PSF}$ is **axial resolution in a PSF map**;

p_x is lateral size of one **pixel** in scanned area;

p_z is axial size of one **pixel** in the scanned area.

D.6.1.2 Lateral resolution in PSF-mapping – $R_{L,PSF}$

Resolution in the lateral direction can be approximated from the FWHM of the **PSF** as the $W_{F,HM}(x,z)$ which is a distance between **pixel** numbers where there is a dip of –6 dB in **pixel** values relative to the maximum value. Because the $W_{F,HM}$ value depends upon the value of the maximum amplitude of the **PSF**-profile, normalization according to the maximum value is necessary. A linear extrapolation is used by application of the formula

$$W_{F,HM,n}(x,z) = W_{F,HM}(x,z) \times \frac{M}{a_{r,max}(x,z) + 1} \quad (D.3)$$

where

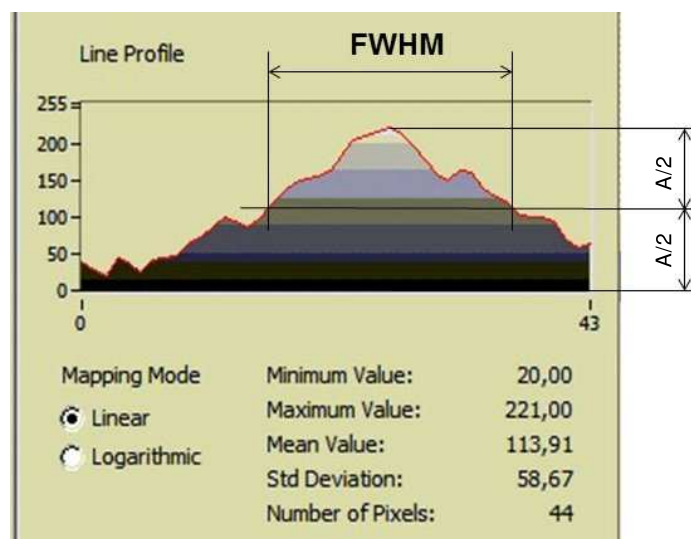
$W_{F,HM,n}(x,z)$ is the normalized value of FWHM at point (x,z) of the **measuring grid**;

$W_{F,HM}(x,z)$ is the measured value of FWHM at point (x,z) of the **measuring grid**;

x,z are coordinates of a point in the **measuring grid**;

$a_{r,max}(x,z)$ is the MER **pixel** value measured for the reflector at the point (x,z) ;

M is number of quantization levels defined by $M = 2^m$ where m is number of **pixel** bits.

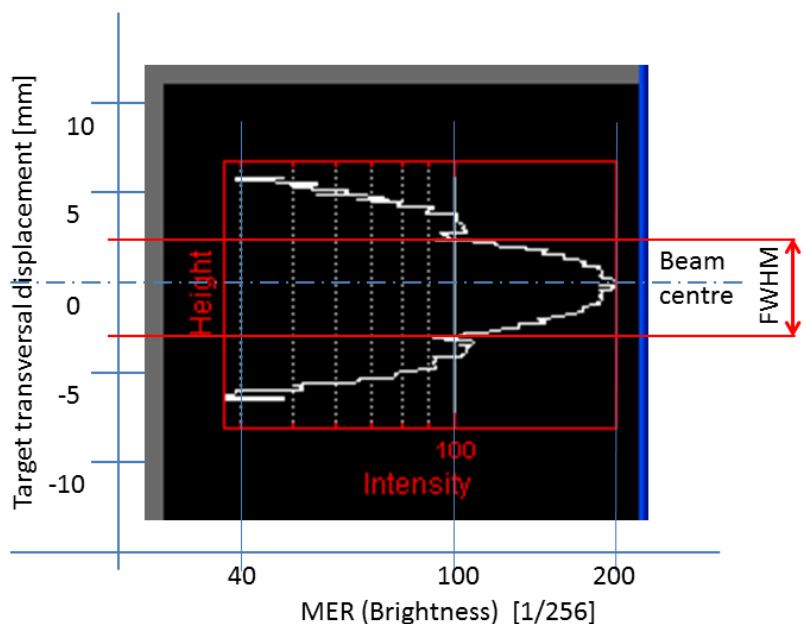


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Figure D.4 – The derivation of FWHM-value from the lateral-amplitude profile of PSF received-echo signal ($a_{r,max}(x_i,z_k) = 221$) with reflector positioned on axis of the ultrasound beam in point (x_i,z_k) of the measuring grid

D.6.1.3 Elevational resolution in PSF-mapping – $R_{E,PSF}$

Resolution in elevational directions can be approximated as the FWHM of a function MER versus point-reflector vertical position at the measured points, where there is a dip of –6 dB below the maximum value of MER; see Figure D.5 and Figure D.11. Because the $W_{F,HM}$ depends upon the value of MER, normalization according to the maximum value is necessary. The normalization procedure applies (see Formula (D.3)).



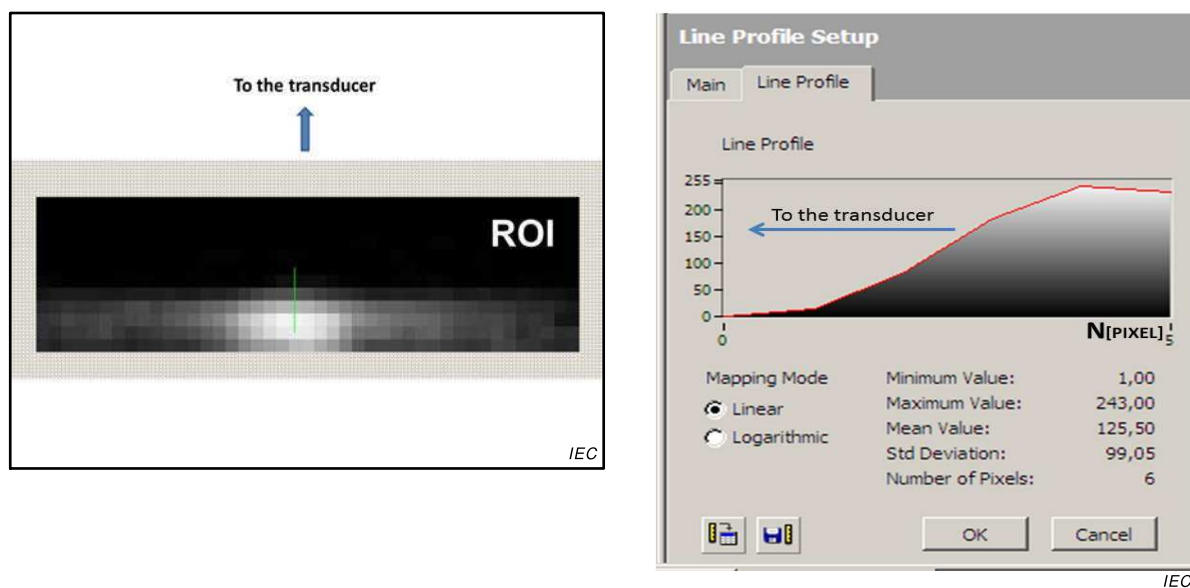
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Figure D.5 – The FWHM-derivation from the elevational (transversal) profile of MER in one point of the measuring grid

The $R_{E,PSF}$ distribution over the scan plane specifies its thickness according to depth of scan and therefore elevational focus and focal-area position. A position of $R_{E,PSF}$ minimum localizes the elevational focal area.

D.6.1.4 Axial resolution in PSF-mapping – $R_{A,PSF}$

Because the echo generated at the front surface of a ball reflector is evaluated for axial direction only (Figure D.6), the **PSF**-mapping process needs a different method to access axial resolution as compared to the $R_{L,PSF}$ and the $R_{E,PSF}$. The Half-Width-at-Half-Maximum (HWHM) is used instead of the FWHM; see Figure D.7. The FWHM may be simply derived as two times HWHM to obtain the $R_{A,PSF}$ (see Formula (D.2)).



The pixel containing MER is crossed by the green line in the axial direction. The line determines a set of pixels to be evaluated.

a) The ROI containing the target sonogram

b) The axial profile line of the evaluated pixels, and basic statistical data

Figure D.6 – The derivation of the reflected-signal axial-profile from the ROI

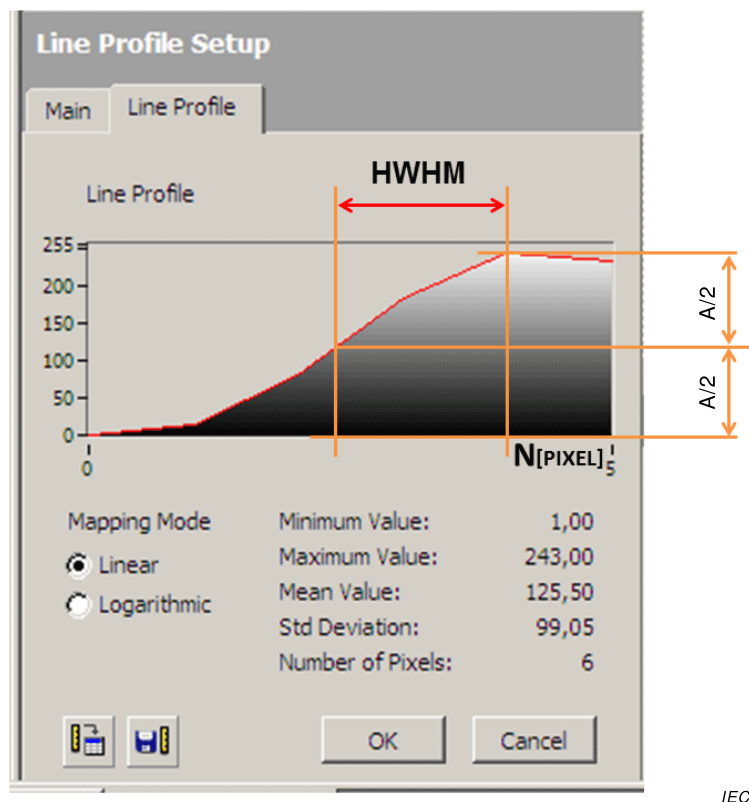
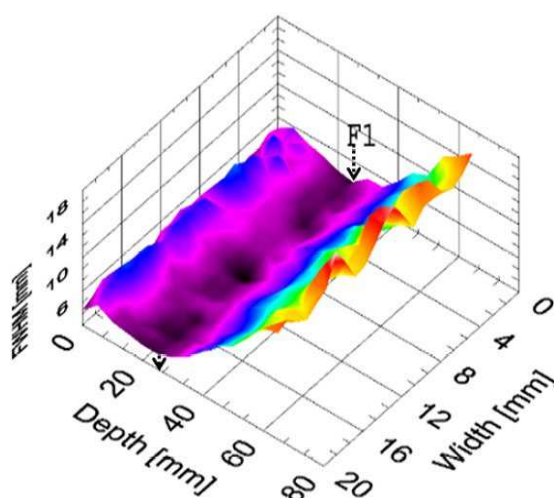


Figure D.7 – The enumeration of the $W_{H, HM}$ parameter from the axial-profile line: $a_{r, \max}(x, z) = 243$

Because the $W_{H, HM}$ value depends upon the value of the maximum of MER, normalization according to the maximum value is necessary. The normalization procedure applies (see Formula (D.3)).

D.6.2 Focal areas in both the azimuth and the elevation directions

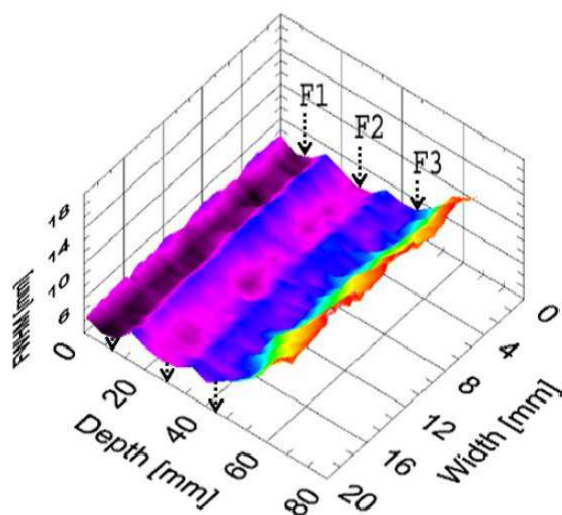
The focal-area position is characterized by local extremes of spatial resolution and sensitivity level. Therefore the focus position is derivable from both the FWHM of **PSF** and the echo-amplitude distribution over the corresponding direction. The focal area has lowest value of the $W_{F, HM}$ (the best spatial resolution) and highest value of the received echo amplitude. In the case of dynamic focusing, not only the position but the number of foci is available from the characteristic. See Figure D.8 and Figure D.9.



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NOTE The focal point F1 (local minimum of $W_{F, HM}$) is at depth approximately 25 mm in the measured area.

Figure D.8 – The distribution of FWHM over a scan area of width 20 mm to depth 80 mm, made with a monofocal scan using a linear 5 MHz transducer



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NOTE The focal-point positions (local minima of the $W_{F, HM}$ values) are observable at depths approximately 10 mm, 27 mm and 42 mm in the measured area.

Figure D.9 – The distribution of FWHM over a scan area of width 20 mm to depth 80 mm, made with the same system as for Figure D.7 but using three focal points (F1, F2 and F3) for the scan

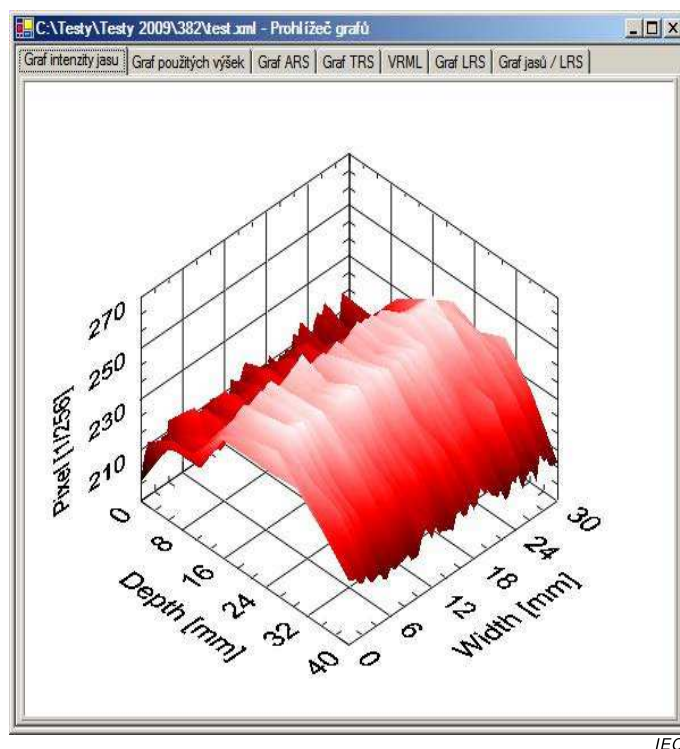
D.6.3 Amplification uniformity in both the axial and the azimuthal directions

The 2D-sonogram is composed of individual **scan lines** that fill the scanning plane in the azimuthal direction. (The 3D-sonogram is created from separate 2D-scan planes.) The MER amplitude varies in the scan plane due to **TGC**, density of the **scan lines**, localization and number of foci, non-uniformity of channel amplifiers and different sensitivity of transducer elements. Some of these effects are desirable, e.g. **TGC**, focusing; some spoil the picture, e.g. defective amplifiers or elements.

All the information can be obtained by analysing the distribution of MER over the scanned area. See Figure D.10.

D.6.4 Preloaded TGC-function

When the **TGC** manual control is adjusted to be depth independent, the manufacturer's preloaded **TGC**-function may be derived from the MER-level distribution along the axial profile. See Figure D.10. The preloaded **TGC**-curve is obviously pre-programmed in the sonograph software and is transducer type-dependent. An aim of this function is to reduce sensitivity for the image area close to the transducer or according to the individual element position in the array of elements in the transducer (e.g. phased array transducers). The echo-amplitude distribution over the scan area contains information about the gain distribution, because the scanned volume in the measuring bath is filled in by degassed working liquid with negligible ultrasound field absorption.



NOTE A linear transducer with nominal frequency 10 MHz, scan width 38 mm, using one focus active in 20 mm depth of the measured area was used.

**Figure D.10 – Plot of the distribution of MER-intensity over the scanning area
30 mm wide and 40 mm deep**

D.6.5 Visualization of ultrasound scanning lines

When the elevation step of the ball-target placement is small enough, a fluctuation of MER-amplitude is observable along the scanning lines in the elevation direction. The maximum intensity is received when the reflector stays on the **ultrasound beam** axis. The same effect is observable on the plot of the FWHM-distribution, only the reflector position on the beam axis

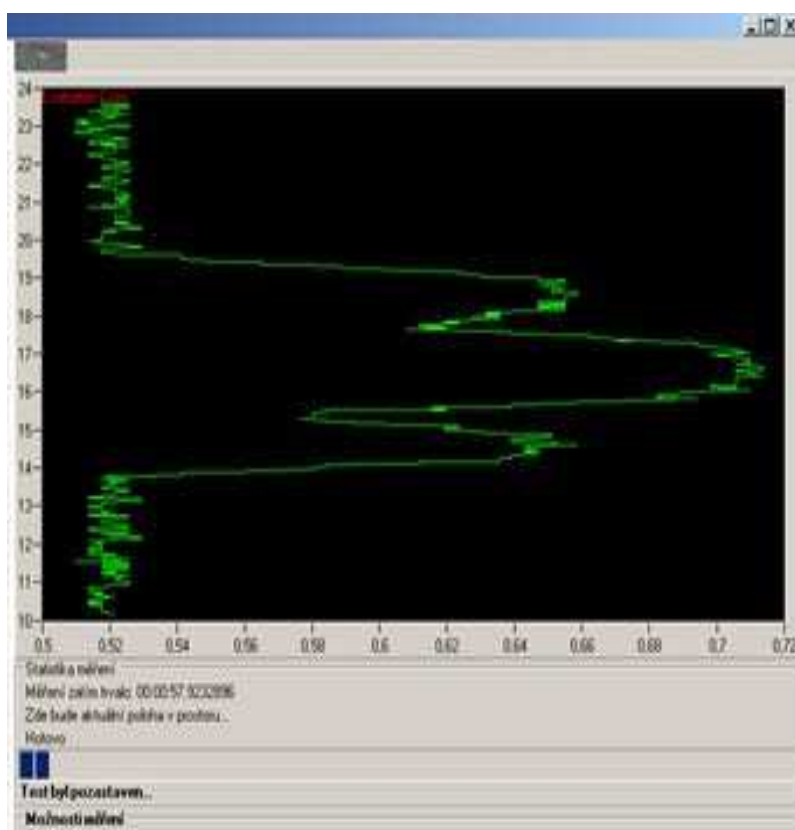
generates drop-down of the parameter value. This method may be used to visualize separate **scan lines**, their density and changes of profile. See Figure D.10.

D.6.6 Elevation width of a scanning plane

This is assessable from the elevational profile of MER-amplitude for the spherical target passing perpendicularly to the scanning plane in each point of the **measuring grid**. The data generates the FWHM of **PSF**-distribution in the azimuthal direction over the scan area.

D.6.7 Side-lobes level in elevational plane

The MER-profile recorded in the elevational direction can display level and beam direction of **side lobes**. See Figure D.11. The side-lobes level is an important parameter of sonograph-imaging quality. Side-lobes generate false reflections creating ghost structures in the sonogram.



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NOTE The abscissa scale is MER $a_{r,\max}(x_i, y, z_k)$ enumerated as a ratio relative to value 256 (M for 8-bit pixel); the ordinate scale is elevational displacement y of the reflector in millimetres.

Figure D.11 – The elevational profile recorded from $a_{r,\max}(x_i, y, z_k)$ values for a spherical target passing perpendicularly to the scanning plane in one point (x_i, z_k) of the measuring grid

In Figure D.11 the main peak value corresponds with target position in the centre of the beam. The two minor peaks are generated by side-lobes in the azimuthal plane. The relative value of MER is on the abscissa; the ordinate gives the target displacement in millimetres.

D.6.8 Accuracy of sonogram geometry and caliper measurement

The system moves the reflector to specified positions with defined **accuracy**. The reflecting point gives point reflections in the sonogram at geometric positions specified with **accuracy** defined by **accuracy** of mechanical positioning system, speed of sound in the water, the

precision of reflection-delay measurement and size of **pixel**. These data may be used to check the **accuracy** of the displayed sonogram geometry, including the dynamic process needed for TM-mode examination.

The method is suitable for an imaging-geometry test of the reflected-signal imaging modes like A-mode, B-mode 2D and 3D, TM, US-elastography. The detailed instructions for the evaluation process are available in 6.4 and Clause 7. The referenced fixed test-objects with reflecting structures may be substituted by a spherical reflector positioned at an exactly defined position. The spherical reflector is approach-angle independent. A partial geometric parameter will be evaluated using suitable software.

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